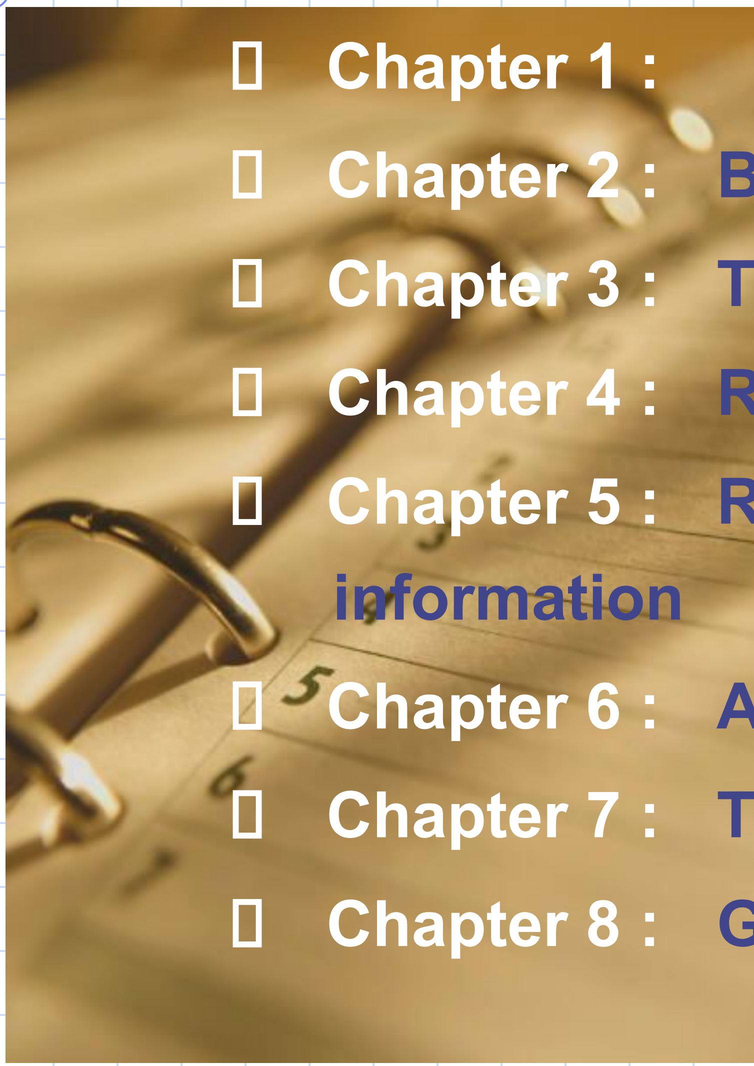
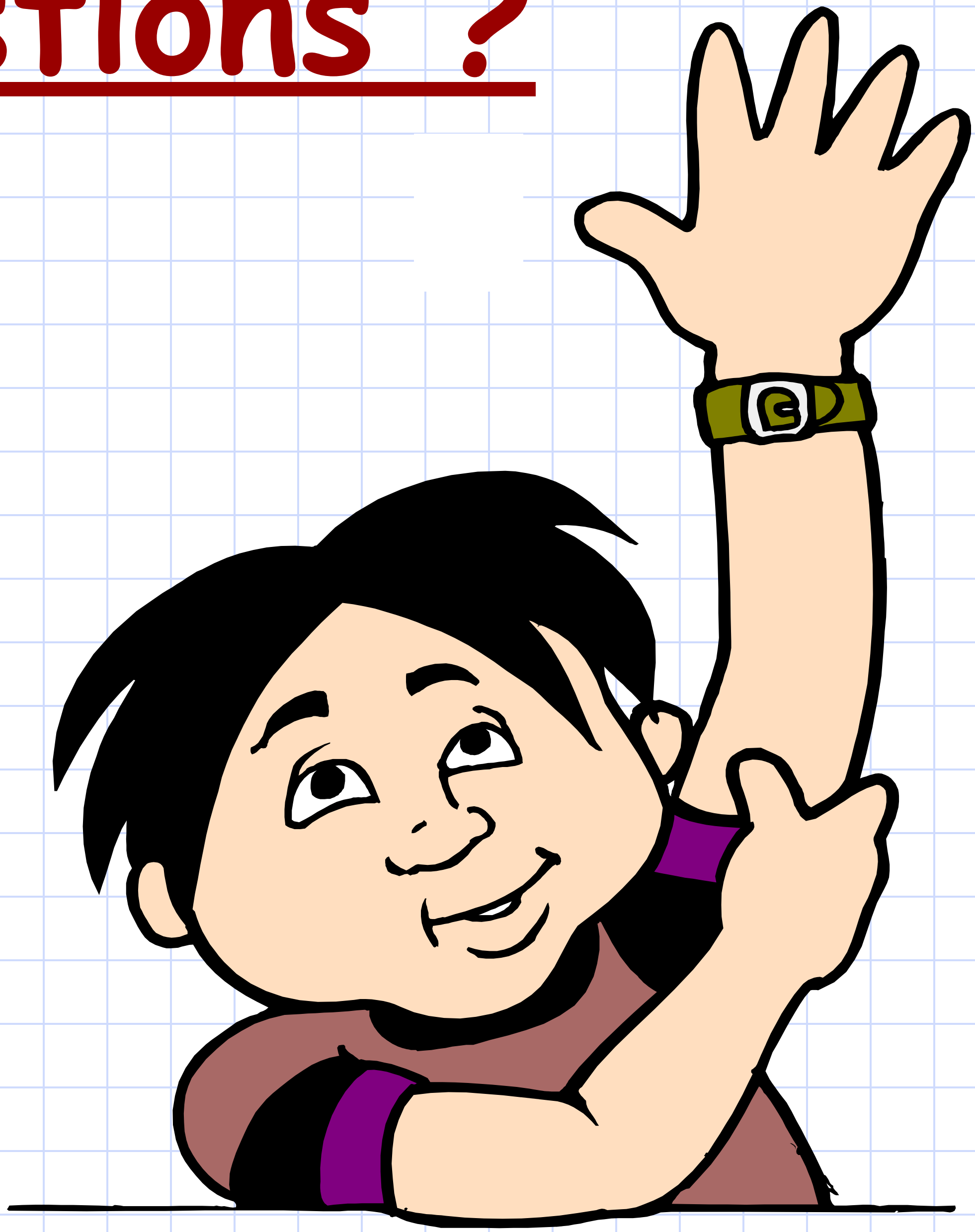
Produced by:
MSC Performance Group
Operations and Engineering Department
Vodafone Egypt

GSM System Survey

Contents



Questions ?



Chapter 1 : Introduction

Banner Advertisement. America 1929

Chapter Objectives

By the End of this Chapter you will:

- Know the History of Telecommunications
- Know the history of GSM evolution
- Know about different GSM Phases

History of Wireless Communication

Date	Place	Activity
1921	Chicago	2 MHz Vehicular Mobile Radio system for Police Applications
1930s	US	Invention of Amplitude Modulation. Half Duplex transmission
1935	US	Invention of Frequency Modulation. Improved Audio Quality
1946	St. Louis	FCC First Domestic Public Land Mobile service at 150 MHz
1969	Europe	Denmark, Finland, Iceland, Norway and Sweden form the first standardization group: Nordic Mobile Telephony (NMT)
1973	Europe	NMT specifies standards to allow mobiles phones to be located within or across their networks. Basis for roaming idea
1979	Chicago	Advanced Mobile Phone System. First Cellular Analog network
1991	Europe	The First Digital Cellular Standard (GSM) is launched

History of GSM

Date	Activity
1982	• Nordic Telecom and Netherlands PTT send a proposal to the ***Conférence Européenne des Postes et Télécommunications*** (CEPT) to specify a common European mobile telecommunication service. • The European Commission (EC) issues a directive, which requires member states to reserve frequencies in the 900 MHz band for GSM.
1986	• Field tests were held in Paris and a GSM permanent nucleus was created and comparative tests of 8 prototypes were performed. • The choice was Time Division Multiple Access (TDMA) or Frequency Division Multiple Access (FDMA).
1987	• A Combination of TDMA and FDMA selected as the transmission tech. for GSM. • September – 13, operators and administrators from 12 areas in the CEPT GSM advisory group sign the charter GSM (Groupe Spéciale Mobile) MoU "Club" agreement, with a launch date of 1 July 1991. • The original French name was later changed to **Global System for Mobile Communications**, but the original GSM acronym stuck. • GSM spec drafted. (Digital Transmission, Time Multiplexing of order 8 and slow Hopping)
1988	• CEPT began producing GSM specifications for phased implementation. • Another five countries signed the MoU

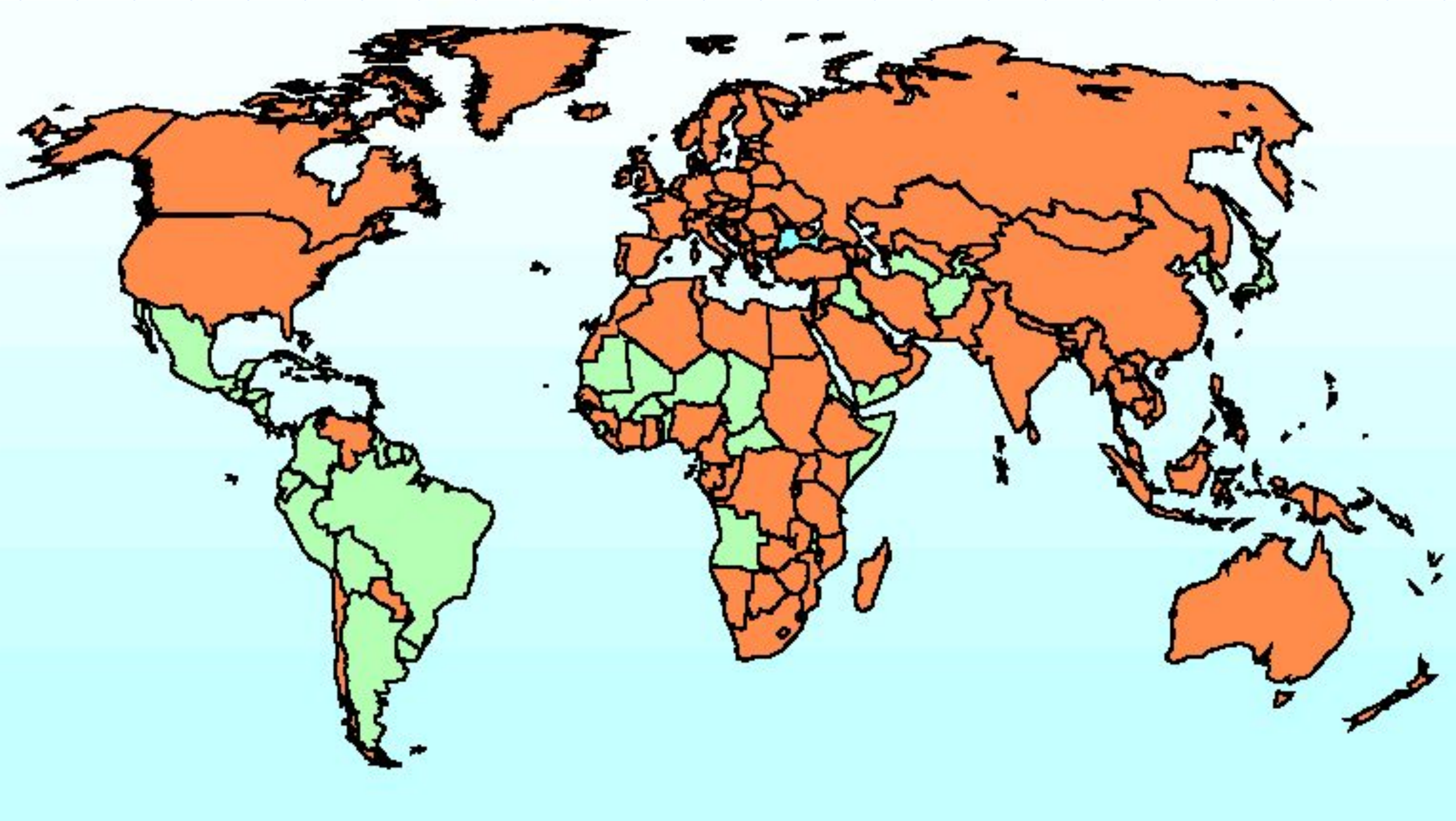
History of GSM

Date	Activity
1989	• The European Telecommunications Standards Institute (ETSI) defined GSM as the internationally accepted digital cellular telephony standard and took over responsibility of GSM specifications.
1990	• Phase 1 specifications were frozen to allow manufacturers to develop network requirements and the first GSM prototype was brought to service.
1991	• The GSM 1800 standard was released. • An addendum was added to the MoU allowing countries outside CEPT to sign.
1992	• Phase 1 specifications were completed. • January - First commercial phase 1 GSM network operator is **Oy Radiolinja Ab** in Finland • December 1992 - 13 networks on air in 7 areas • First International roaming agreement was signed between **Telecom Finland** and **Vodafone in UK.**
1993	• Australia became the first non-European country to sign the MoU. The MoU now had a total of 70 signatories. • GSM demonstrated for the first time in Africa at Telkom '93 in Cape Town • GSM networks were launched in Norway, Austria, Ireland, Hong Kong and Australia. • The number of GSM subscribers reached one million. • The first commercial DCS 1800 system was launched in the U.K. • December 1993 - 32 networks on air in 18 areas

History of GSM

Date	Activity
1996	• GSM MoU is formally registered as an Association registered in Switzerland • December 1996 120 networks on air in 84 areas • 8K SIM launched • Pre-Paid GSM SIM Cards launched • Option International launches world's first GSM/Fixed-line modem • Nov 1996 - Sole Governmental operator in **Egypt** goes online.
1997	• First dual-band GSM 900-1900 phone launched by Bosch
1998	• At the beginning of 1998 the MoU has a total of 253 members in over 100 countries and there are over 70 million GSM subscribers worldwide. GSM subscribers account for 31% of the world's mobile market • Vodacom Introduces Free Voice Mail • GSM SIM Cracked in USA • 21 May 1998. **Egypt** privatizes its GSM operator. • Iridium Live 11/98 • 125m GSM 900/1800/1900 users worldwide (12/98) • **1 Dec 1998. Click GSM commercial launch.**

GSM Coverage worldwide



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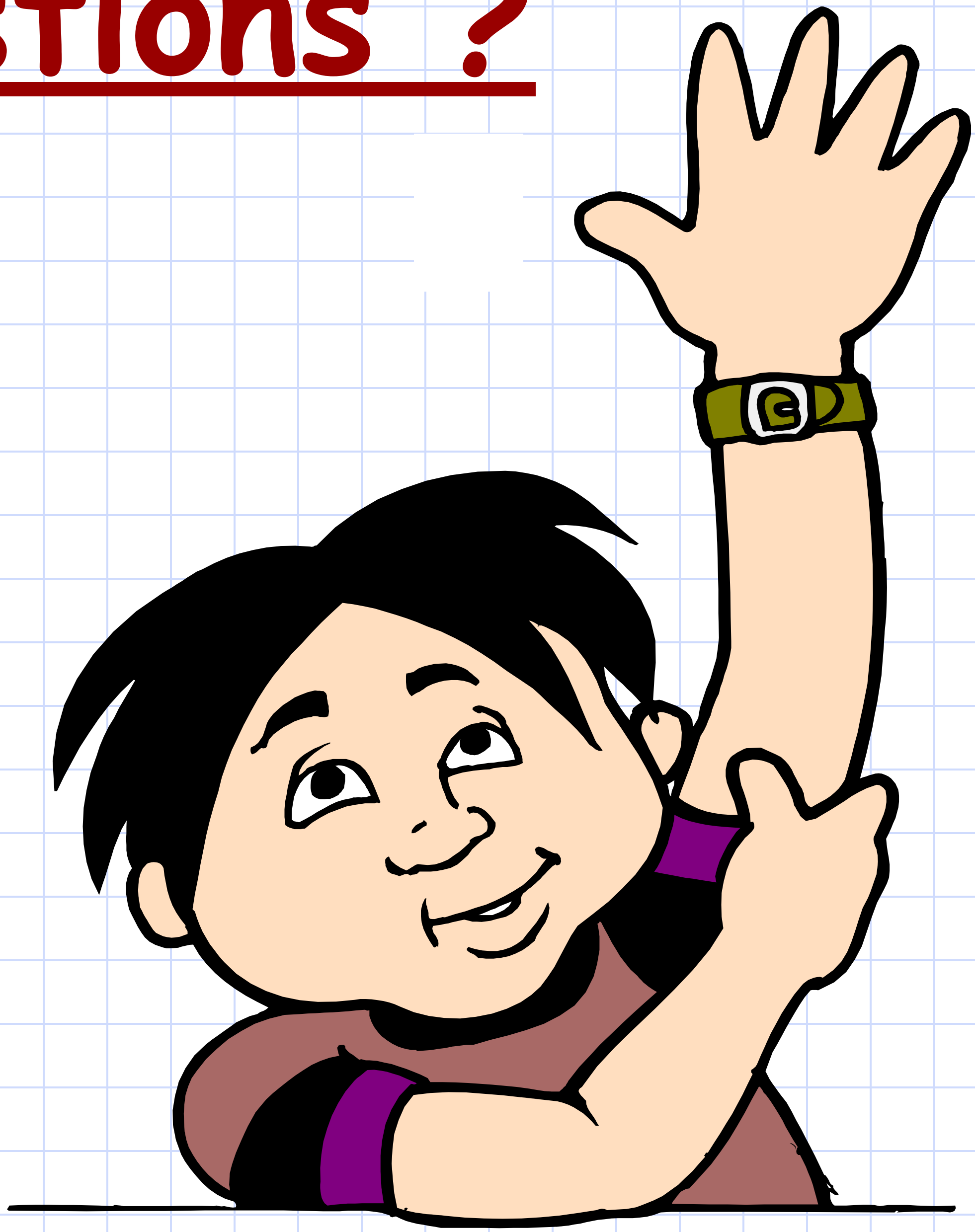


Chapter 2 : GSM Network Elements

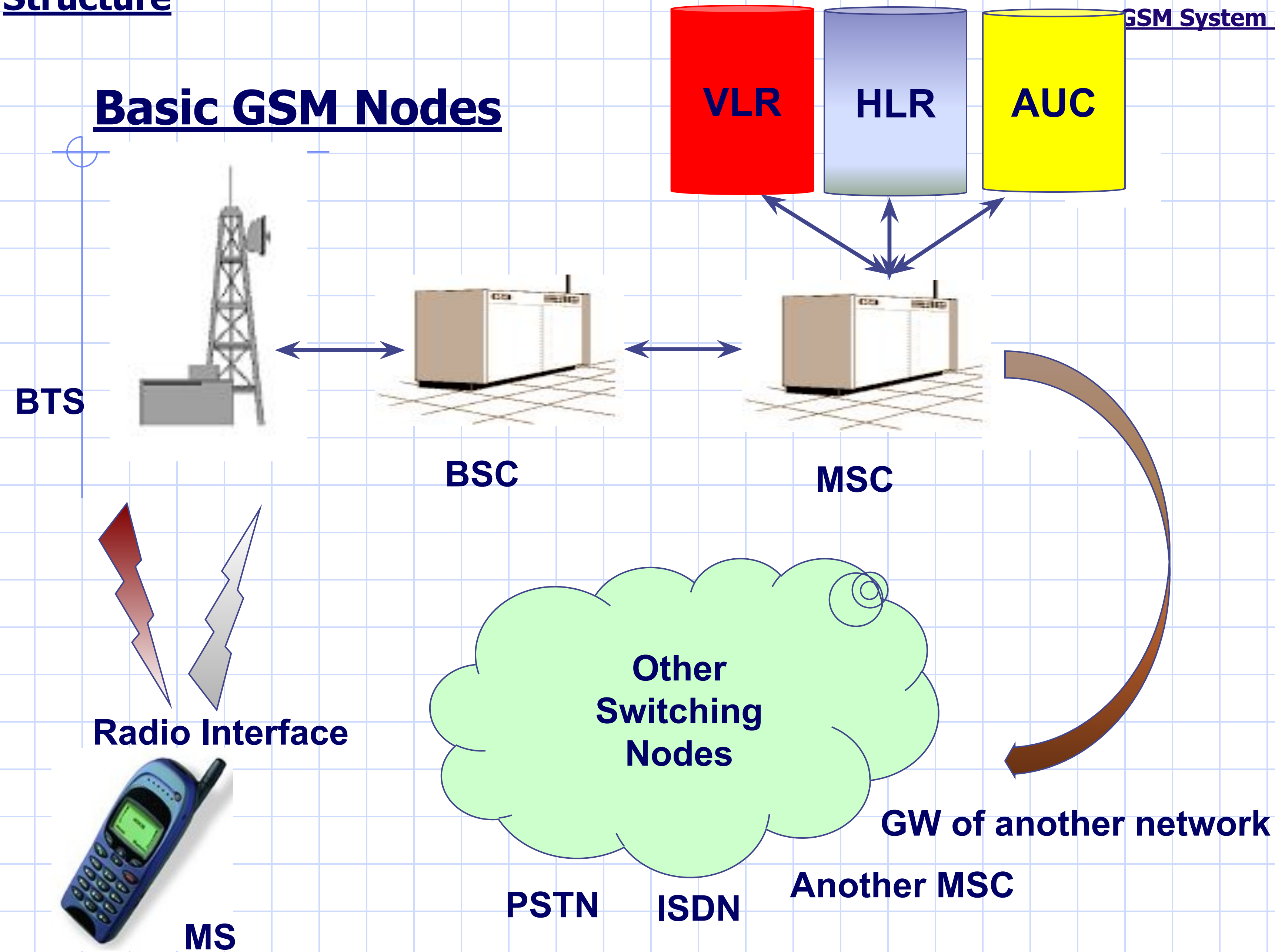
We hear Music .. But we don’t see the musicians ..

Chapter Objectives

By the End of this Chapter you will:

- Know the role of the Basic nodes that compose the GSM network
- Know how they are interconnected together
- Know the different Identities used in GSM World

Basic GSM Nodes



The Mobile Services Switching Center (MSC)

- Administers its Base Station Controller(s) BSC(s).
- Switches calls to/from mobile subscribers.
- Records charging and accounting details.
- Provides the gateway functionality to other networks.

The Home Location Register (HLR)

- Controls the routing of mobile terminal calls and SMS.
- Stores for each mobile subscriber:

    Basic subscriber categories.

    Supplementary services.

    Current location (Location Area).

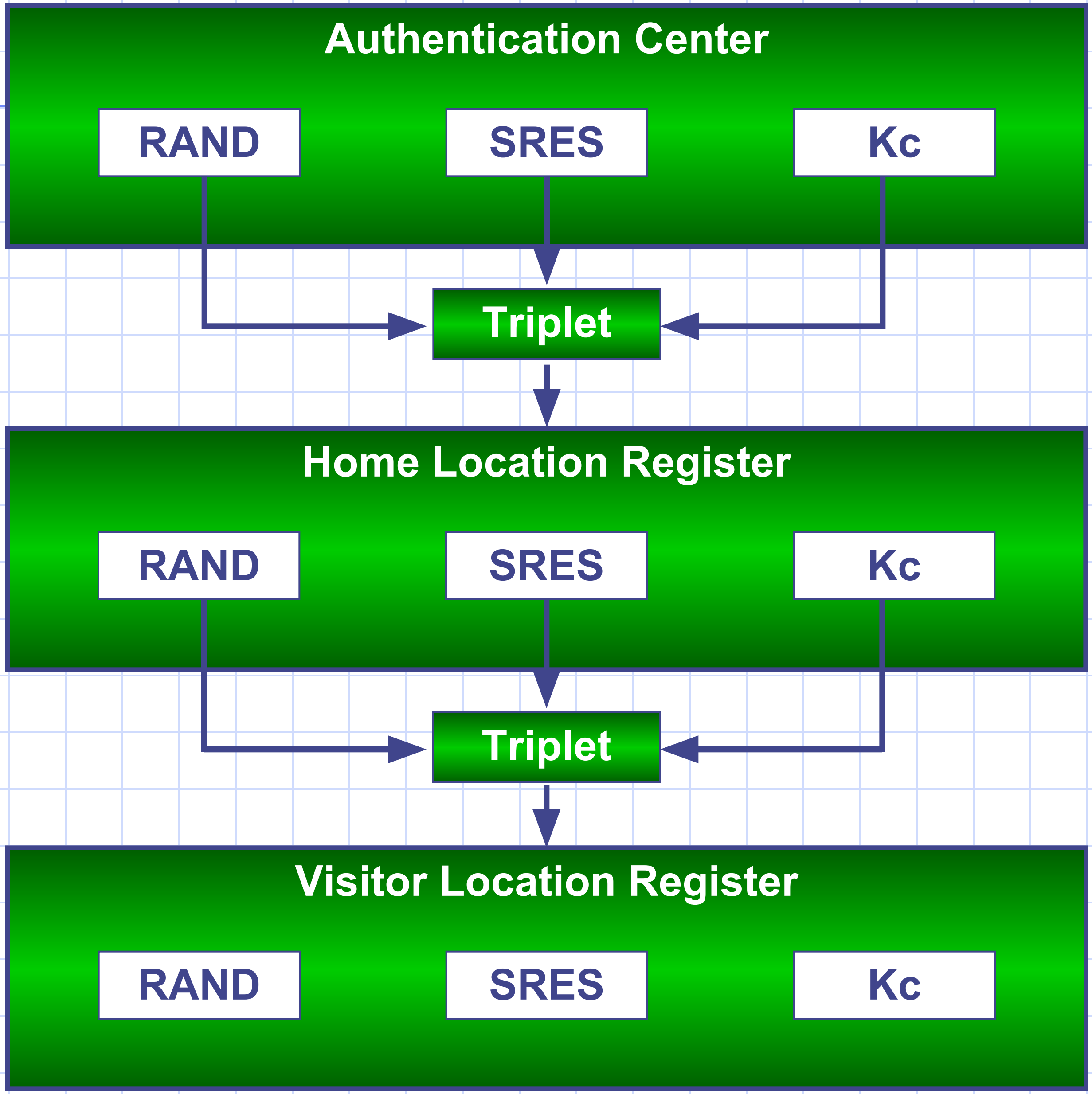
    Allowed/barred services.

    Authentication data.

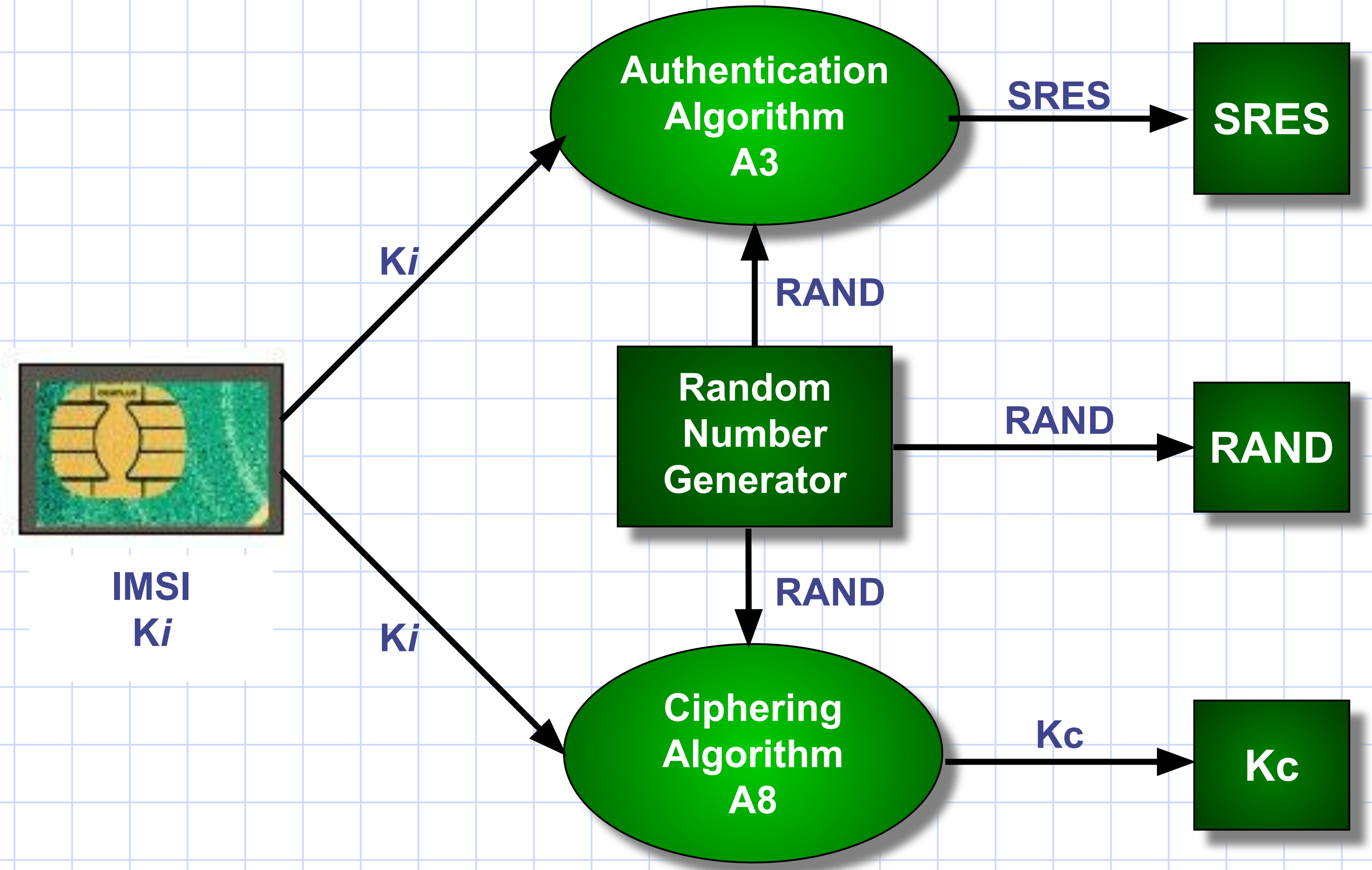
Basic GSM Network Structure

GSM System Survey

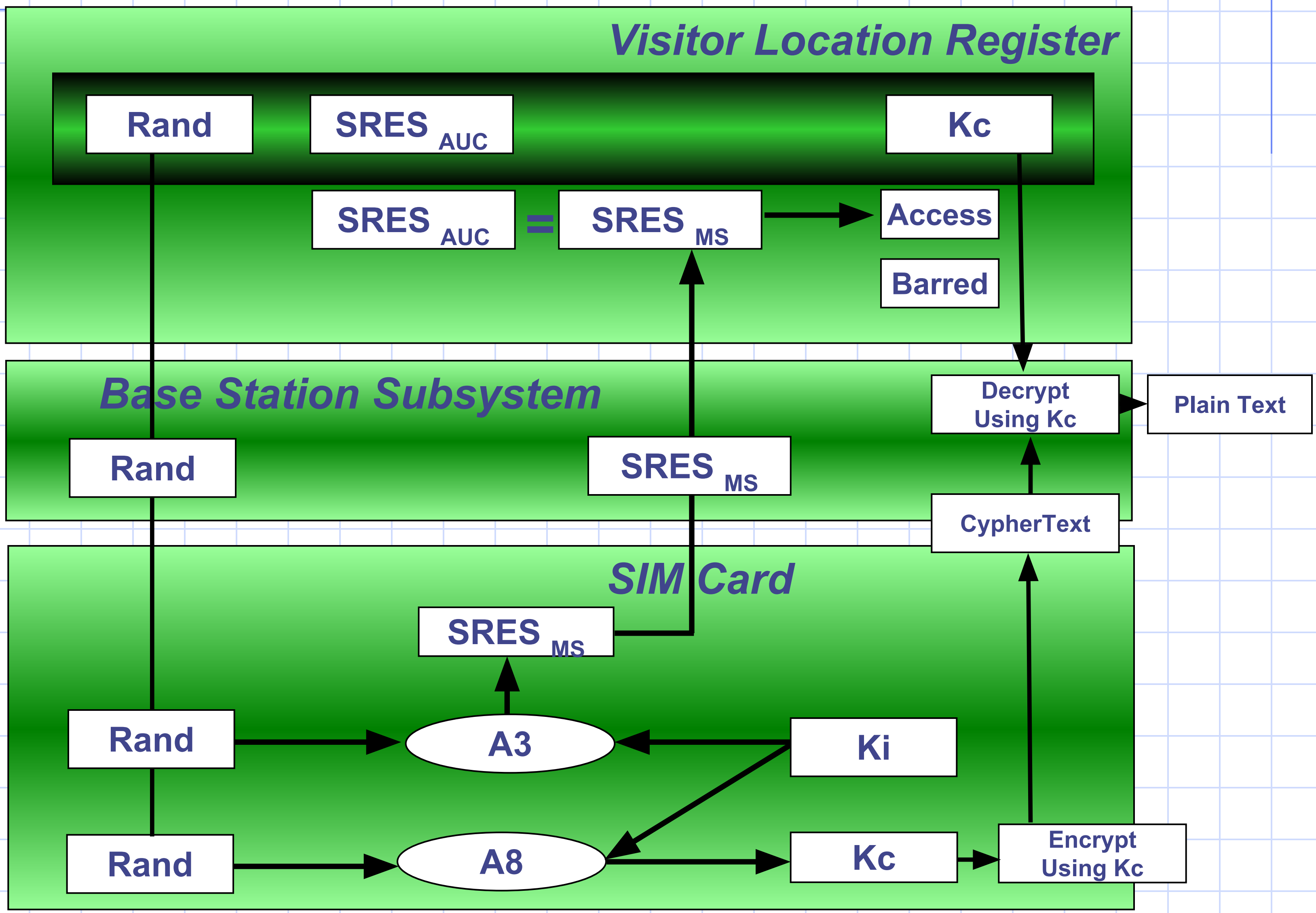
Triplets



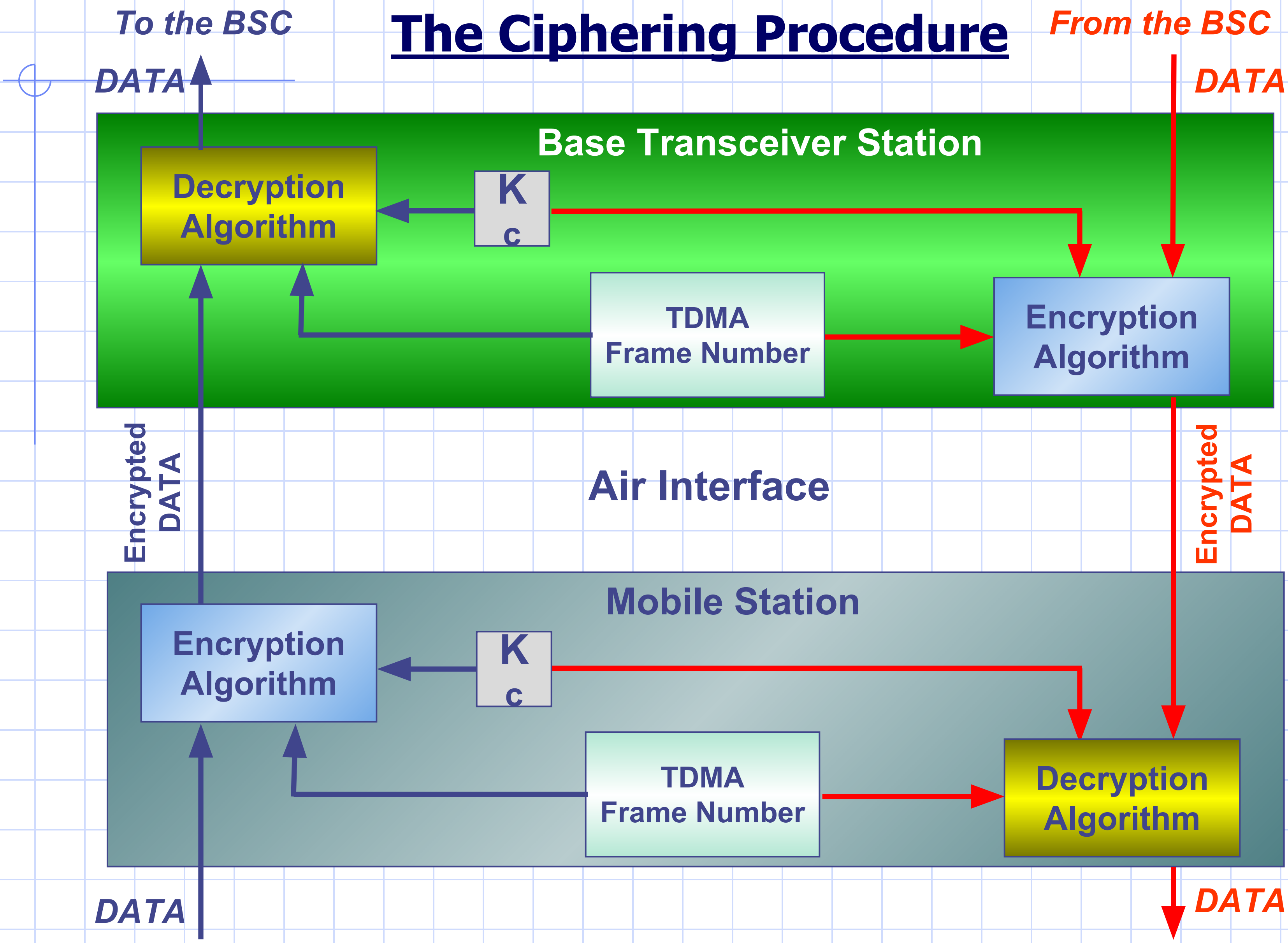
Producing Triplets



The Authentication Procedure



The Ciphering Procedure



The Visitor Location Register (VLR)

- It stores a copy of the HLR Profile for all currently registered mobile subscribers who are covered by cells belonging to the MSC coverage area.
- The VLR is always integrated with the MSC.
- The VLR stores the Location area of the MS (which is not stored in the HLR).

The Base Station Controller (BSC)

- ❖ Manages the Radio Communication with the mobile station over the air interface.
- ❖ Controls the handover of calls in progress Between BTSs
- ❖ Supervises the transmission network and the operation of each BTS

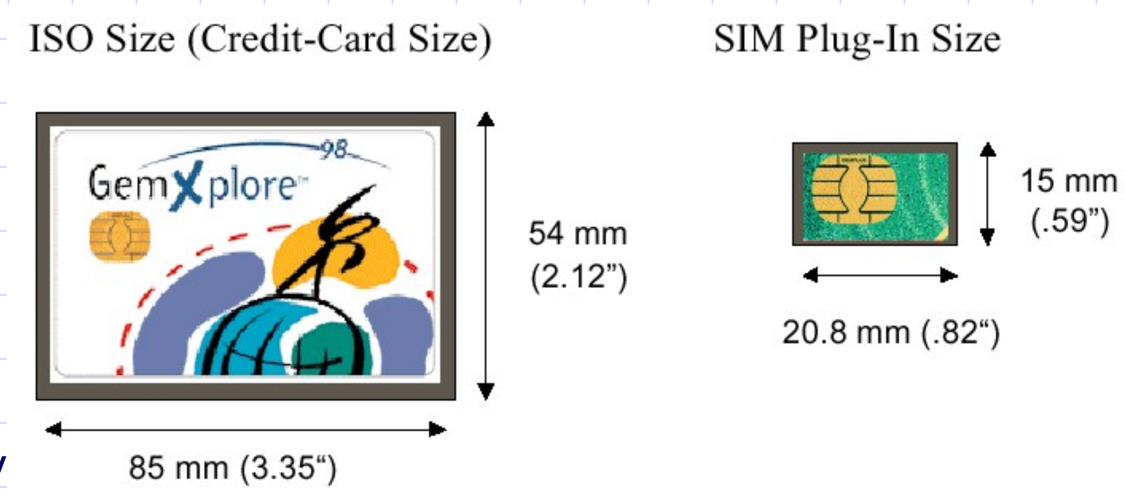
The Base Transceiver Station (BTS)

- Consists of the radio transmitters, receivers and the antenna system required to provide the coverage area for one cell.
- Converts the GSM radio signals into a format that can be recognized by the BSC.
- Records and passes to the BSC the Signal strength measurements.
- Performs the network end of the ciphering/encryption process.

The Mobile Station (MS)

+

The Mobile Equipment is said to be a Mobile Station if the Subscriber Identity Module (SIM Card) is added to it



The SIM Card contains:

A processor and memory

that stores:

- The international mobile subscriber Identity IMSI
- The Authentication and ciphering keys.

GSM Identities

Mobile Station ISDN Number (MSISDN)

20	10	1100477		44	385	196099
CC	NDC	SN		CC	NDC	SN

VodaFone Egypt MSISDN

VodaFone UK MSISDN

CC : Country Code
NDC : Network Destination Code
SN : Subscriber Number

GSM Identities

International Mobile Subscriber Identity (IMSI)

234	15	1234567890		602	02	1234567890
MCC	MNC	MSIN		MCC	MNC	MSIN

Vodafone UK IMSI

Vodafone Egypt IMSI

MCC : Mobile Country Code
MNC : Mobile Network Code
MSIN : Mobile Station Identification Number

GSM Identities

International Mobile Equipment Identity (IMEI)

6 Digits	2 Digits	6 Digits
TAC	FAC	SN

IMEI

TAC: Type Approval Code,
The first two digits are the code for the country approval

SN: Serial Number

Final Assembly Codes (FAC)

01,02	AEG
07,40	Motorola
10,20	Nokia
30	Ericsson
40,41,44	Siemens
47	Optional International
50	Bosch
51	Sony
51	Siemens
51	Ericsson
60	Alcatel
70	Sagem
75	Dancall
80	Philips
85	Panasonic

GSM Identities

Temporary Mobile Subscriber Identity Number (TMSI)

The TMSI can be allocated to the mobile subscriber in order to be used instead of his IMSI during all radio communications. The purpose is to keep subscriber information confidential on the air interface.

The TMSI is relevant on the local MSC/VLR level only and is changed at certain events or time intervals. Each local operator can define its own TMSI structure.

GSM Identities

Mobile Station Roaming Number (MSRN)

When a mobile terminating call is to be set-up, the HLR of the called subscriber requests the MSC/VLR to allocate an MSRN to the called subscriber.

This MSRN is returned via the HLR to the GMSC.

The GMSC routes the call to the MSC/VLR exchange where the called subscriber is currently registered.

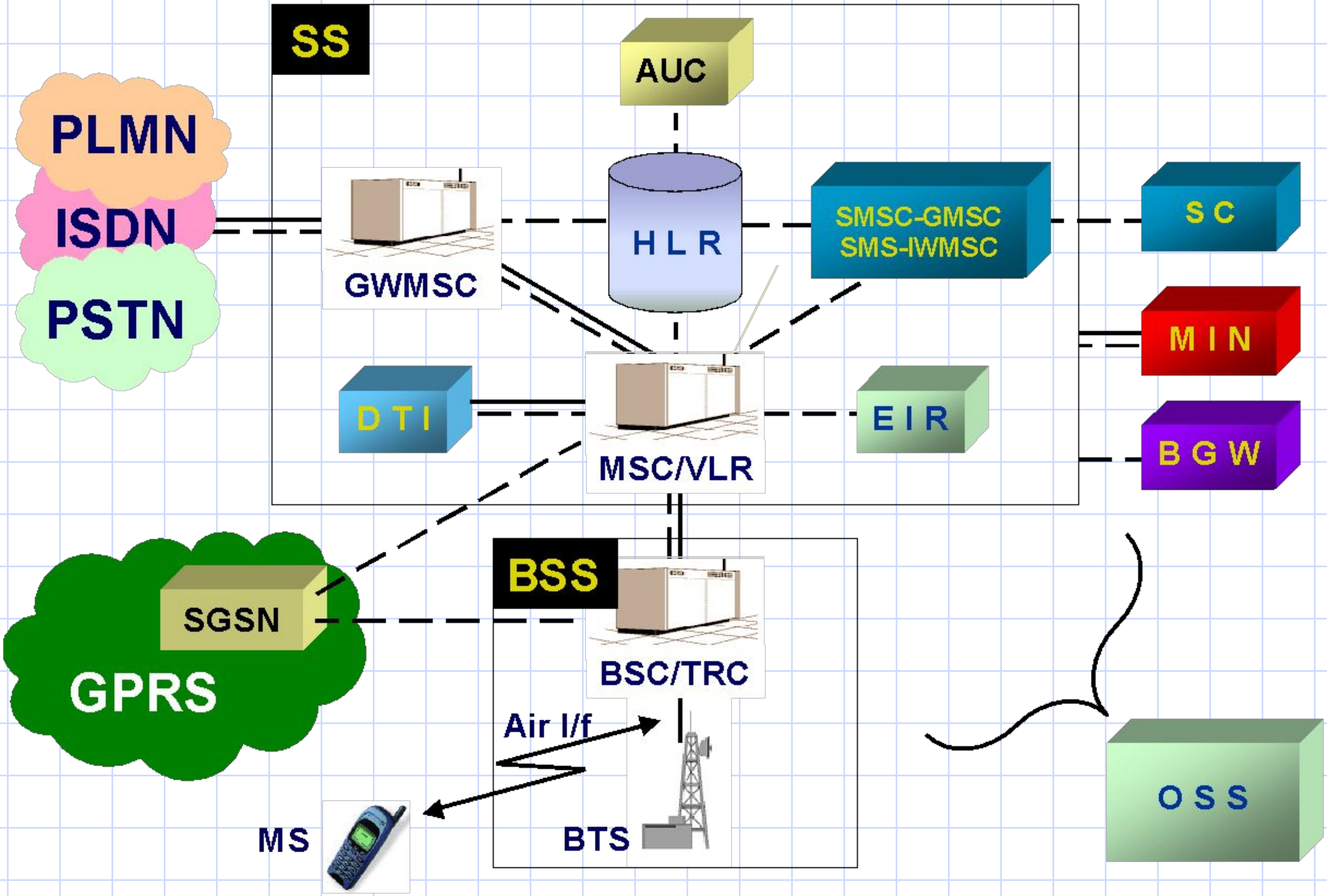
The routing is done using the MSRN. When the routing is completed, the MSRN is released.

The interrogation call routing function (request for MSRN) is part of the MAP.

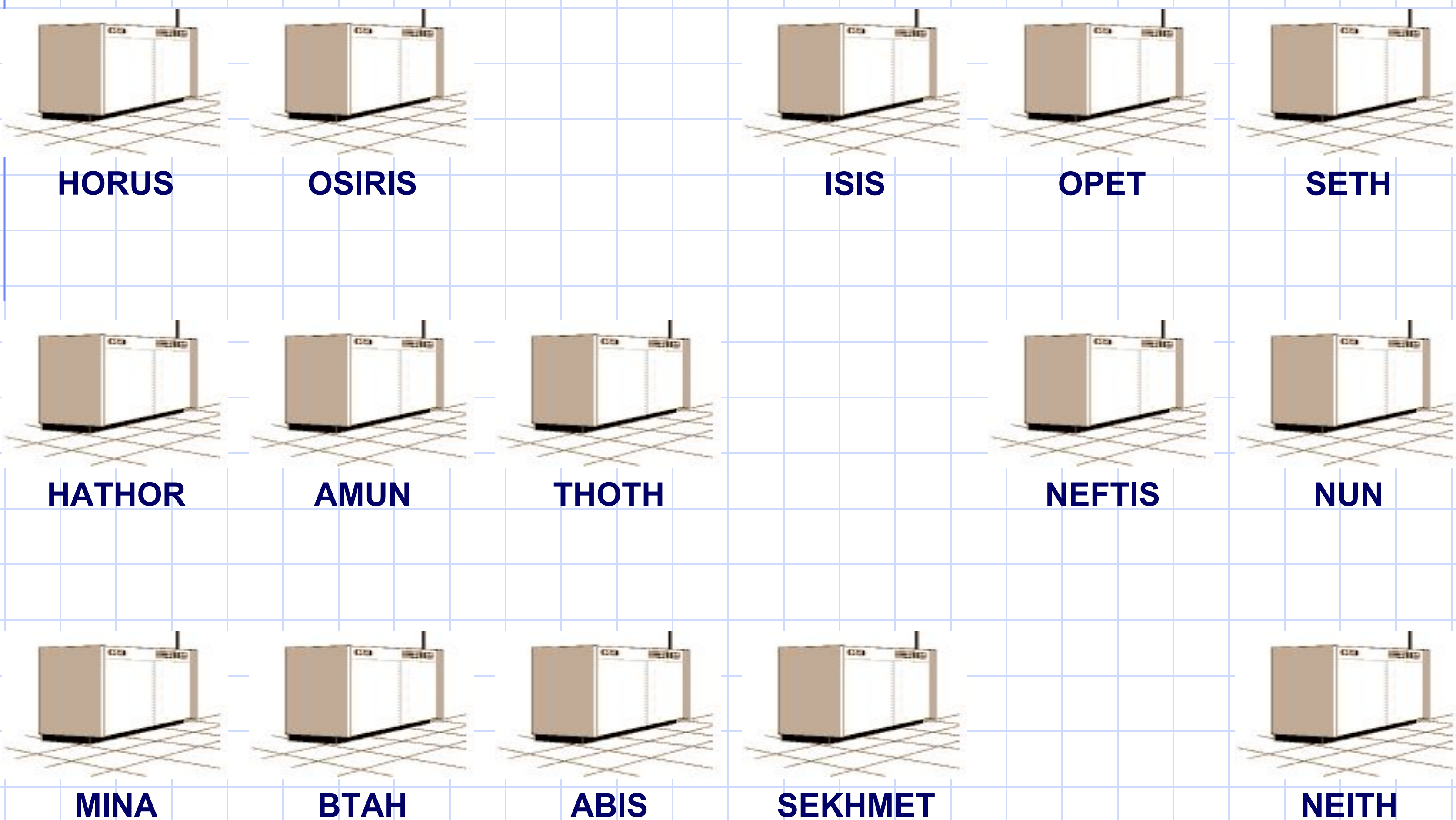
All data exchanged between GMSC-HLR-MSC/VLR for the purpose of interrogation is sent over S7 signaling.

The MSRN is built up like an MSISDN.

GSM Network Structure



Vodafone Egypt Network Structure



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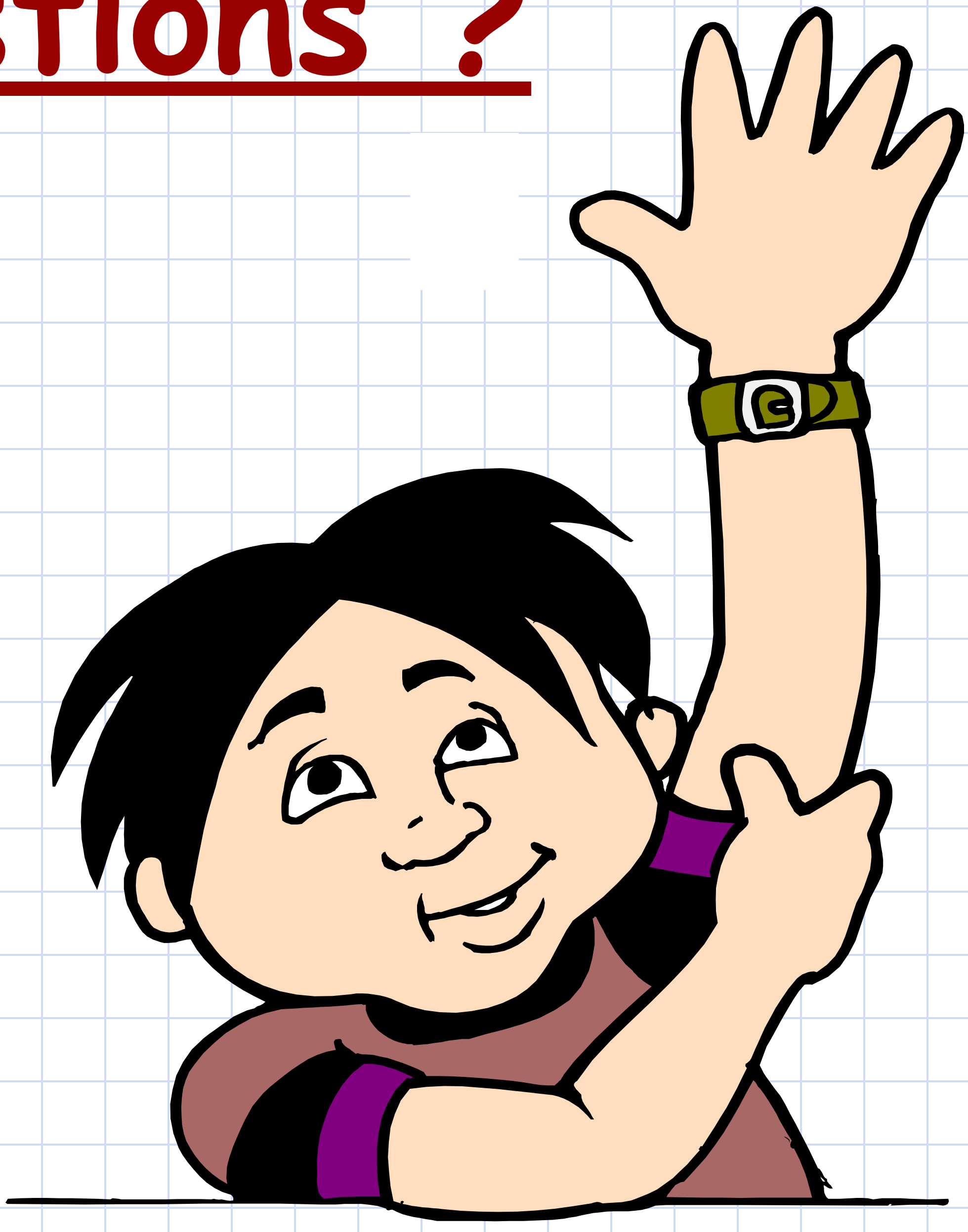


<u>Chapter 3 : Transmission System</u>

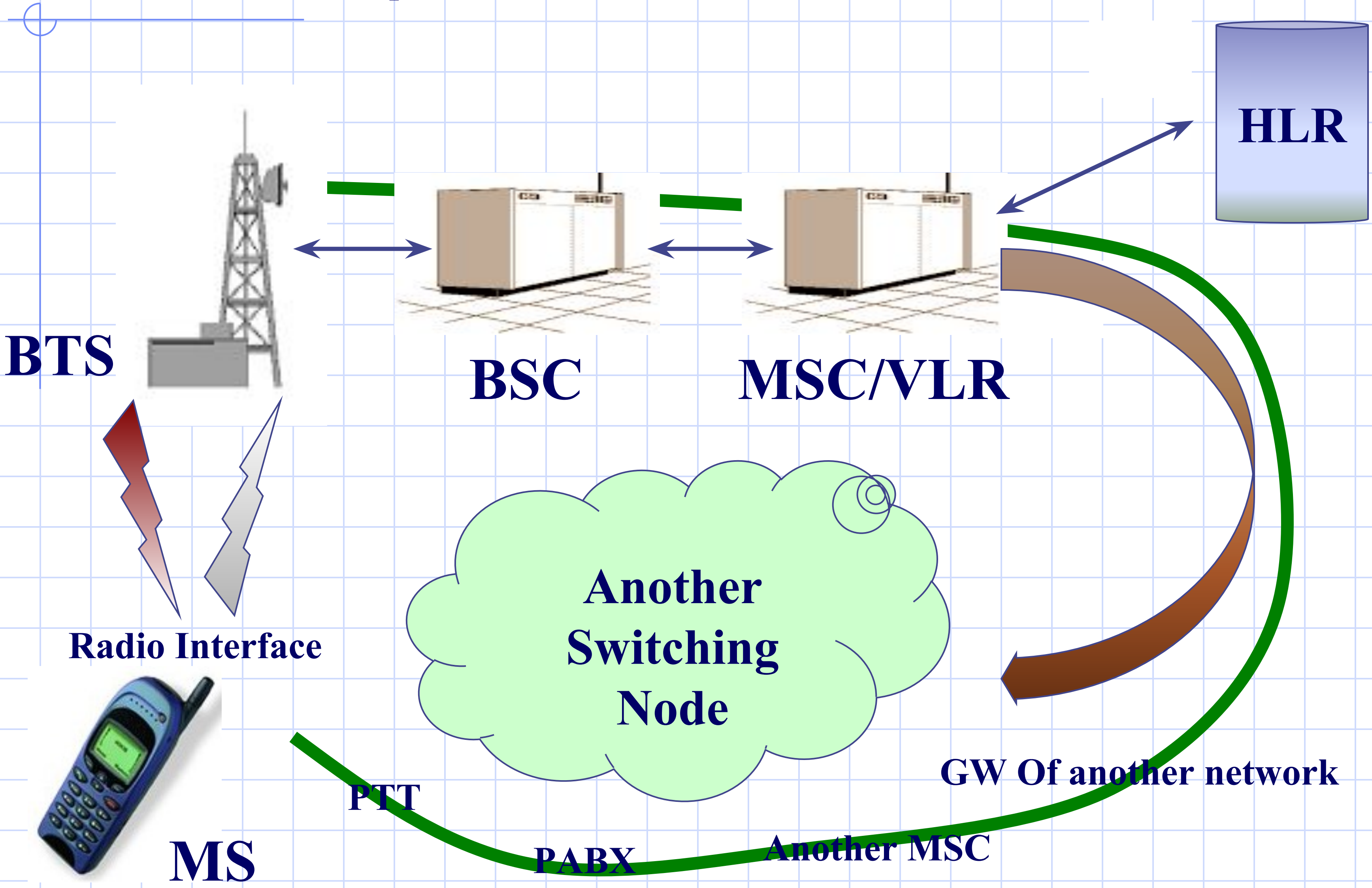
We mean by a transmission system, how the nodes are interconnected

Chapter Objectives

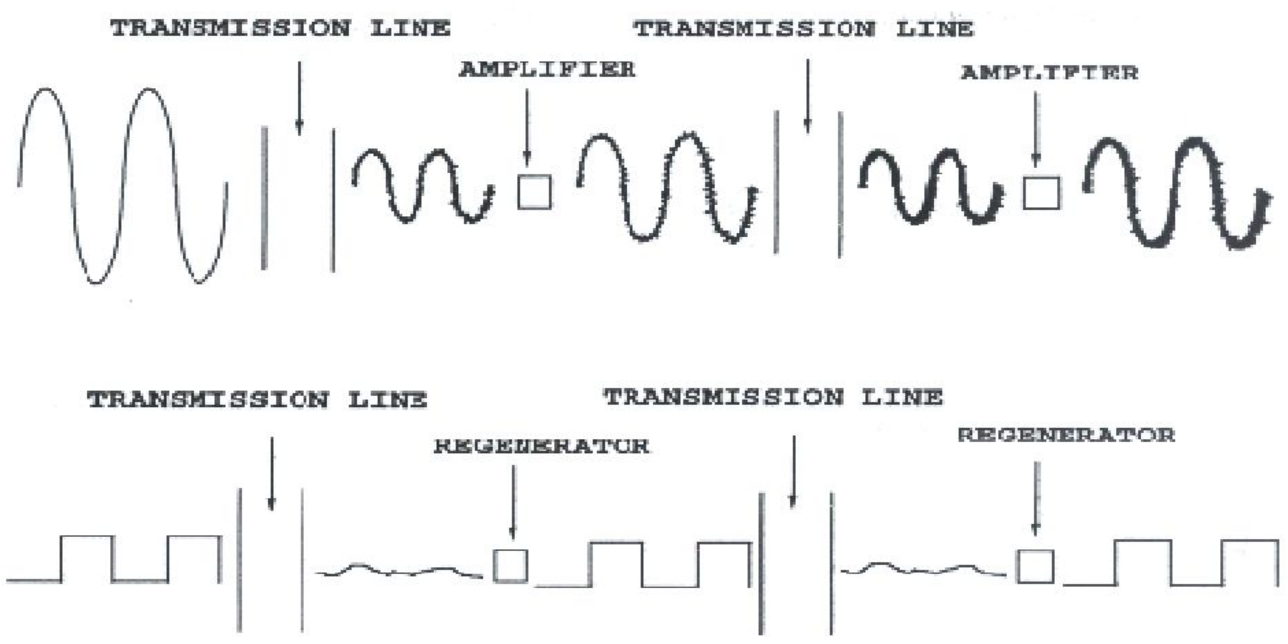
By the End of this Chapter you will:

- Know How do we use TE infrastructure in connecting our Nodes
- Know the basic role of the DXX

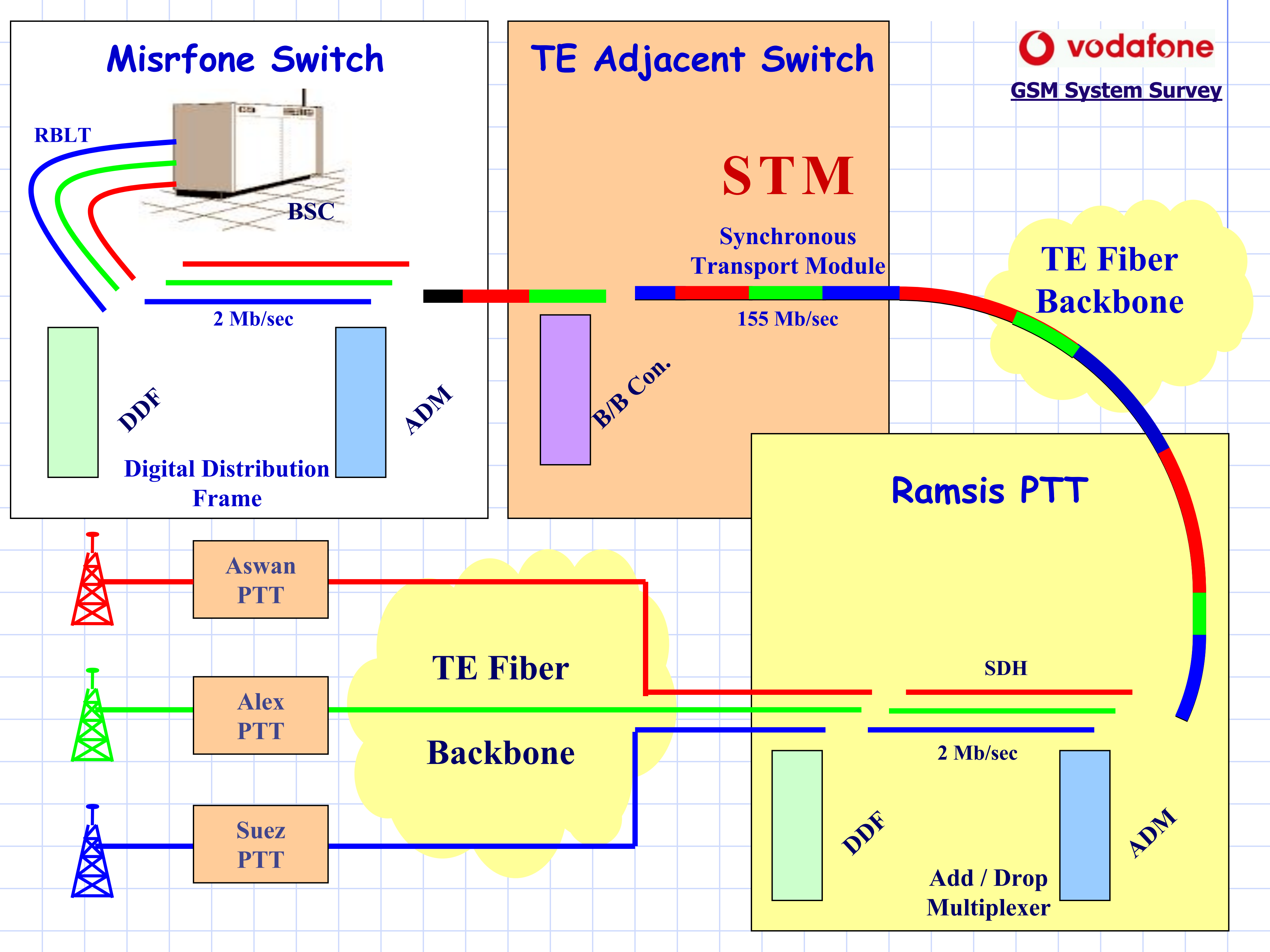
Review the System



Digital Versus Analog Transmission



Digital Communication offers better noise performance



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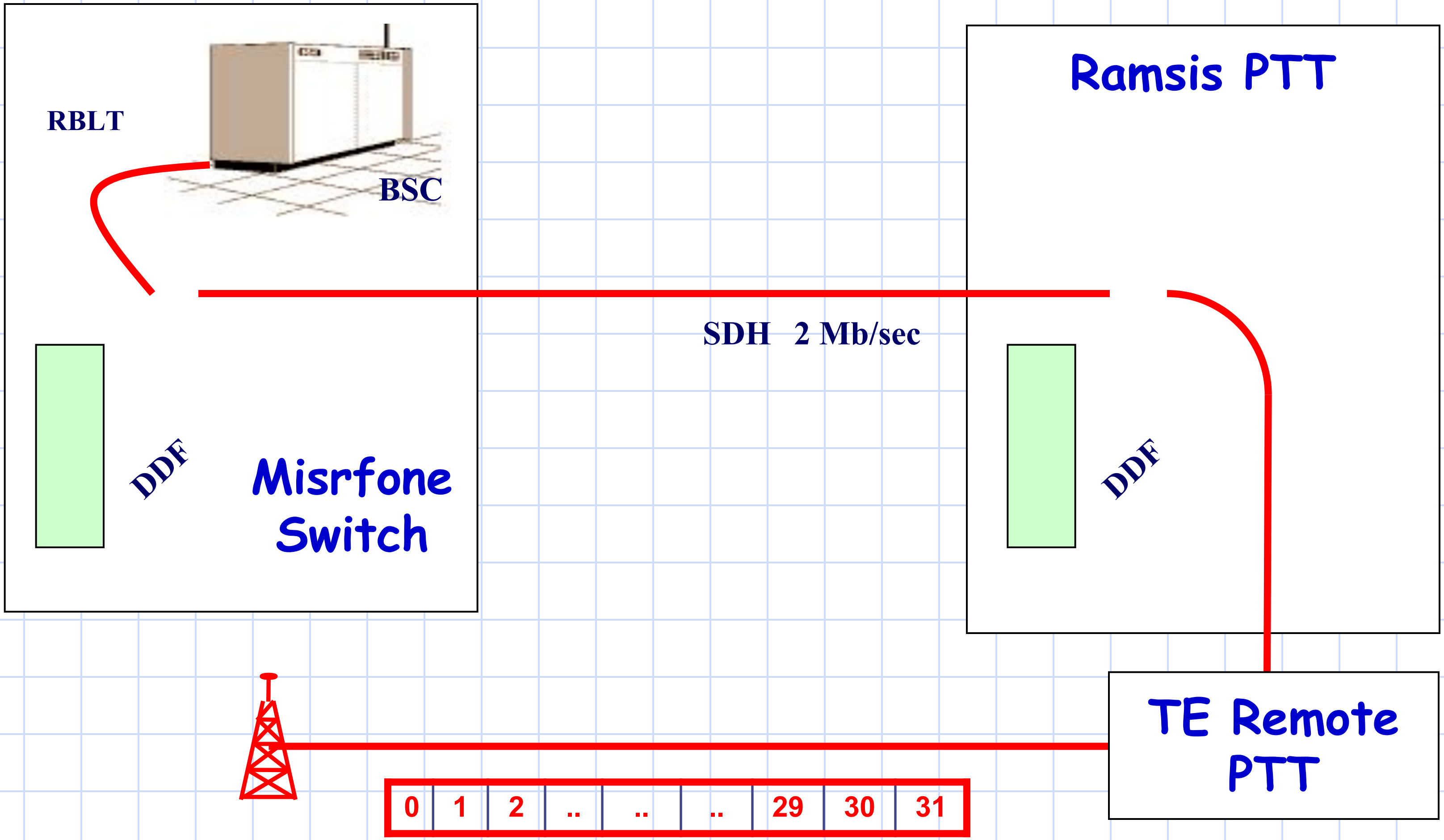
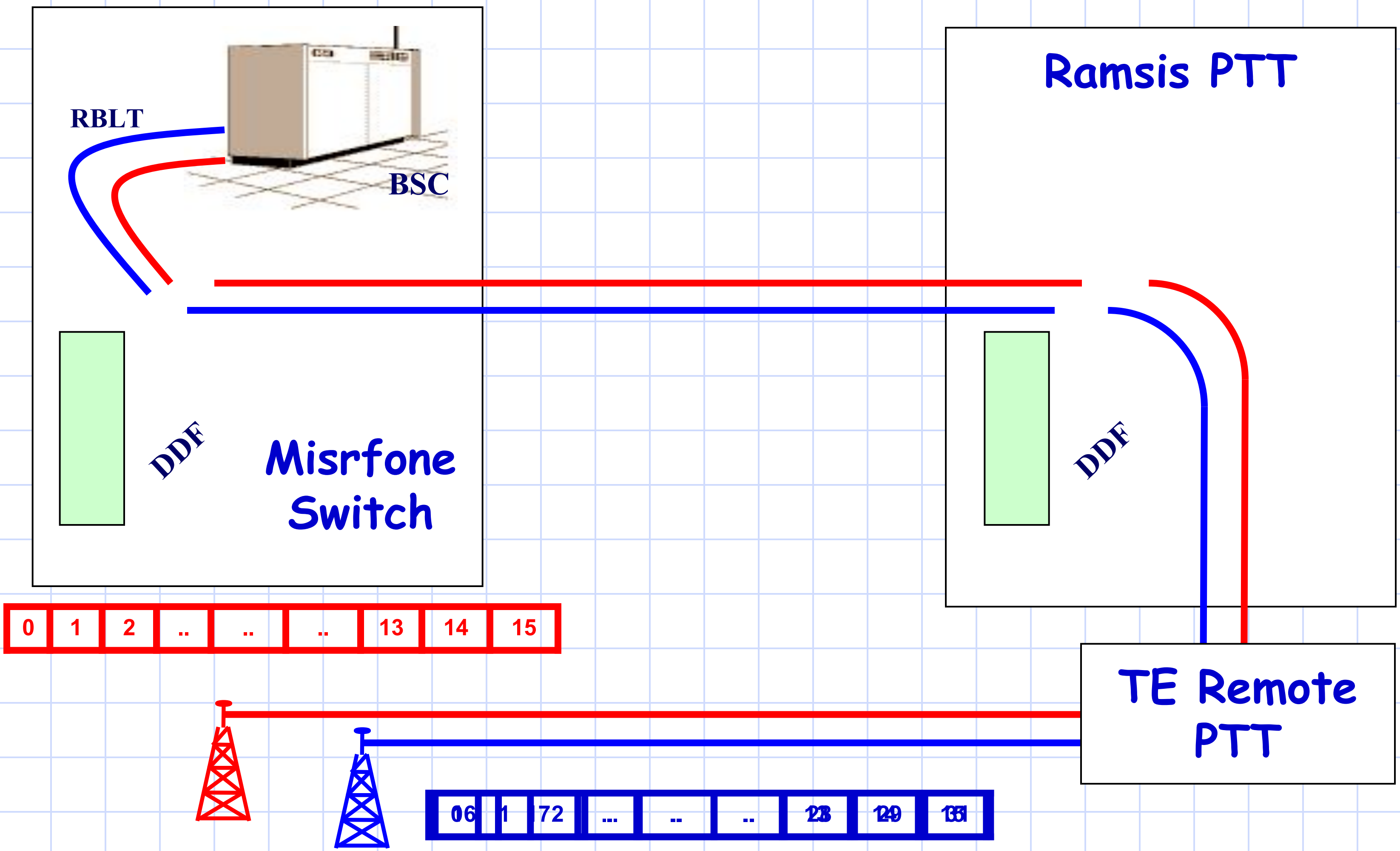
Misrfone Switch
RBLT
BSC
2 Mb/sec
DDF
ADM
Digital Distribution Frame
TE Adjacent Switch
STM
Synchronous Transport Module
155 Mb/sec
B/B Con.
TE Fiber Backbone
Ramsis PTT
SDH
2 Mb/sec
DDF
ADM
Add / Drop Multiplexer
Aswan PTT
Alex PTT
Suez PTT
TE Fiber
Backbone

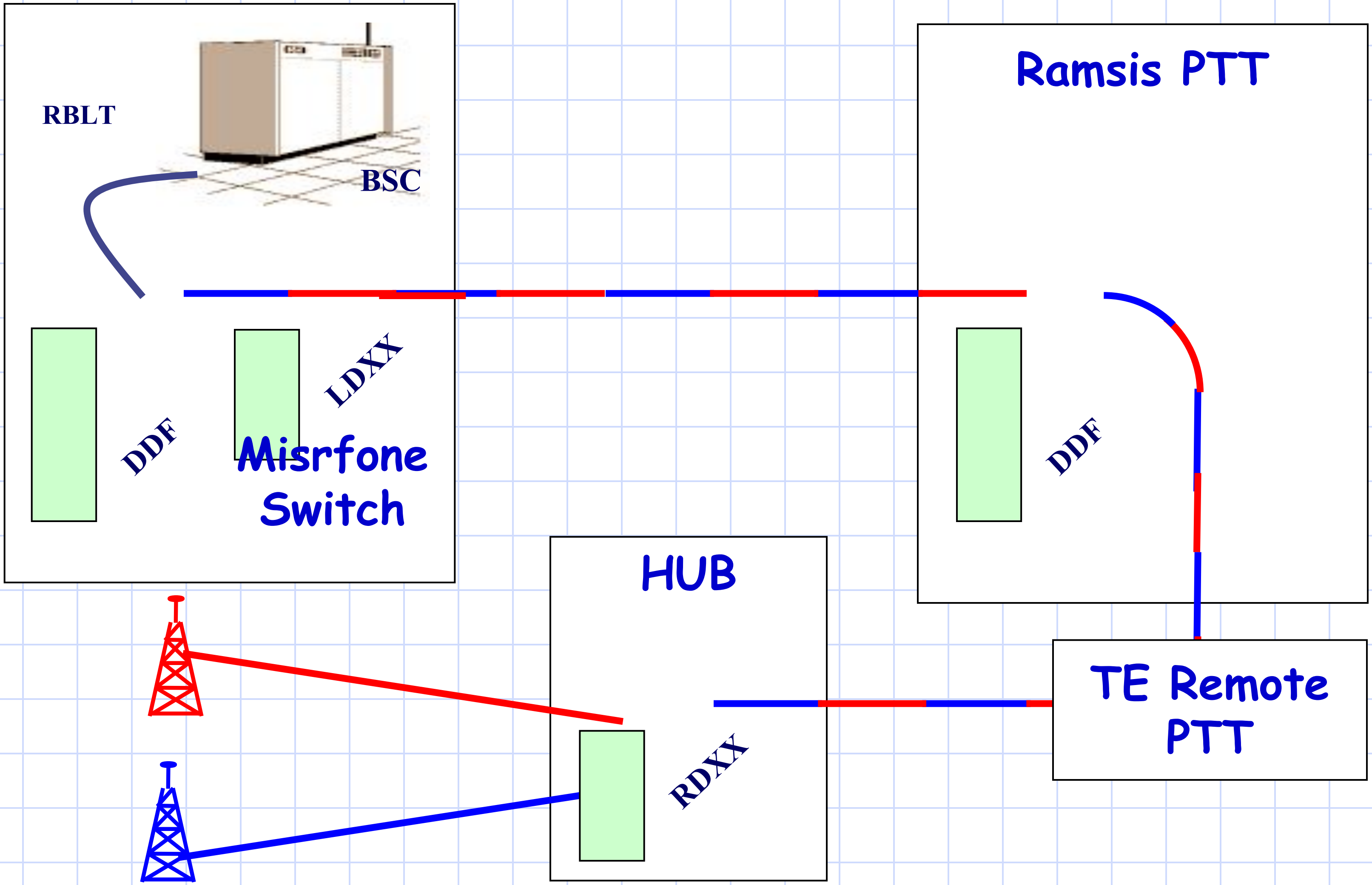
Transmission system



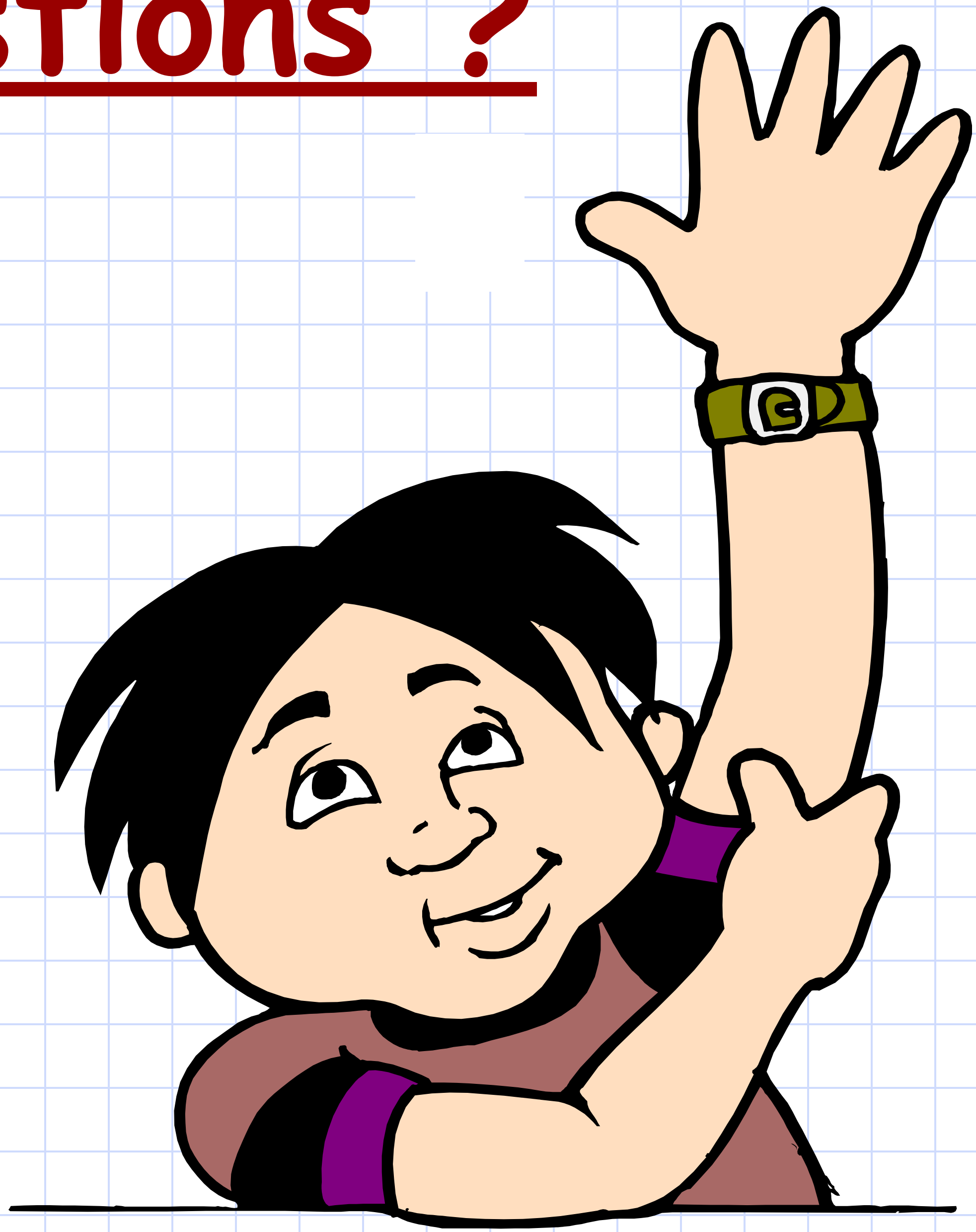
Insufficient use of resources



Introducing DXX



Questions ?



Break

Chapter 4 : Radio Coverage

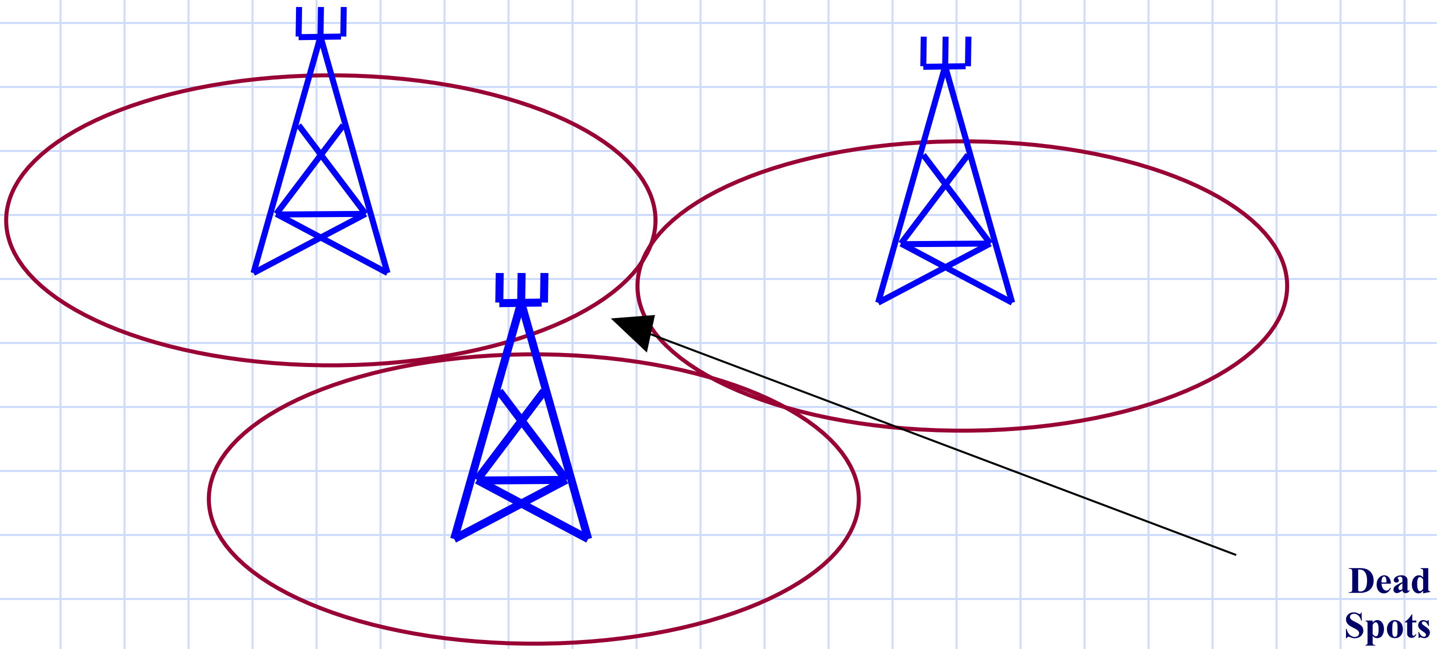
A visible pattern of sound waves

Chapter Objectives

By the End of this Chapter you will:

- Know The Geometrical Theoretical shape of the cells
- Know the frequency band allocated for GSM
- Know what is meant by frequency Reuse
- Know when to use different cluster sizes

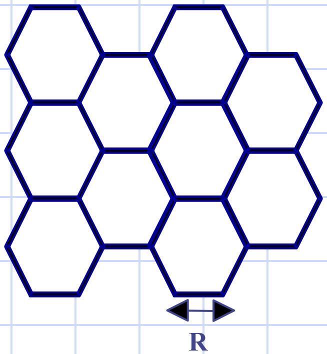
Cell Geometry

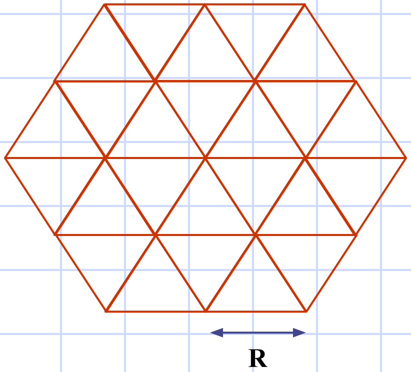


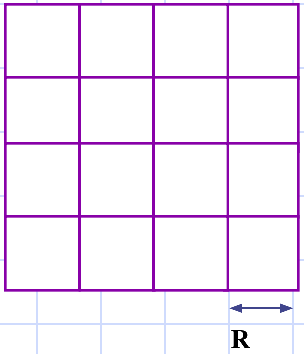
Problem of omni directional antennas

Cell Geometrical Shape

To solve the dead spot problem



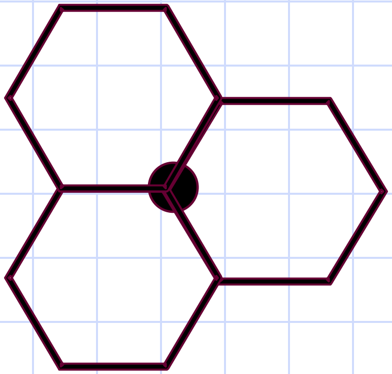




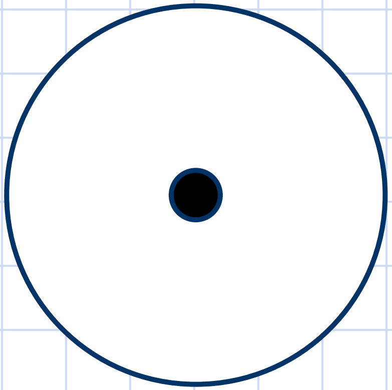
Tradeoffs

- The number of cells required to cover a given area.
- The cell transceiver power.

Transceiver Antenna

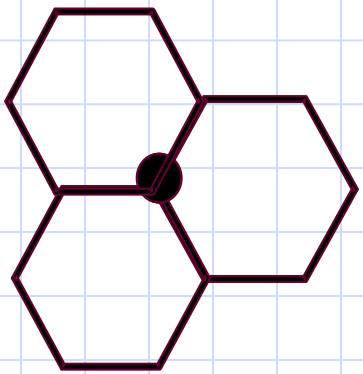


Sectorial Antenna



Omni-Directional Antenna

Sectorial Antenna

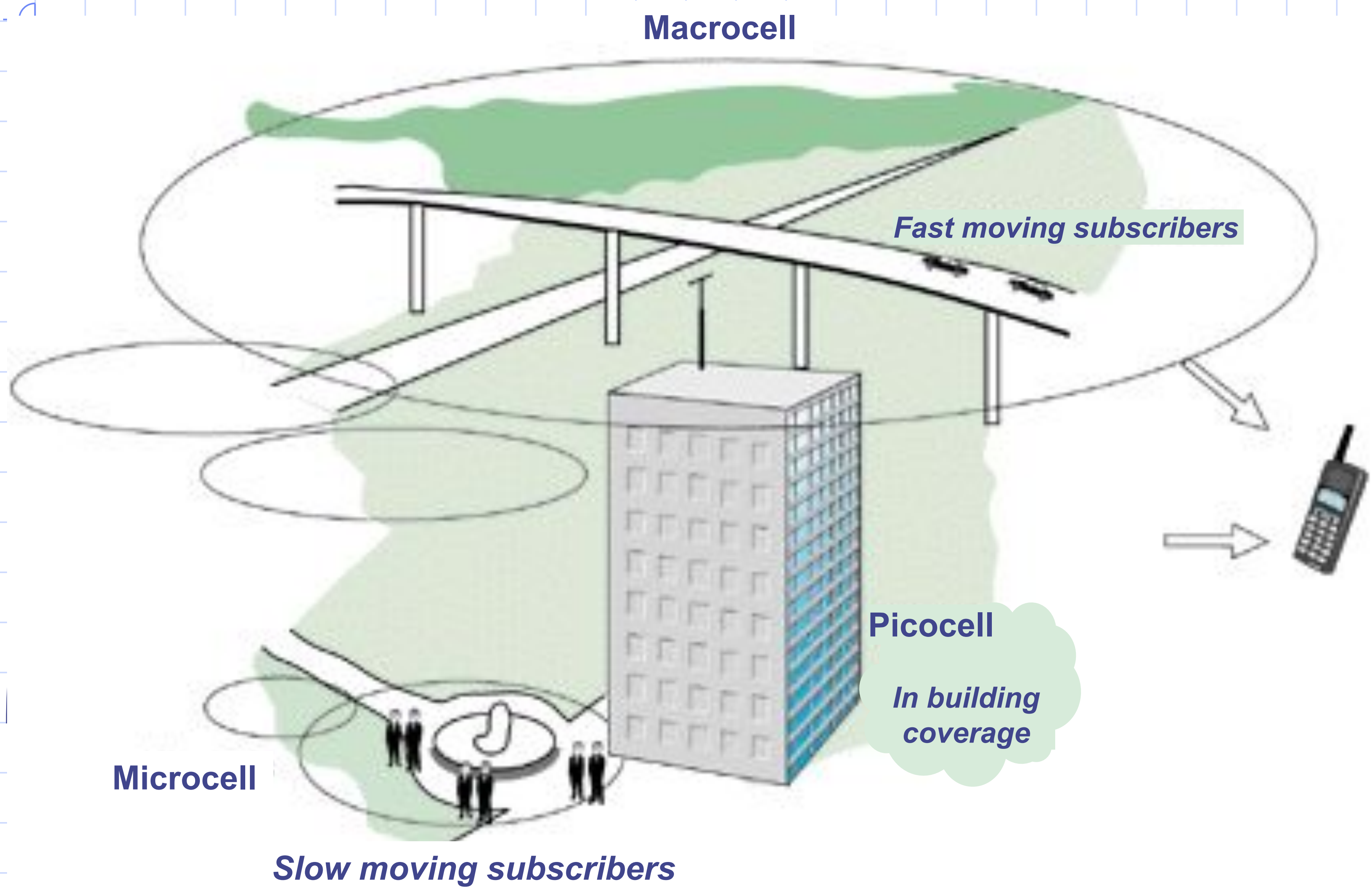


Sectorial Antenna

The cells will take the form of overlapping circles.

Due to the obstacles in the coverage area the actual shape of the cells would be Random.

Cell Classification



GSM Coverage Plan

To provide coverage for a large service area of a mobile network we have two Options:

(A) Install one transceiver with high radio power at the center of the service area

Drawbacks

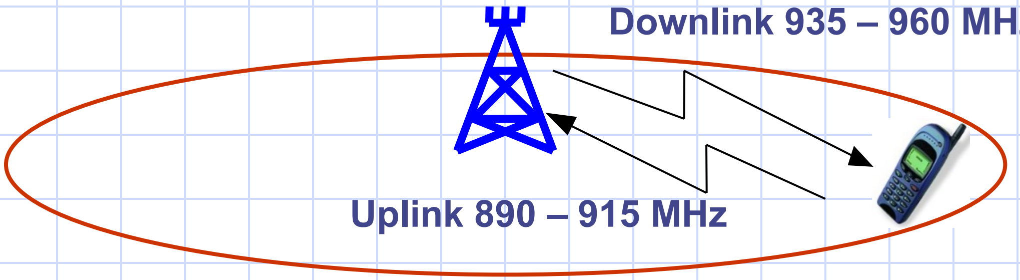
- The mobile equipments used in this network should have high output power in order to be able to transmit signals across the coverage area.
- The usage of the radio resources would be limited.

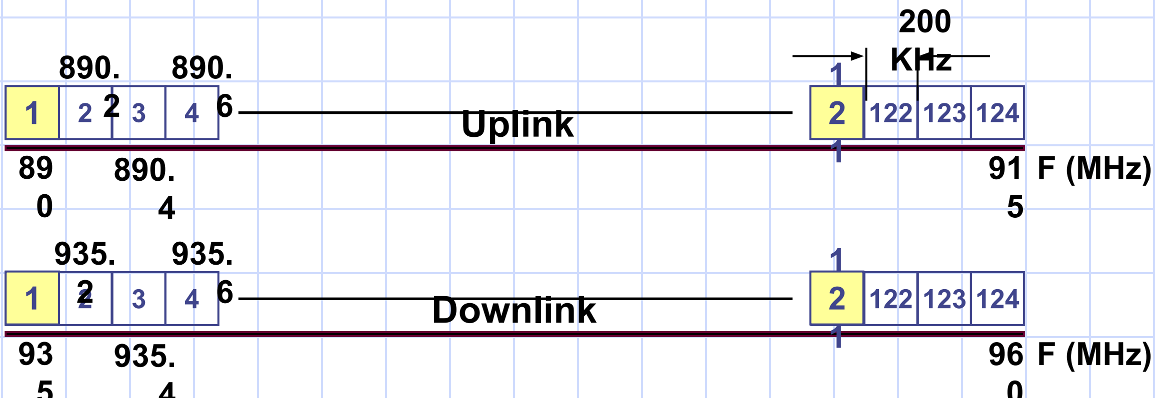
(B) Divide the service area into smaller areas (cells)

Advantages

- Each cell as well as the mobile handsets will have relatively small power transceivers.
- The frequency spectrum might be "reused" in two far separated cells. This yields:
  - ❖Unlimited capacity of the system.
  - ❖Good interference characteristics

Spectrum Allocation (GSM 900)



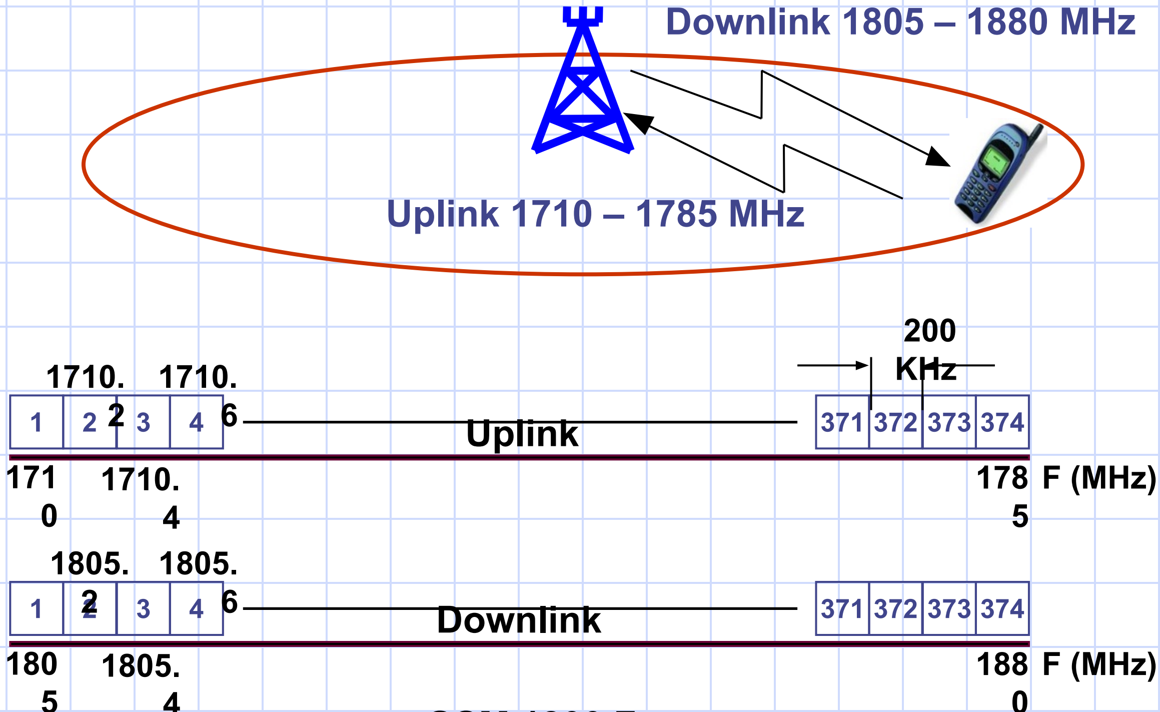


GSM 900 Frequency Allocation

ARFCN

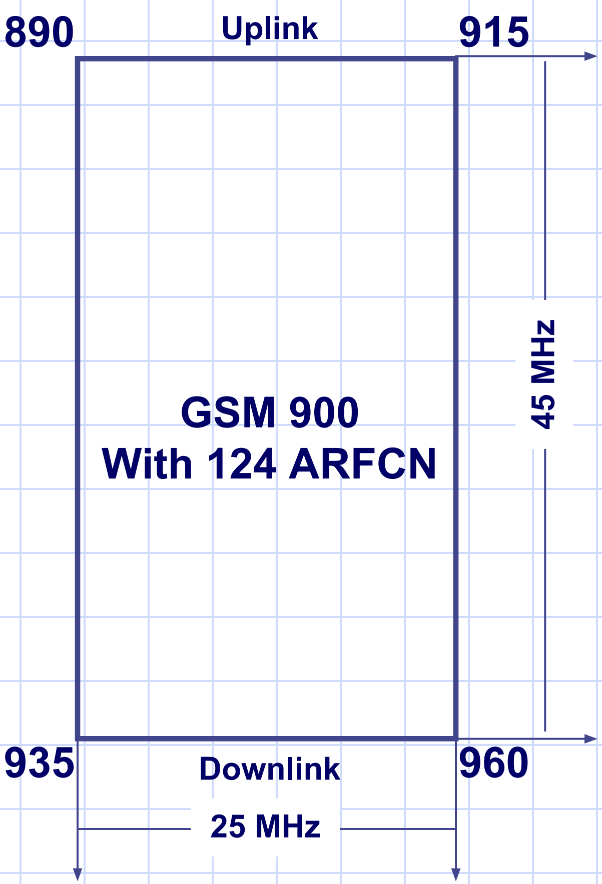
Absolute Radio Frequency Channel Number

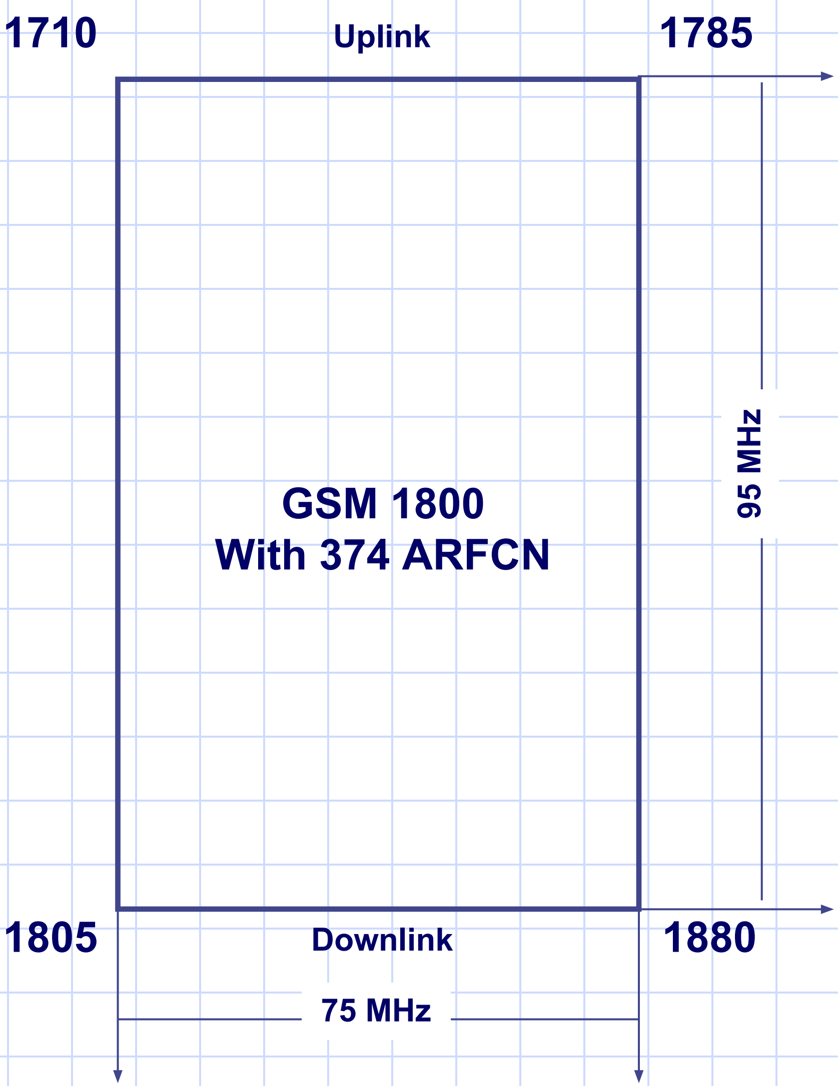
Spectrum Allocation (GSM 1800)



GSM 1800 Frequency Allocation

Comparison


890
Uplink
915
GSM 900
With 124 ARFCN
45 MHz
935
Downlink
960
25 MHz


1710
Uplink
1785
GSM 1800
With 374 ARFCN
95 MHz
1805
Downlink
1880
75 MHz

System	GSM 800	P-GSM 900	E-GSM 900	GSM 1800	GSM 1900
Frequencies: • Uplink • Downlink	 824-849 MHz 869-894 MHz	 890-915 MHz 935-960 MHz	 880-915 MHz 925-960 MHz	 1710-1785 MHz 1805-1880 MHz	 1850-1910 MHz 1930-1990 MHz
Wavelength	37.5 cm	~ 33 cm	~ 33 cm	~ 17 cm	~ 16 cm
Bandwidth	25 MHz	25 MHz	35 MHz	75 MHz	60 MHz
Duplex Distance	45 MHz	45 MHz	45 MHz	95 MHz	80 MHz
Carrier Separation	200 kHz	200 kHz	200 kHz	200 kHz	200 kHz
Radio Channels[1]	125	125	175	375	300
Transmission Rate	270 kbits/s	270 kbits/s	270 kbits/s	270 kbits/s	270 kbits/s

Table 3-1 Frequency-related specifications

Frequency Reuse

Why do we need frequency reuse?

Total no of channels (frequencies) = 124

Every channel can be shared between a maximum of 8 subscribers.

Maximum no of simultaneous calls = 8 X 124 = 992 !!

The frequency reuse is performed by dividing the whole available frequencies between a group of neighboring cells which is called frequency reuse pattern or a “Cluster”, and then repeat this cluster over the whole network on 2 conditions:

- The group of frequencies allocated to a given cell must not be used in the adjacent cells.
- Enough distance between the cells where the same group of frequencies are reused.

Radio Coverage

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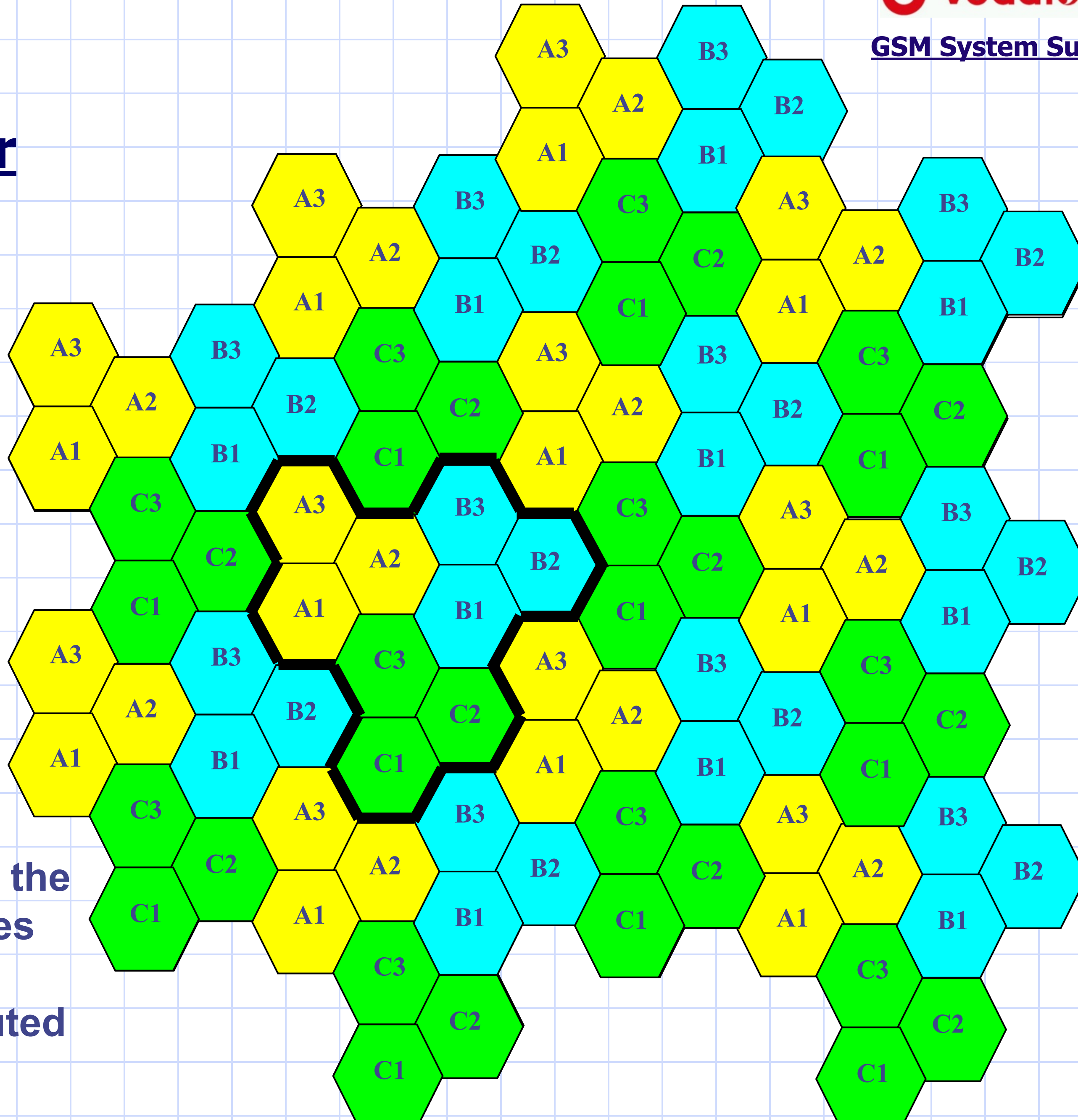
3/9 Cluster



3/9 cluster in which the available frequencies are divided into 9 groups and distributed between 3 sites

Radio Coverage

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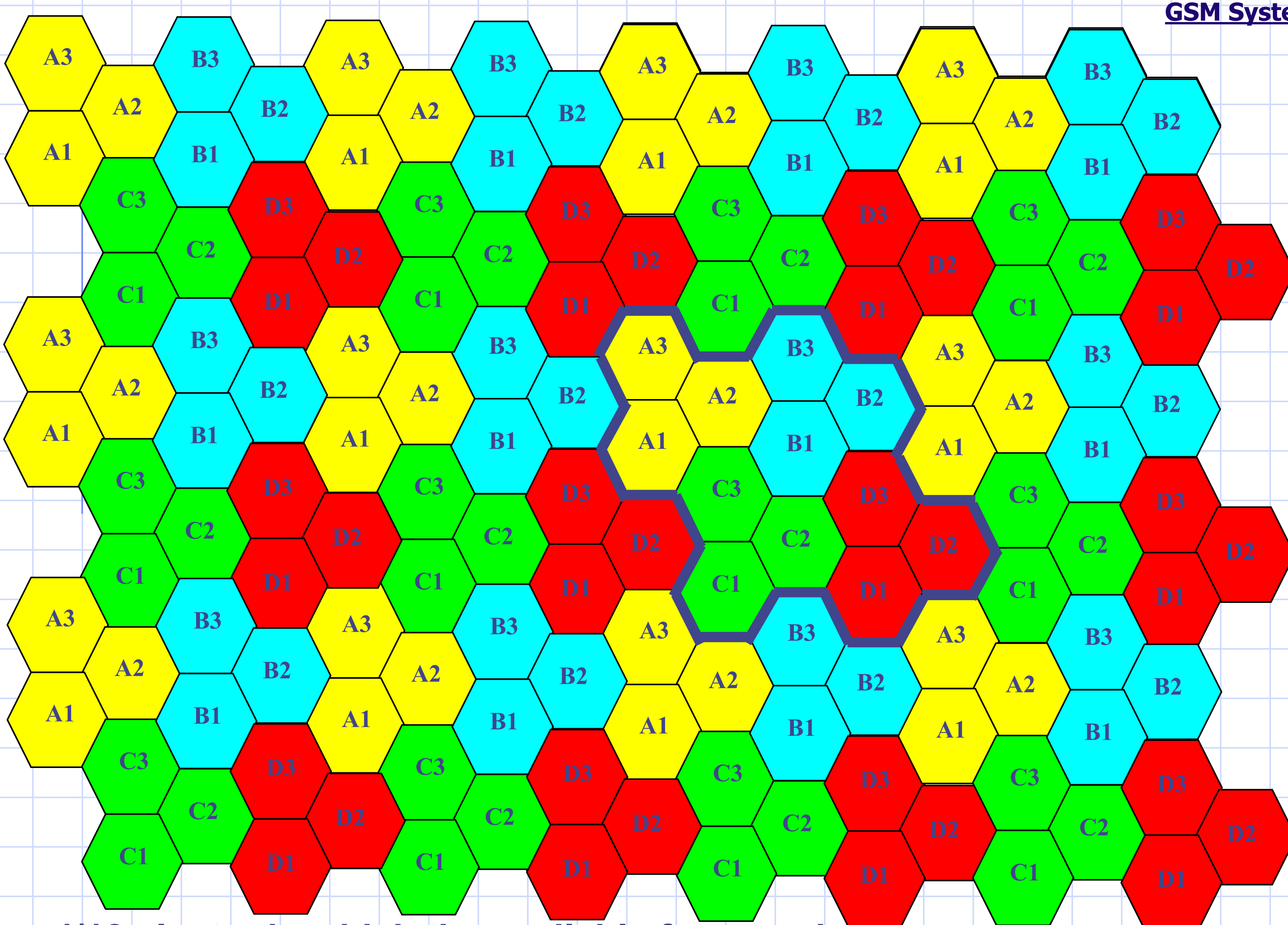

4/12 cluster in which the available frequencies are divided into 12 groups and distributed between 4 sites

4 / 12 Cluster

Radio Coverage

7 / 21 cluster in which the available frequencies are divided into 21 groups and distributed between 7 sites

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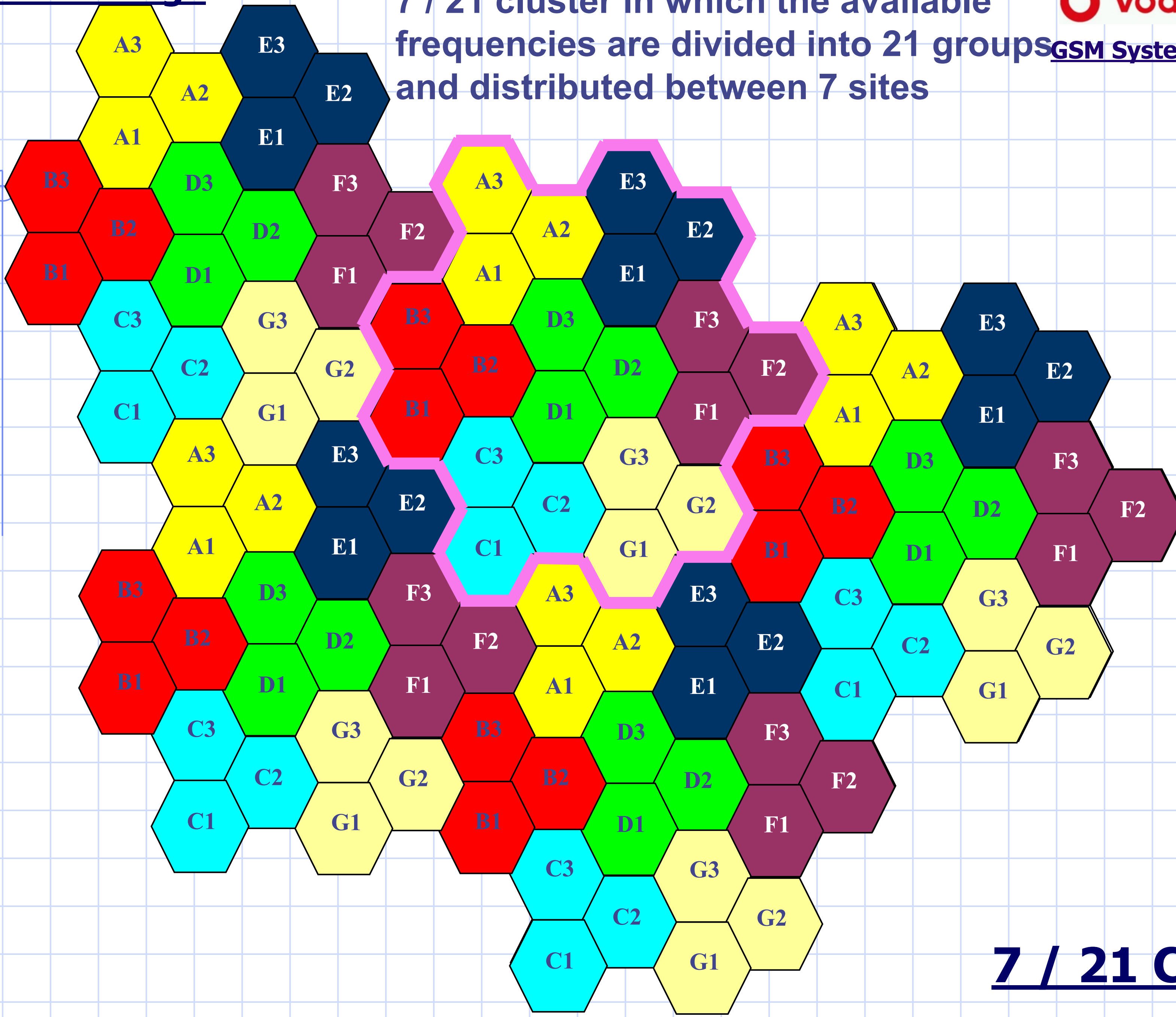


7 / 21 Cluster

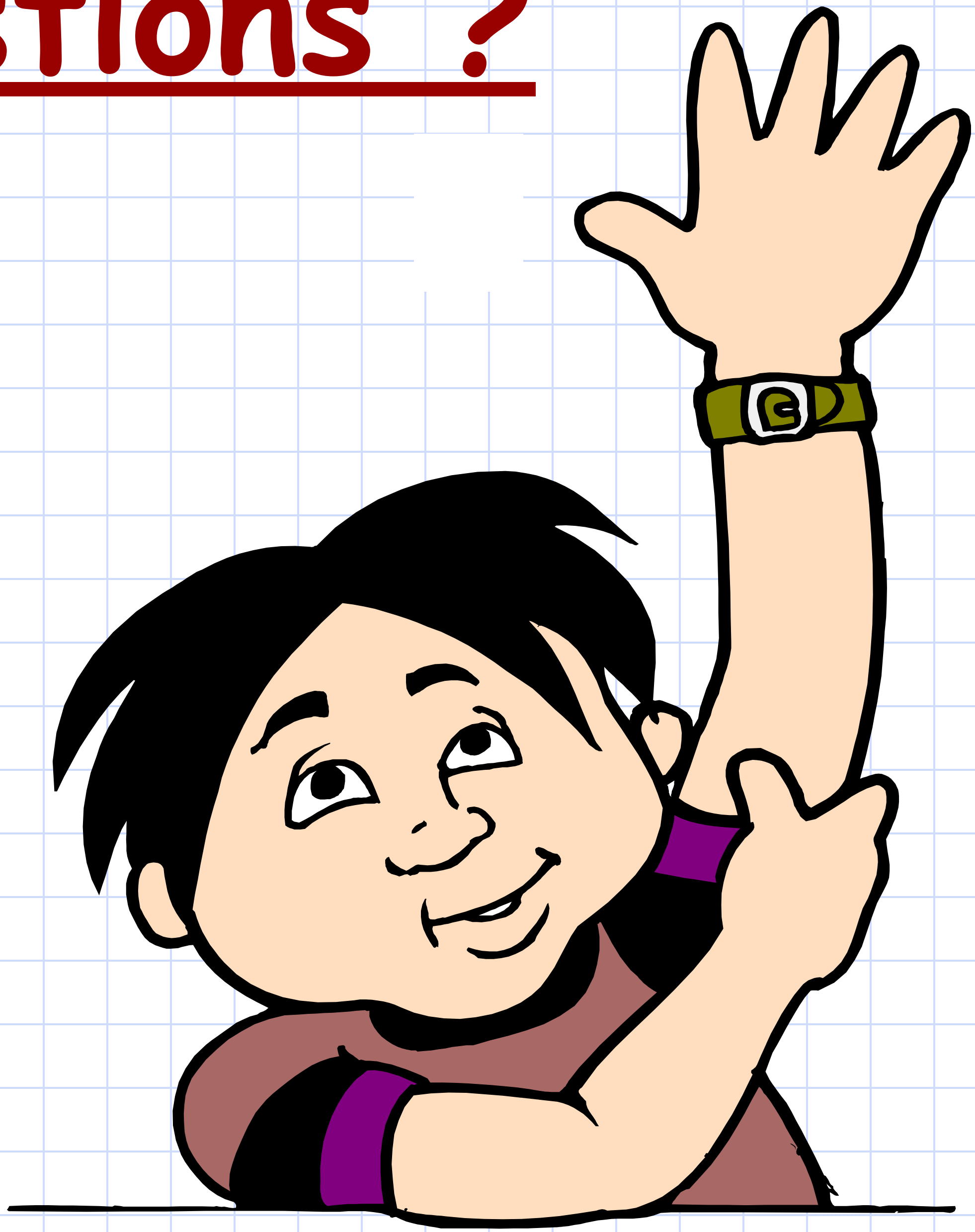
Which Cluster Size to use?

Carrier to interference ratio

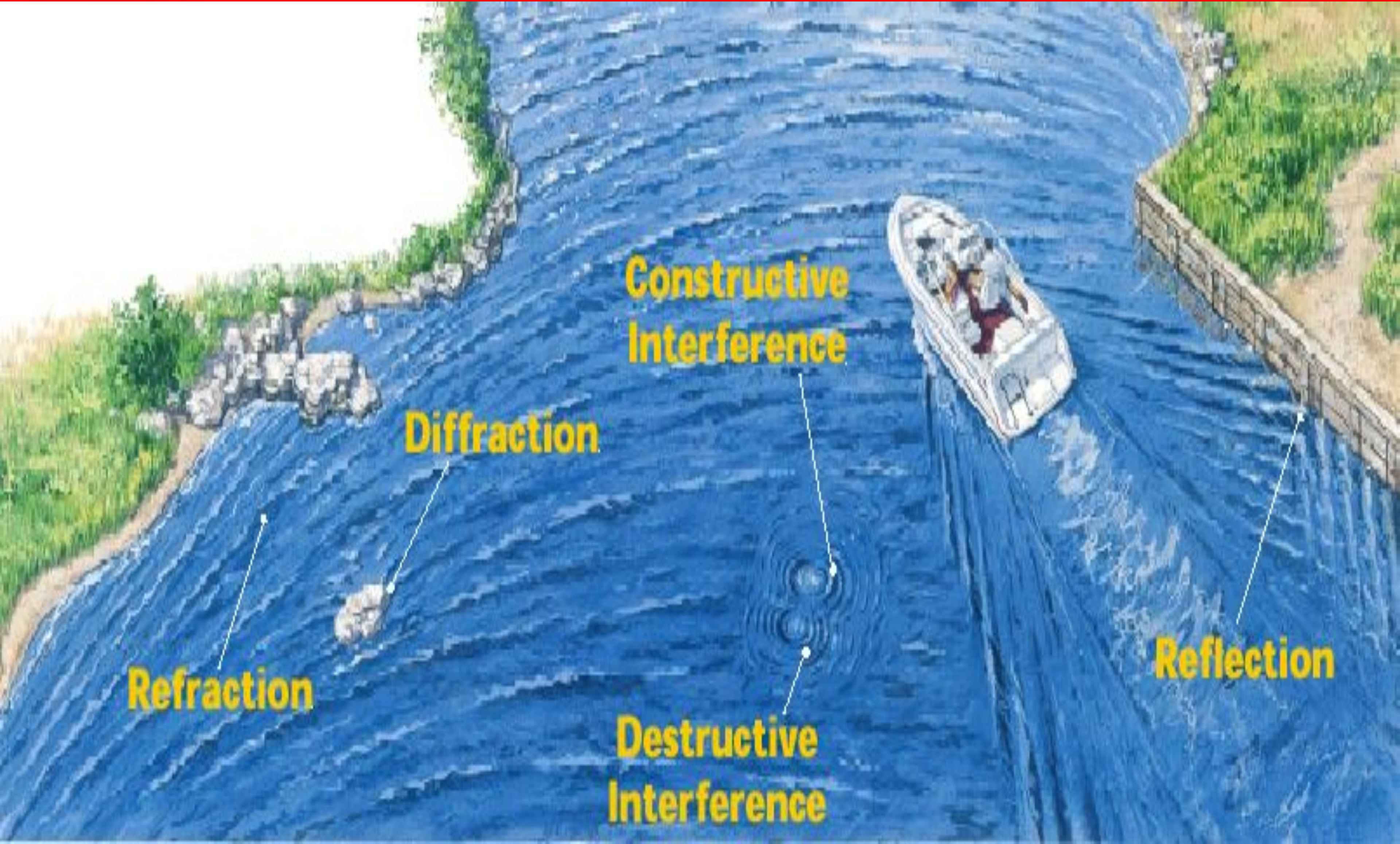
It's the difference in power level between the carrier in a given cell and the same carrier received from the nearest cell that reuses the same frequency.

	Number of frequencies per site	Traffic Channels	C/I Ratio
3/9	High	High	Low
4/12	Medium	Medium	Medium
7/21	Low	Low	High

Questions ?



Chapter 5 : Radio problems & Digital Information



Chapter Objectives

By the End of this Chapter you will:

- Know the Fading Problem and how it is Solved
- Know the Time Dispersion Problem and how it is Solved
- Know the Time Delay Problem and how it is Solved
- Know how the Digital information is coded

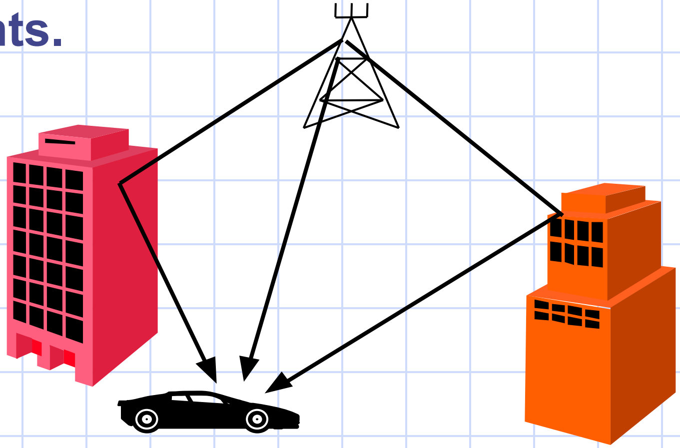
Fading Problems

1. Shadowing (Normal fading):

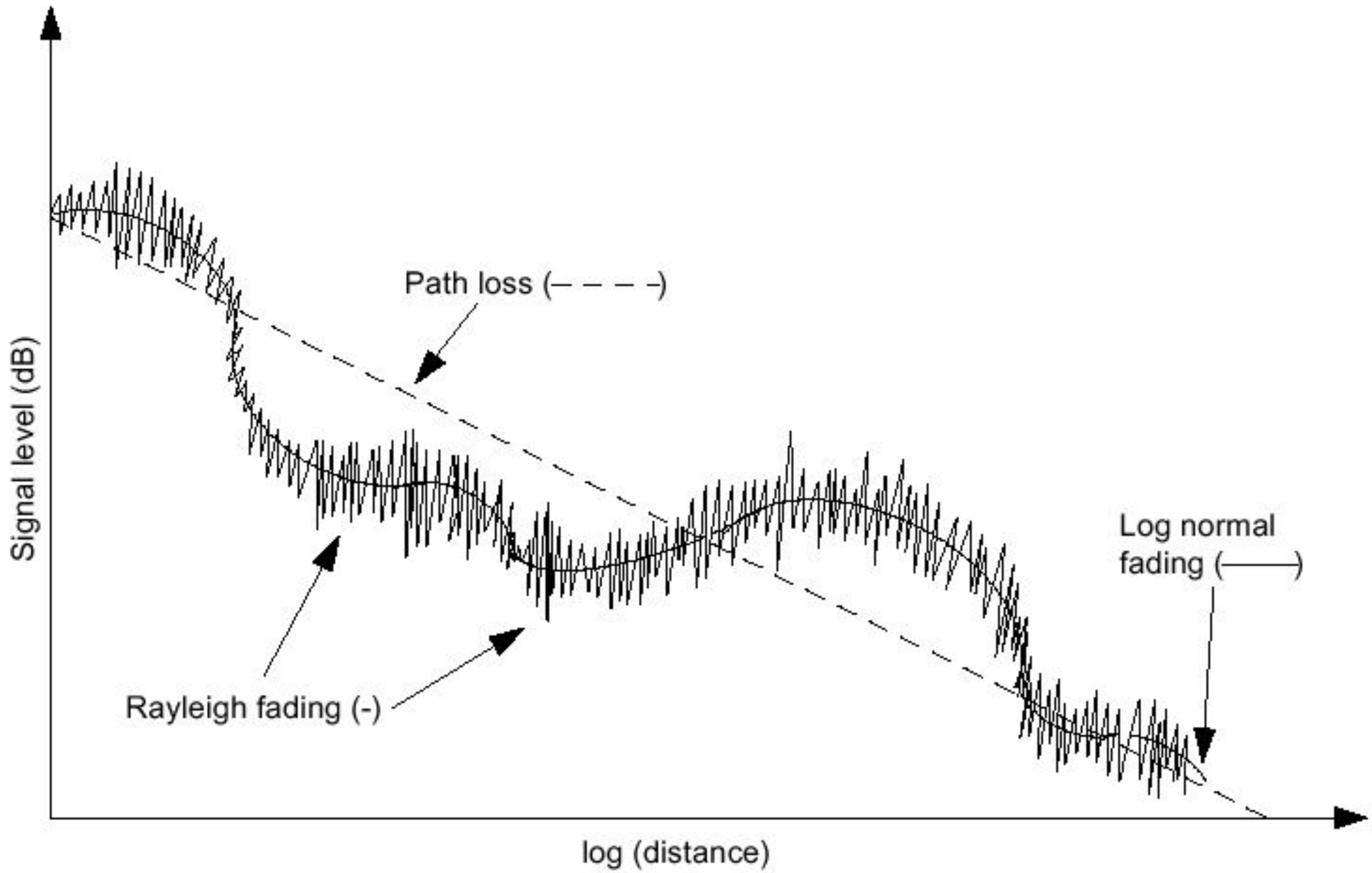
The reason for shadowing is the presence of obstacles like large hills or buildings in the path between the site and the mobile.
The signal strength received fluctuates around a mean value while changing the mobile position resulting in undesirable beats in the speech signal.

2. Rayleigh Fading (Multi-path Fading)

The received signal is coming from different paths due to a series of reflection on many obstacles. The difference in paths leads to a difference in paths of the received components.

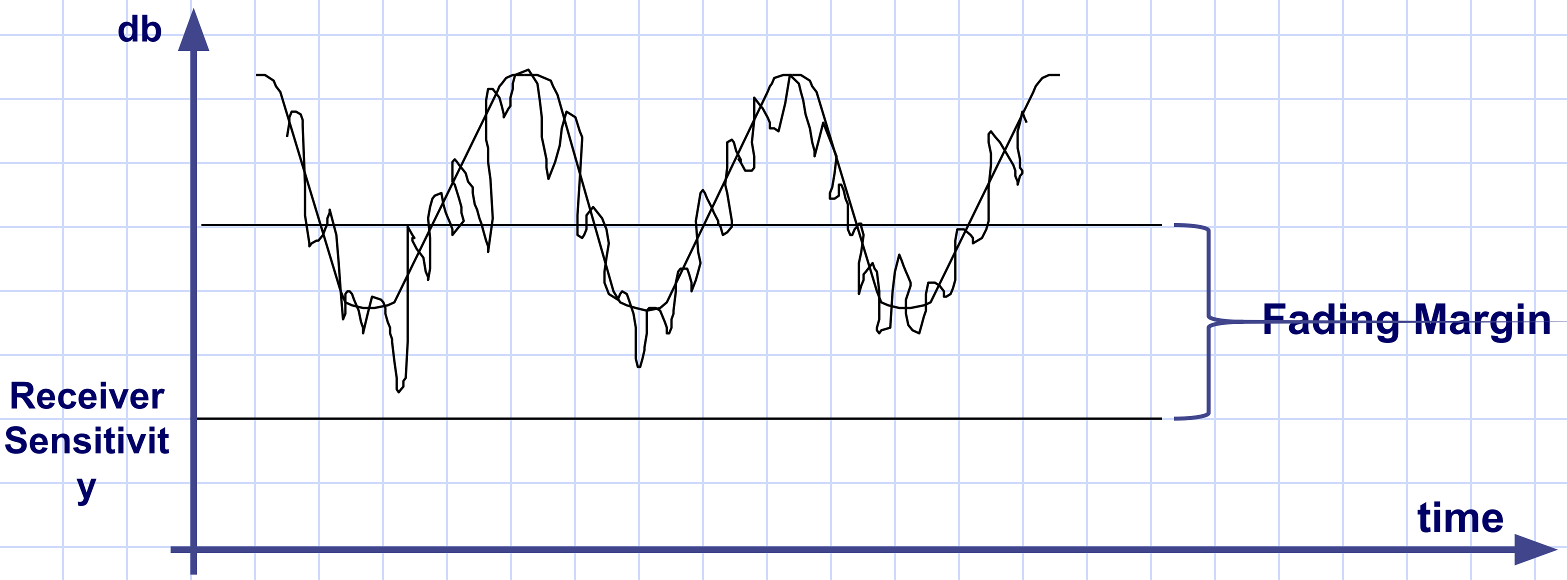


Fading Problems



Fading Problems Solutions

1. Increase the fading Margin

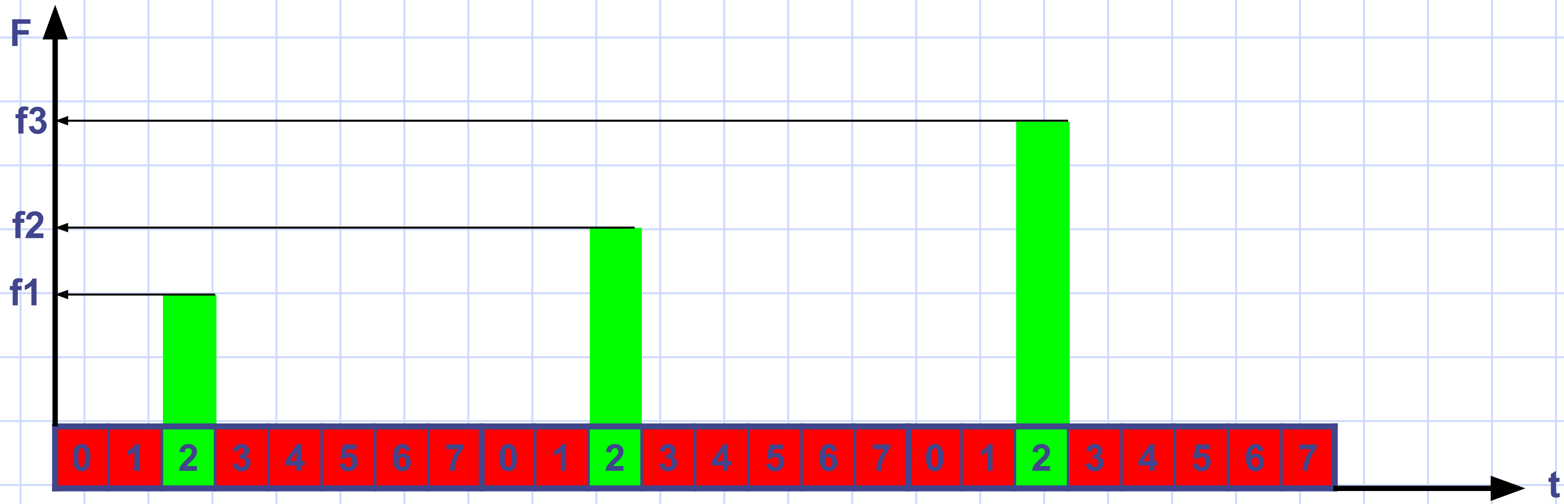


Fading Problems Solutions

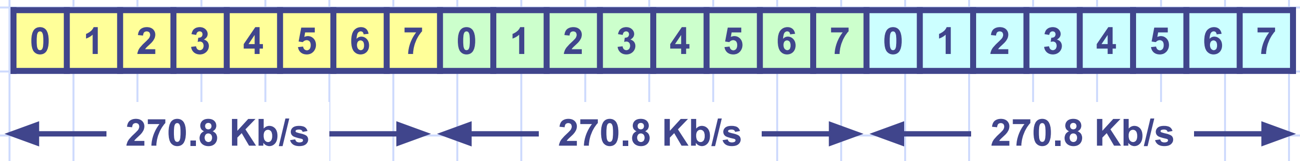
2. Antenna diversity (Space Diversity)

The cell transceiver will use two receiving antennas instead of one. They will be separated by a distance of about 5 meters, and they will receive radio signals independently, so they will be affected differently by the fading dips and the better signal received will then be selected.

3. Frequency hopping (frequency Diversity)



Time dispersion problem



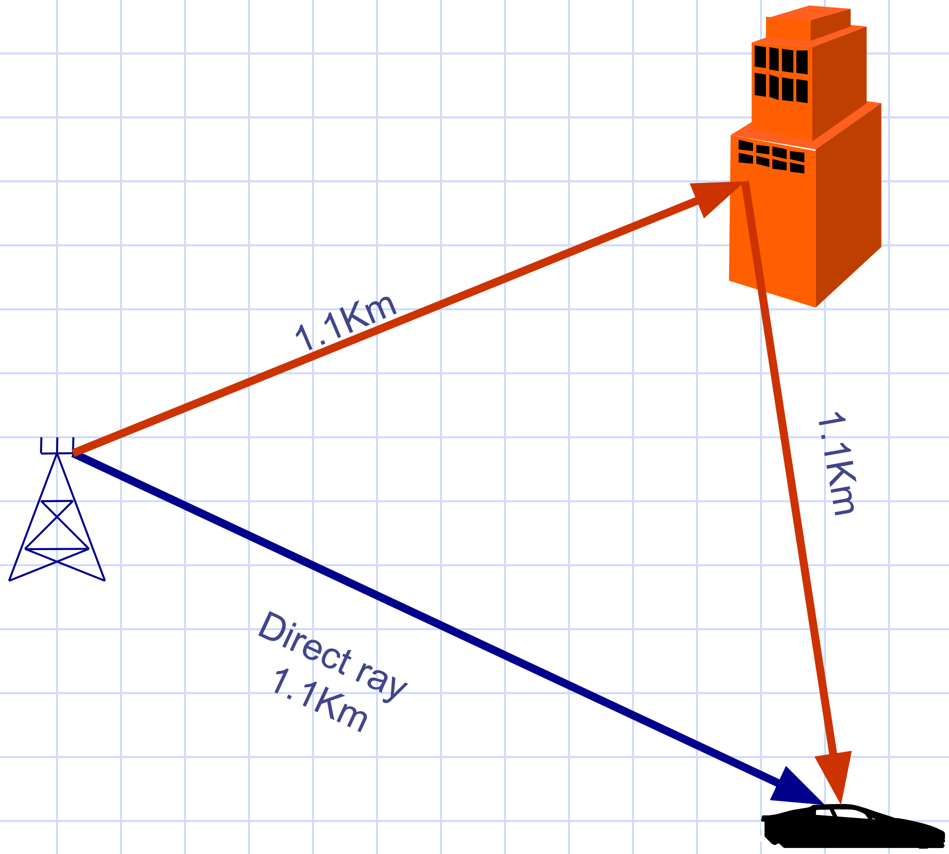
Frame Rate = 270.8 Kb/s

One BIT Duration = 3.7 μ sec

Bit stream is moving with the velocity of light which equals 3 x 10 ^{5} Km/sec

Then, when bit 2 is transmitted, bit one will cut a distance = 3.7 x 10 $^{-6}$ x 3 x 10 ^{5} = 1.1 km

Time dispersion problem



There would be an interference between the a bit in the reflected ray and 1 bit later in the direct way .

Time dispersion problem Solution

1. Increase the Carrier to reflection ratio

The C/R ratio is defined as the difference in signal strength between the signal received from the RBS and the strongest reflected signal .

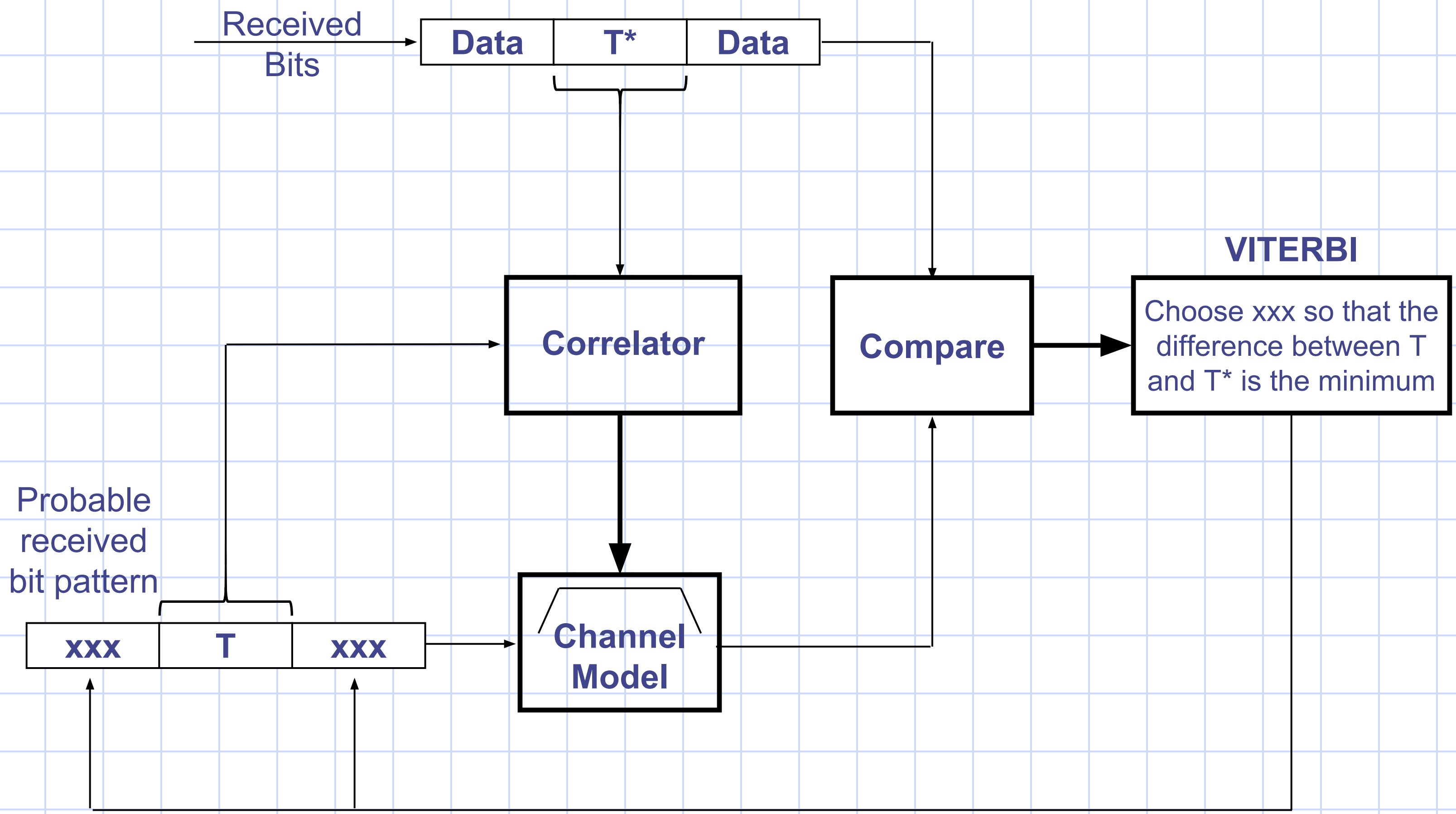
$$C/R \propto \text{Relative position to the BTS}$$

Result

Planner should choose the proper position of the site to make the C/R maximum everywhere in the coverage area of the site.

Time dispersion problem Solution

2. Use Viterbi Equalizer (Adaptive Equalization)



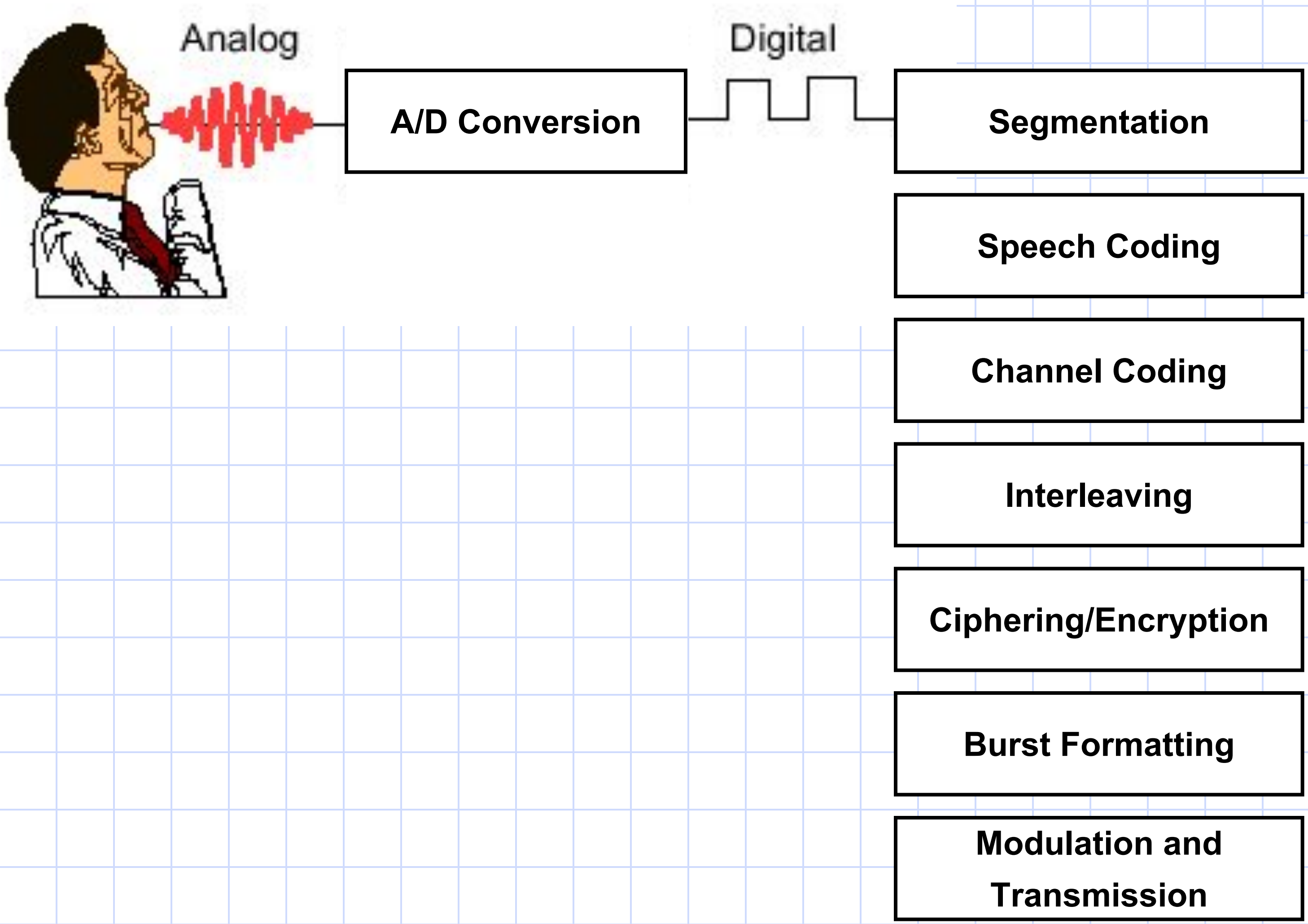
Time Delay/ Time Alignment problem

If a mobile subscriber is located far from the site, then its transmitted bursts will arrive at the cell transceiver with a significant delay that may lead to overlapping with the bursts sent on the next time slot.

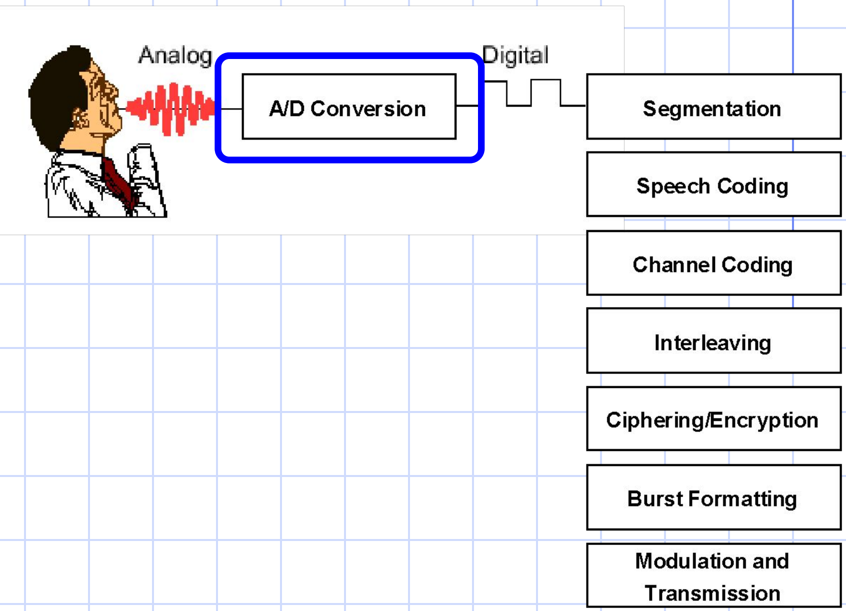
Solution to time delay problem

The site will send a "Timing Advance" value to the mobile station that is moving away, telling it to send its bursts with a certain amount of time ahead of the synchronization time.
The timing advance has values from 0 to 63 depending on how far the mobiles located. The size of a cell is limited by this parameter to a maximum radius of 35 Km.

GSM Transmission Process



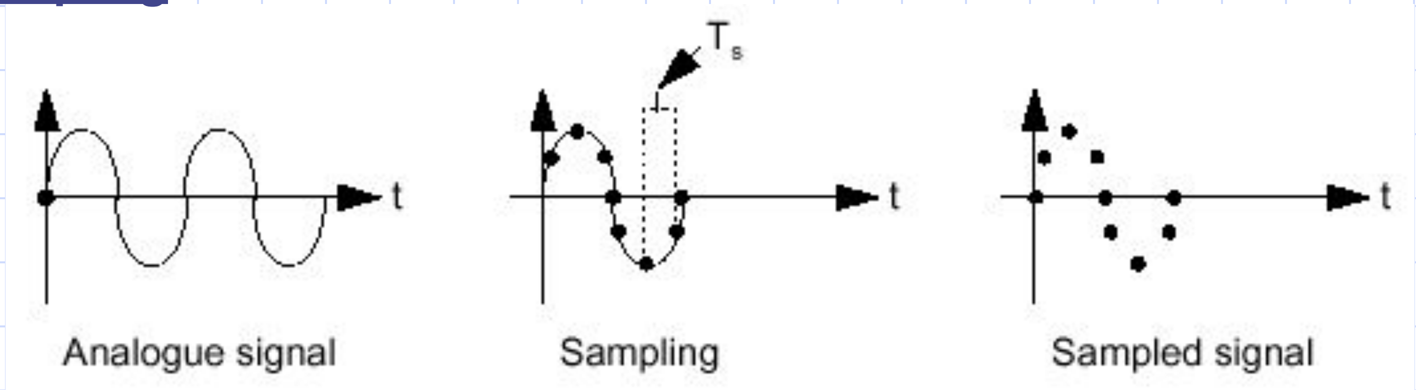
Analog to Digital Conversion



Analog to digital conversion takes place in 3 steps:

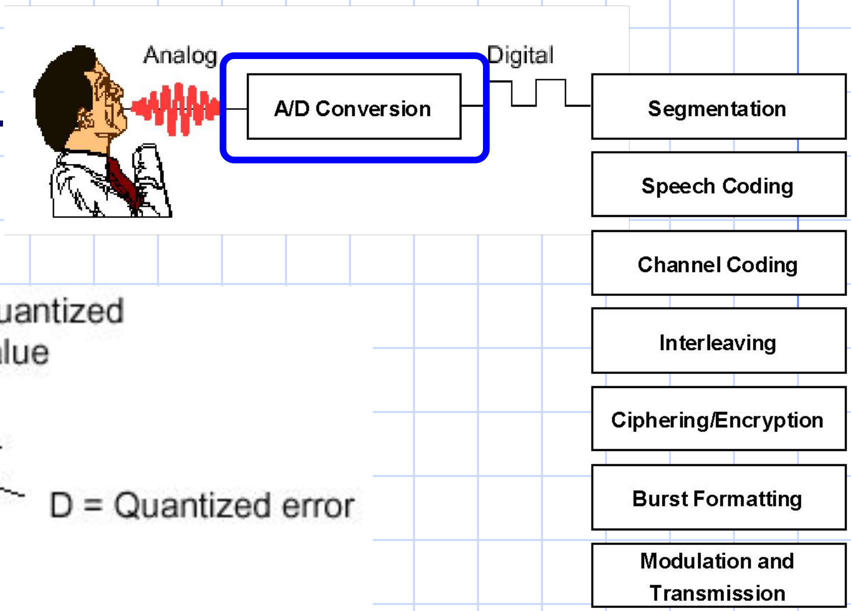
1. Sampling
2. Quantization
3. Coding

1. Sampling

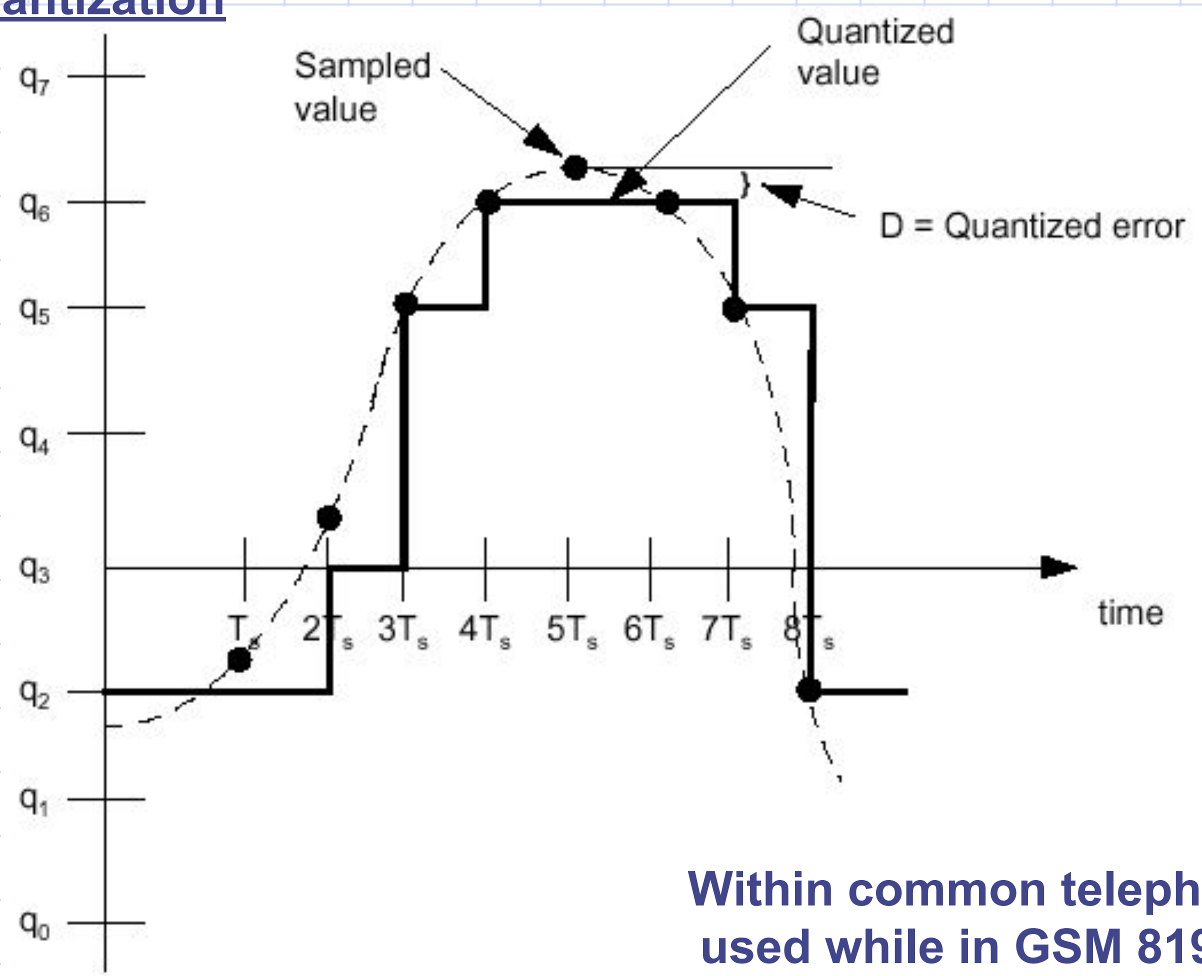


Telecommunication systems use Sampling rate = 8 Kbit/s

Analog to Digital Conversion

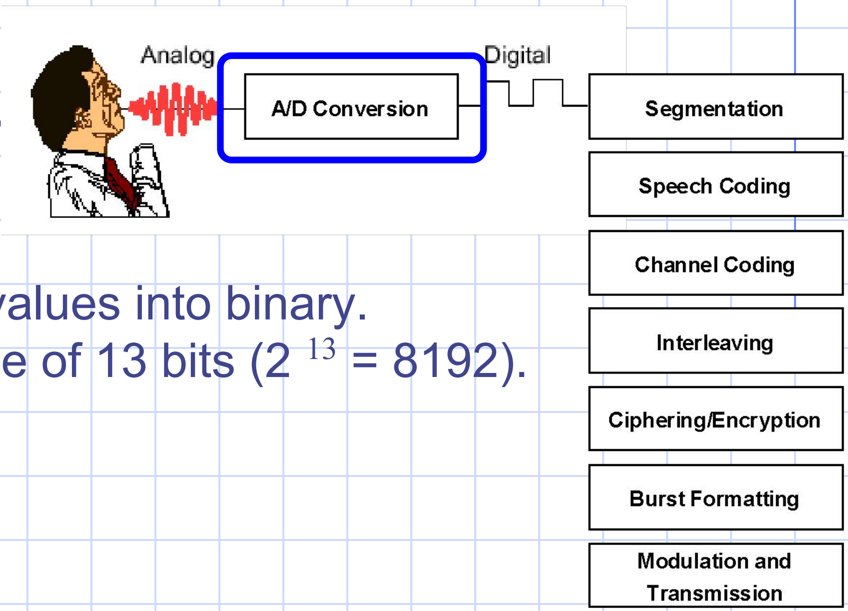


2. Quantization



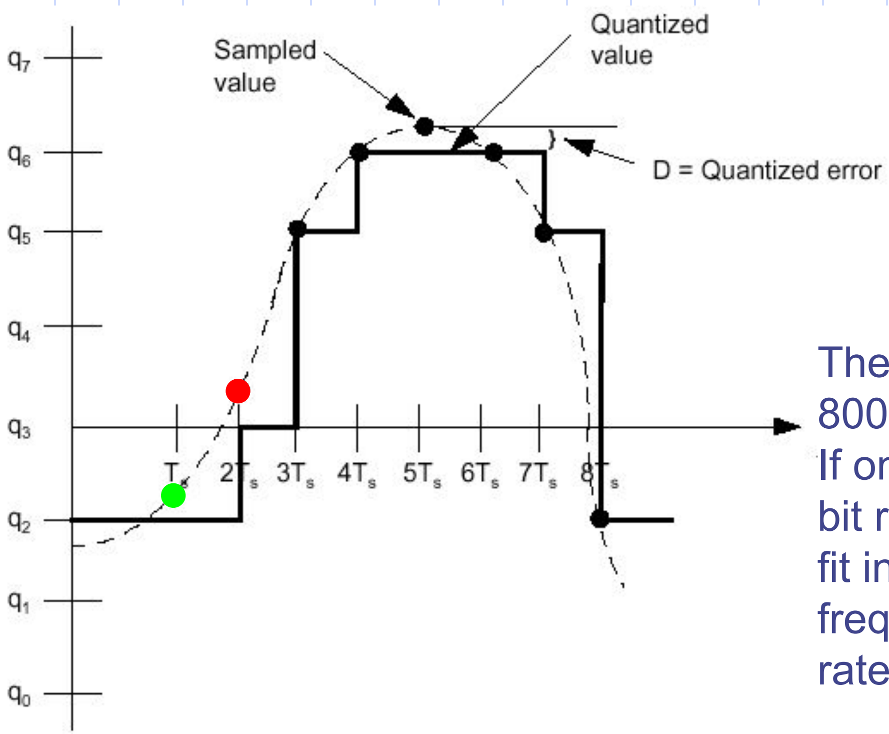
Within common telephony, 256 levels are used while in GSM 8192 levels are used.

Analog to Digital Conversion

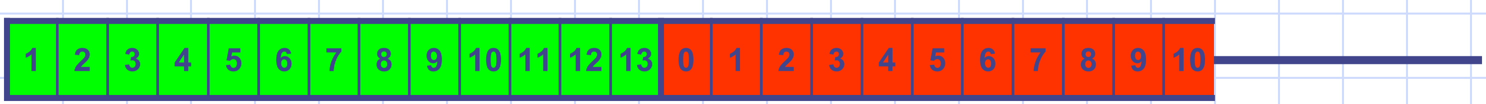


3. Coding

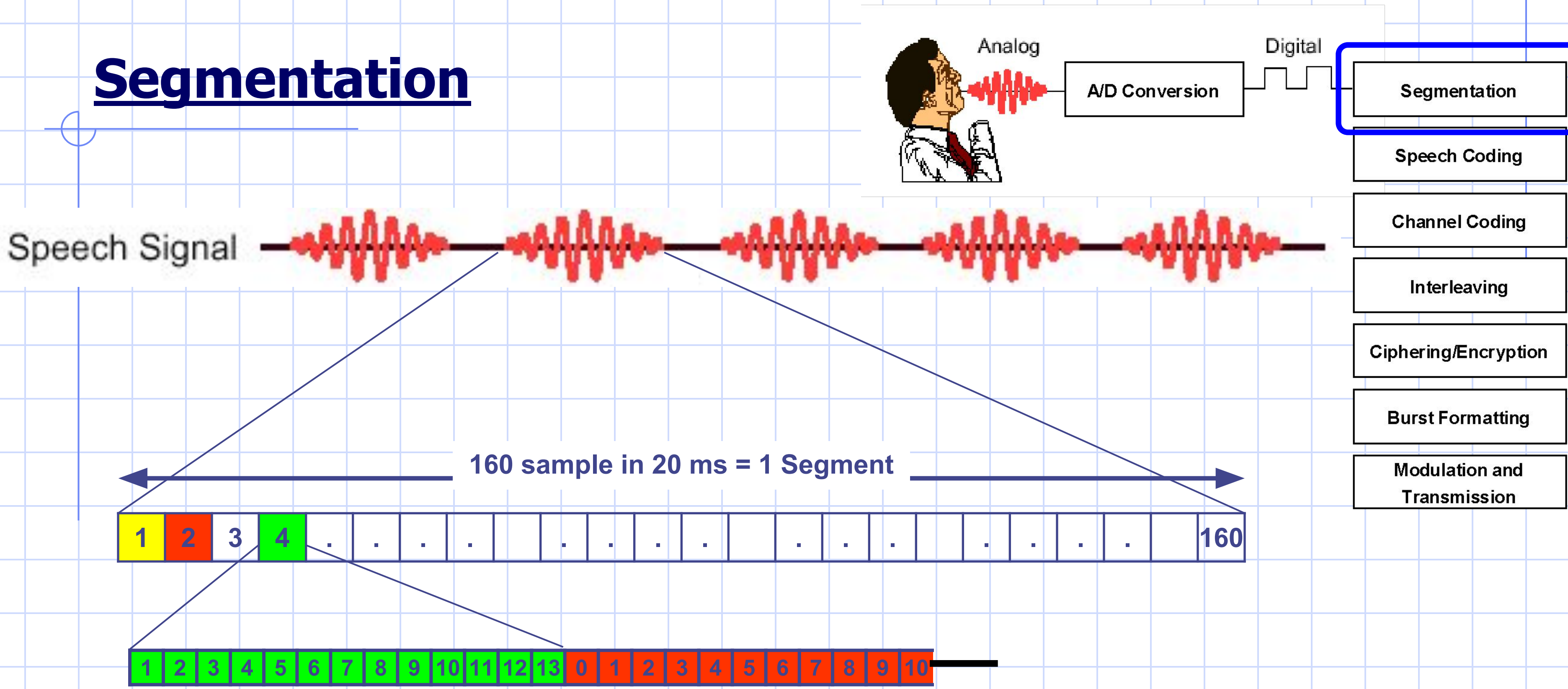
Coding involves converting the quantized values into binary.
Every value is represented by a binary code of 13 bits ($2^{13} = 8192$).



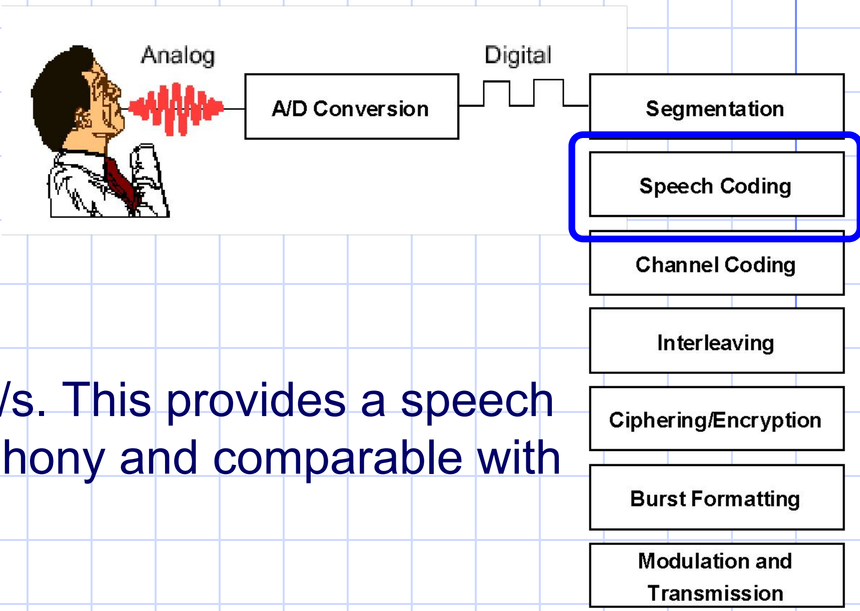
The output rate of the A/D Conversion process is:
8000 Samples/Sec x 13 bits/Sample = 104 Kb/s
If one frequency will be used for 8 calls, then the bit rate will be 8 x 104 kb/s = 832 kb/s this will not fit in the 200 KHz channel allocated for one frequency. Coding should be used to reduce the rate.



Segmentation



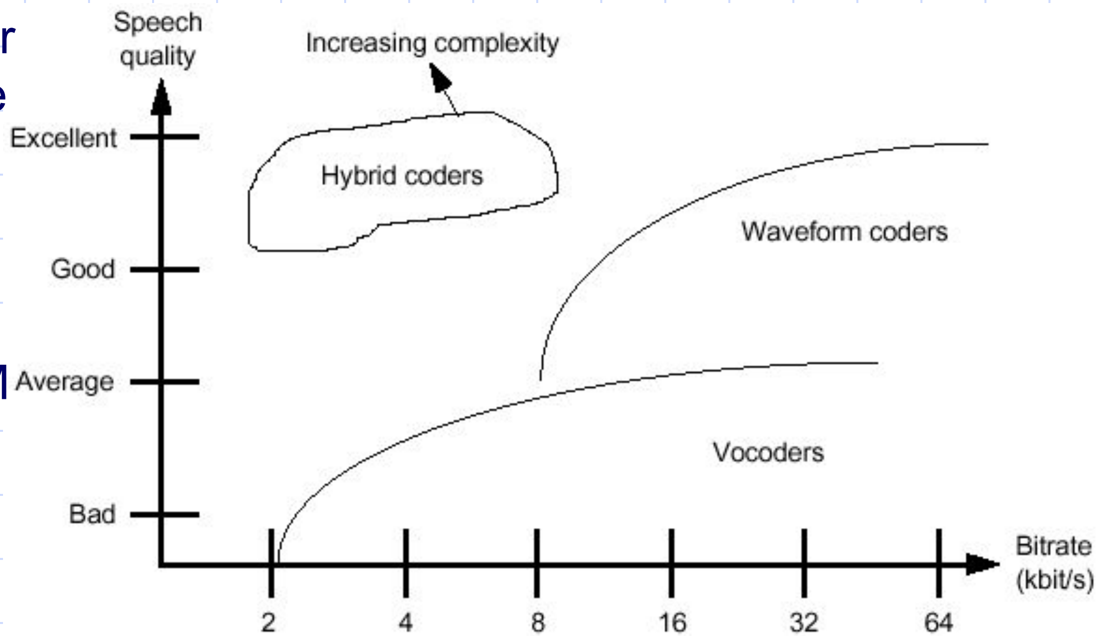
GSM Speech Coding



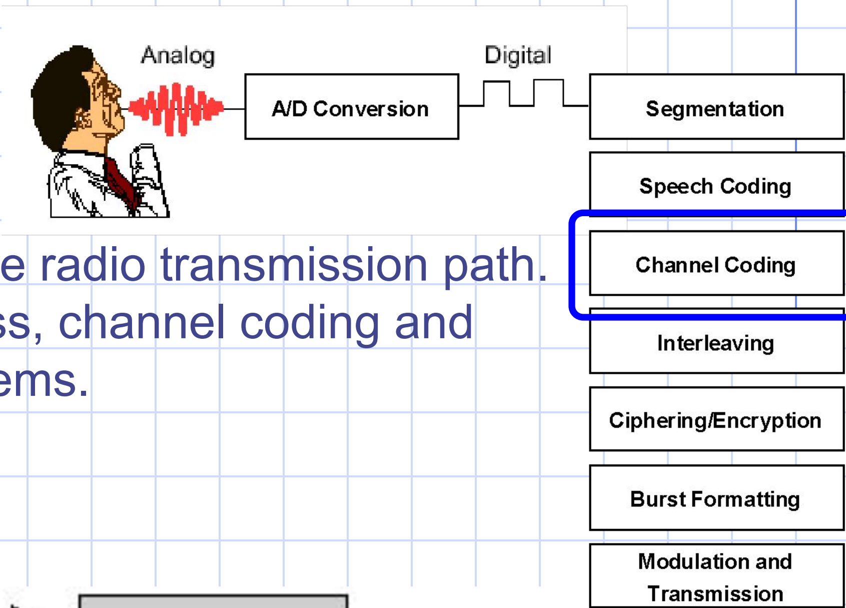
Instead of using 13 bits per sample as in A/D conversion, GSM speech coding uses 260 bits to encode one segment.

This calculates as 260 bits / 20 ms = 13 kb/s. This provides a speech quality which is acceptable for mobile telephony and comparable with wire line PSTN phones.

Many types of speech coders are available. Some offer better speech quality, at the expense of a higher bit rate (waveform coders). Others use lower bit rates, at the expense of lower speech quality (vocoders). The hybrid coder used by GSM provides good speech quality with a low bit rate, at the expense of speech coder complexity.

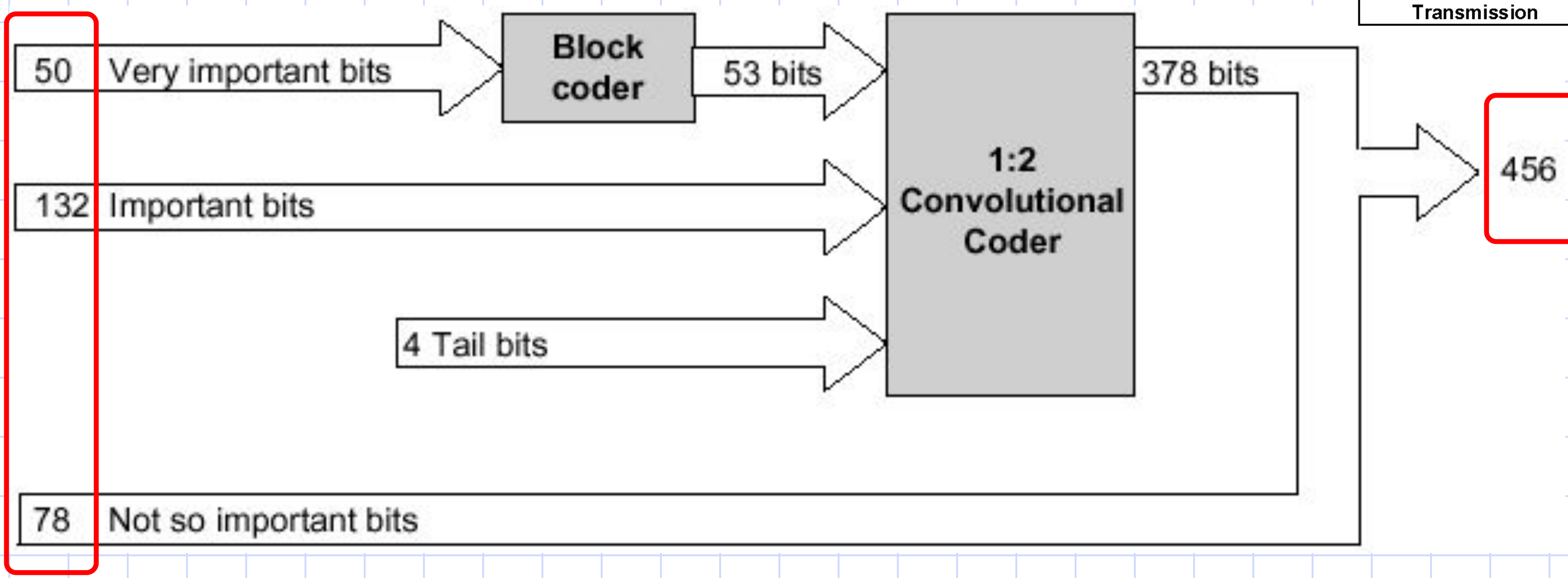


Channel Coding

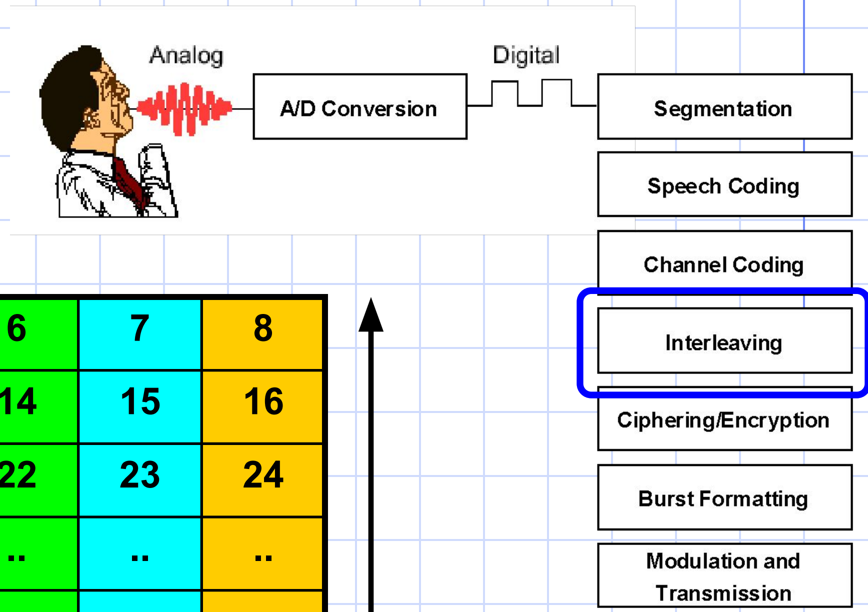


speech coding does not consider the problems which may be encountered on the radio transmission path. The next stages in the transmission process, channel coding and interleaving, help to overcome these problems.

Input is 260 bits representing 1 segment



Interleaving



1. First Level Interleaving

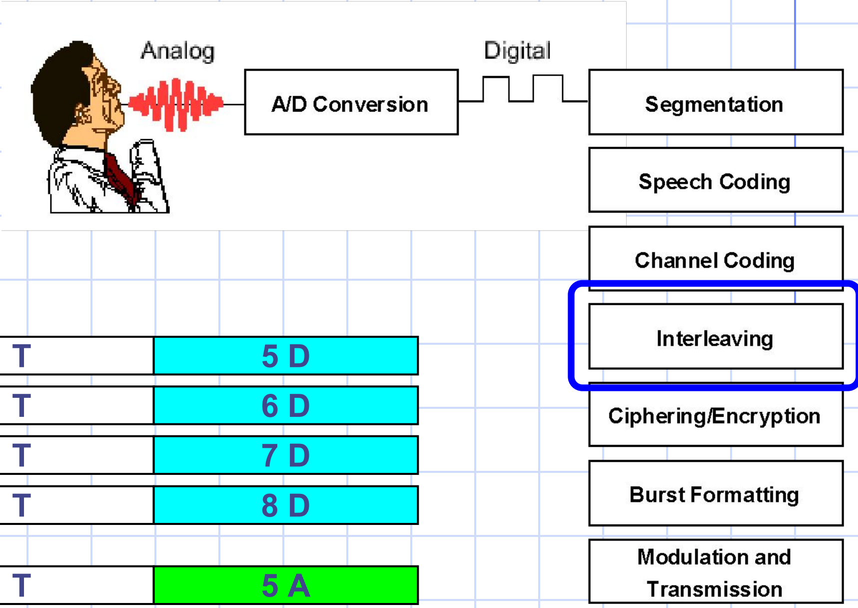
1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	..	..	..	..
..	..	..	..	..	..	..	..
..	..	..	..	..	..	..	..
..	..	..	..	..	..	..	..
..	..	..	..	..	..	..	..
				429	430	431	432
433	434	435	436	437	438	439	440
441	442	443	444	445	446	447	448
449	450	451	452	453	454	455	456

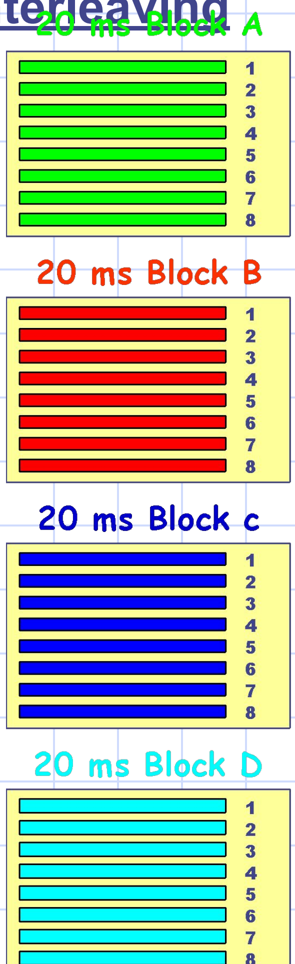
57 Bits

8 Groups

Interleaving

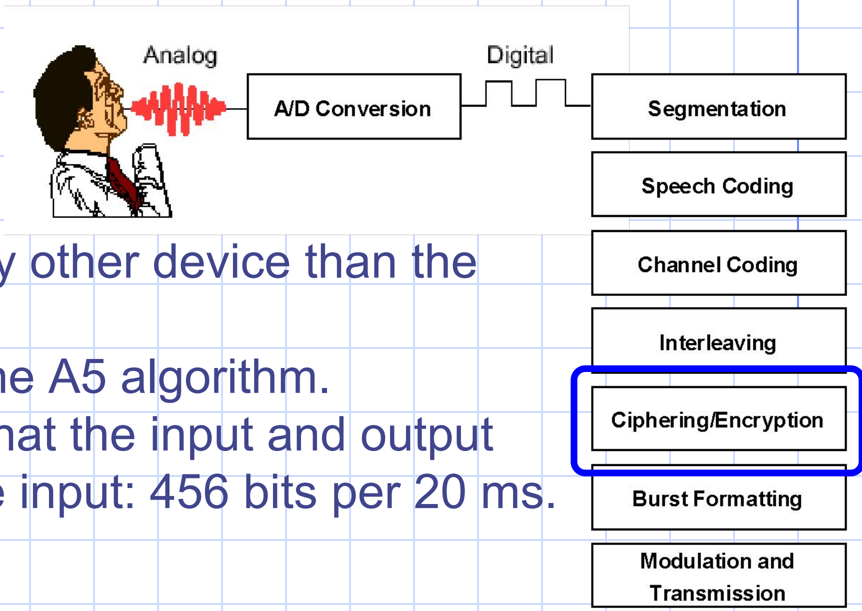
2. Second Level Interleaving





1 A	T	5 D
2 A	T	6 D
3 A	T	7 D
4 A	T	8 D
1 B	T	5 A
2 B	T	6 A
3 B	T	7 A
4 B	T	8 A
1 C	T	5 B
2 C	T	6 B
3 C	T	7 B
4 C	T	8 B
1 D	T	5 C
2 D	T	6 C
3 D	T	7 C
4 D	T	8 C

Ciphering and Encryption

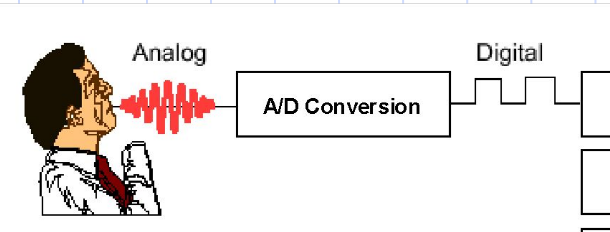


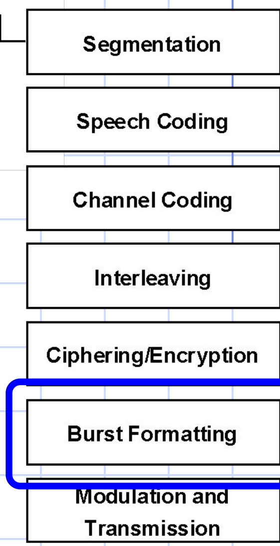
The purpose of ciphering is to encode the burst so that it cannot be interpreted by any other device than the intended receiver.

The ciphering algorithm in GSM is called the A5 algorithm.

It does not add bits to the burst, meaning that the input and output to the ciphering process is the same as the input: 456 bits per 20 ms.

Burst Formatting





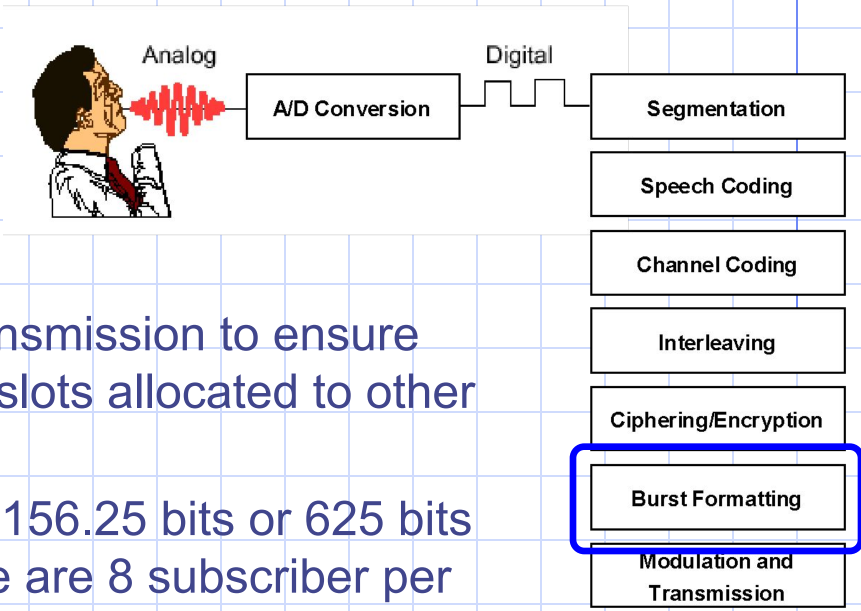
Every transmission from an MS/BTS must include some extra information such as the training sequence. The process of burst formatting is to add these bits (along with some others such as tail bits) to the basic speech/data being sent.

In GSM, the input to burst formatting is the 456 bits received from ciphering. Burst formatting adds a total of 136 bits per block of 20 ms, bringing the overall total to 592.

Now, the 592 bits will be sent on 4 bursts, each containing 2 x 57 bits + 136 / 4 = 148 bits.

However, each time slot on a TDMA frame is 0.577 ms long. This provides enough time for 156.25 bits to be transmitted (each bit takes 3.7 us),
The rest of the space, 8.25 bit times, is empty and is called the Guard Period (GP). This time is used to enable the MS/BTS “ramp up” and “ramp down”.

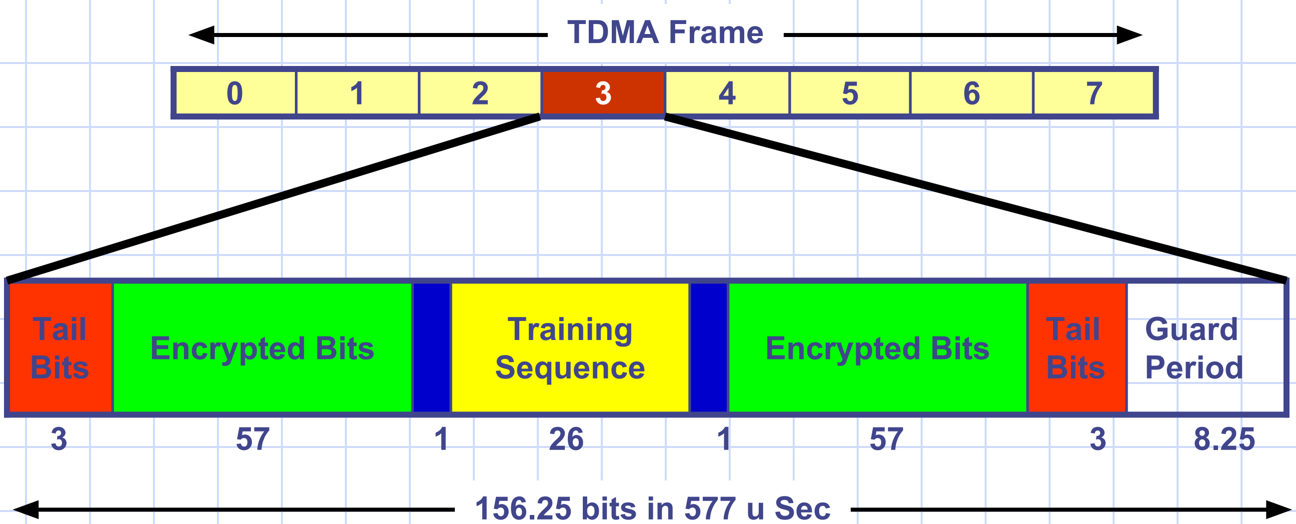
Burst Formatting



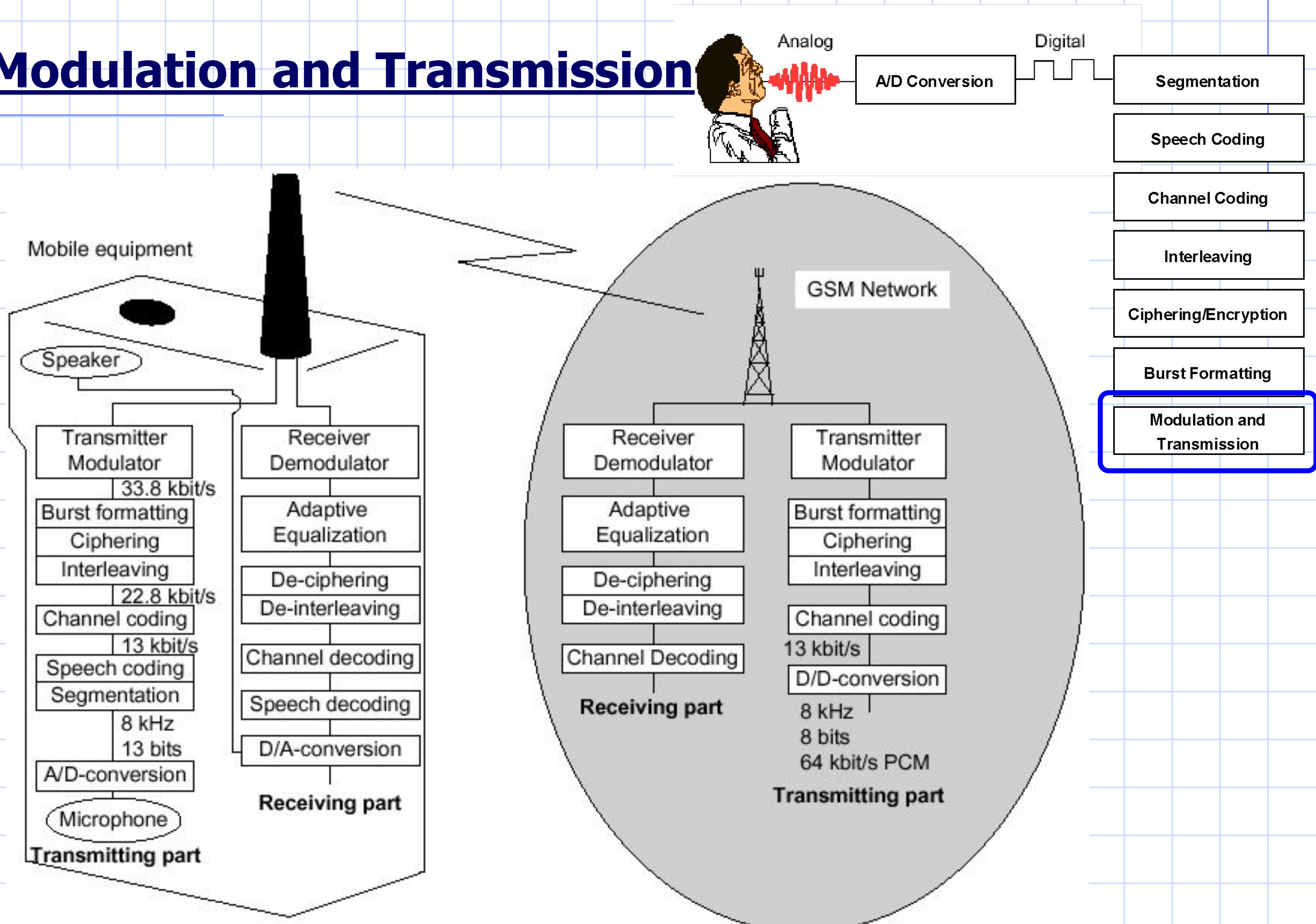
To ramp up means to get power from the battery/power supply for transmission.

Ramping down is performed after each transmission to ensure that the MS is not transmitting during time slots allocated to other MSs.

The output of burst formatting is a burst of 156.25 bits or 625 bits per 20 ms. When it is considered that there are 8 subscriber per TDMA frame, the overall bit rate for GSM can be calculated to be 270.9 kbits/s.



Modulation and Transmission



Chapter 6 : Air Interface

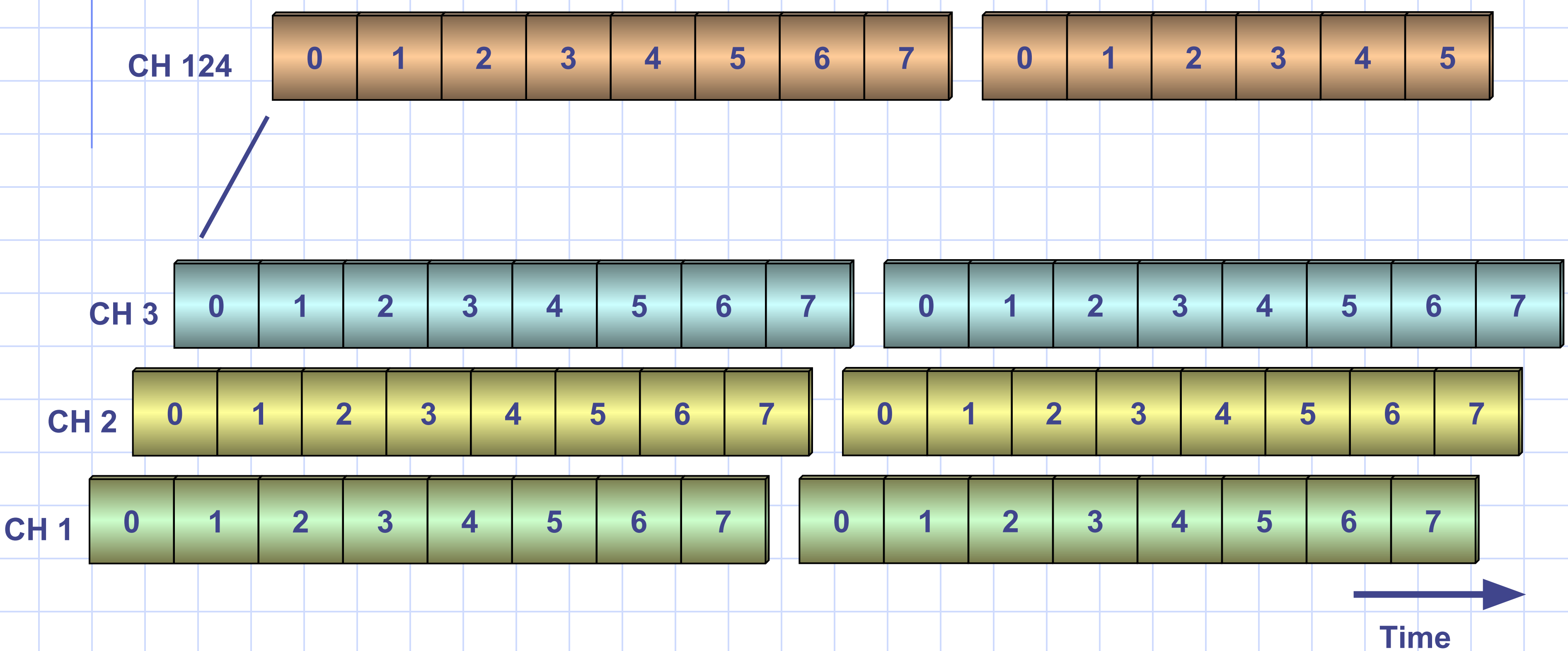
Chapter Objectives

By the End of this Chapter you will:

- Know How the Physical and Logical Channels are classified
- Know the different types of Traffic Channels
- Know the different types of Control Channels
- Know the Structure of each Control Channel
- Know how control and traffic data is Mapped in the air interface
- Know the terminology of different TDMA Frames

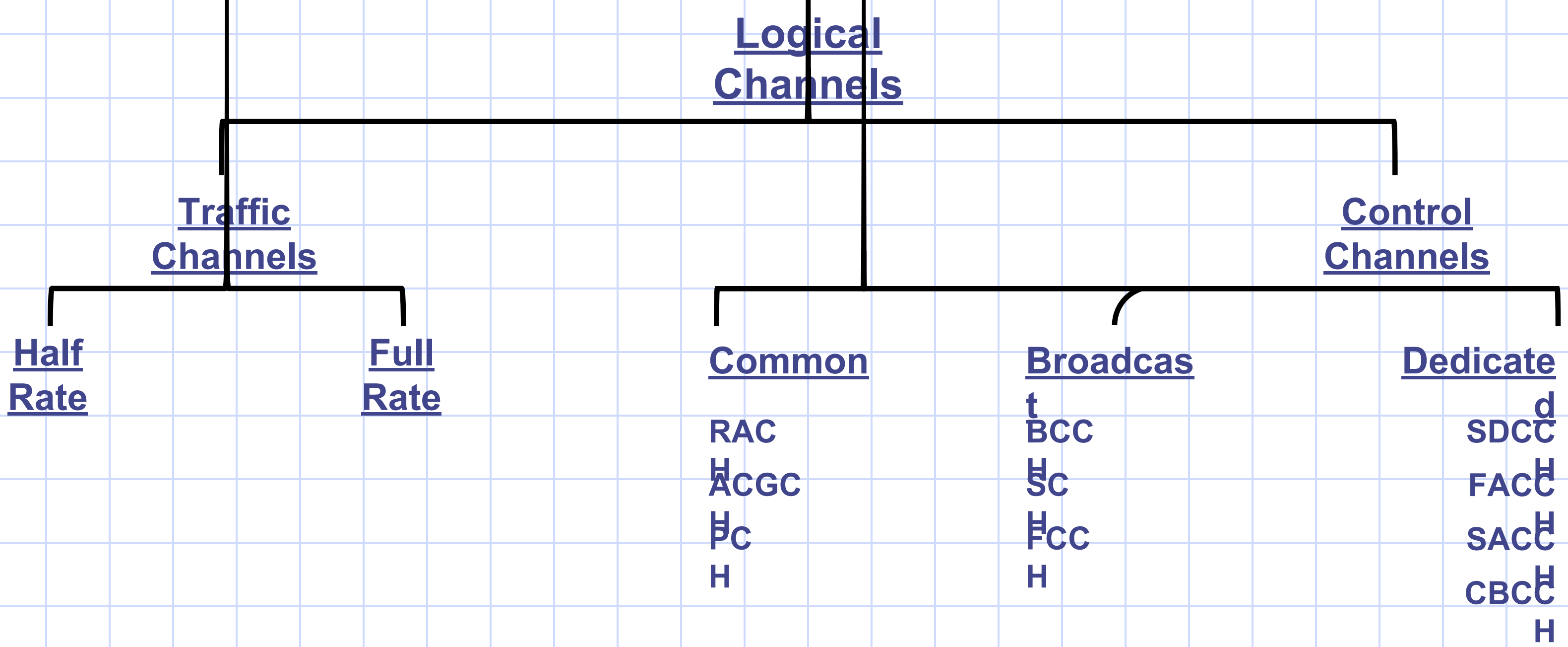
Physical Channels

GSM band is divided into 124 RF channels, and each channel is divided into 8 time slots using TDMA. These time slots are called "physical channels".



Logical Channels

A physical channel may be occupied by a traffic channel or a control channel, both of them are classified as "logical channels".



Traffic Channels

Carries either encoded speech or user data up and down link between a single mobile and a single BTS.

Enhanced Full Rate = 15.1 Kbit/S

Full Rate = 13 Kbit/S

Half Rate = 6.5 Kbit/S

Broadcast Channels

From Single BTS to all the mobiles in the area

Frequency Correction Control CHannel (FCCH)

Carries information for frequency correction of the mobile

Synchronization CHannel (SCH)

Carries 2 important pieces of information

- TDMA frame number (max = 2715684)
- Base station identity Code (BSIC)

Broadcast Control CHannel (BCCH)

Carries Cell specific data

Broadcast CHannels (BCH's)			
Logical Channel	*Direction*	*BTS*	*MS*
Frequency Correction CHannel (FCCH)	Downlink, point to multipoint	Transmits a carrier frequency.	Identifies BCCH carrier by the carrier frequency and synchronizes with the frequency.
Synchronization CHannel (SCH)	Downlink, point to multipoint	Transmits information about the TDMA frame structure in a cell (e.g. frame number) and the BTS identity (Base Station Identity Code (BSIC)).	Synchronizes with the frame structure within a particular cell, and ensures that the chosen BTS is a GSM BTS - BSIC can only be decoded by an MS if the BTS belongs to a GSM network.
Broadcast Control CHannel (BCCH)	Downlink, point to multipoint	Broadcasts some general cell information such as -Location Area Identity (LAI), -maximum output power allowed in the cell and -the identity of BCCH carriers for neighboring cells.	Receives LAI and will signal to the network as part of the Location Updating procedure if the LAI is different to the one already stored on its SIM. MS sets its output power level based on the information received on the BCCH. The MS stores the list of BCCH carrier frequencies on which Rx. level measurement is done for Handover decision.

Common Control Channels

To or from a certain BTS to a single mobile

Paging CHannel (PCH) ↓

It's used to page (search) for a specific mobile

Random Access CHannel (RACH) ↑

Request allocation of SDCCH

Access Grant CHannel (AGCH) ↓

Allocate SDCCH to the mobile station.

Common Control Channels (CCCH)			
Logical Channel	*Direction*	*BTS*	*MS*
Paging CHannel (PCH)	Downlink, point to multi-point	Transmits a paging message to indicate an incoming call or short message. The paging message contains the identity number of the mobile subscriber that the network wishes to contact.	At certain time intervals the MS listens to the PCH. If it identifies its own mobile subscriber identity number on the PCH, it will respond.
Random Access CHannel (RACH)	Uplink, point to point	Receives access-request from MS for call setup/ loc. update/ SMS	Answers paging message on the RACH by requesting a signaling channel.
Access Grant CHannel (AGCH)	Downlink, point to point	Assigns a signaling channel (SDCCH) to the MS.	Receives signaling channel assignment (SDCCH).

Dedicated Control Channels

Standalone Dedicated Control CHannel (SDCCH)

Carries system signaling during:

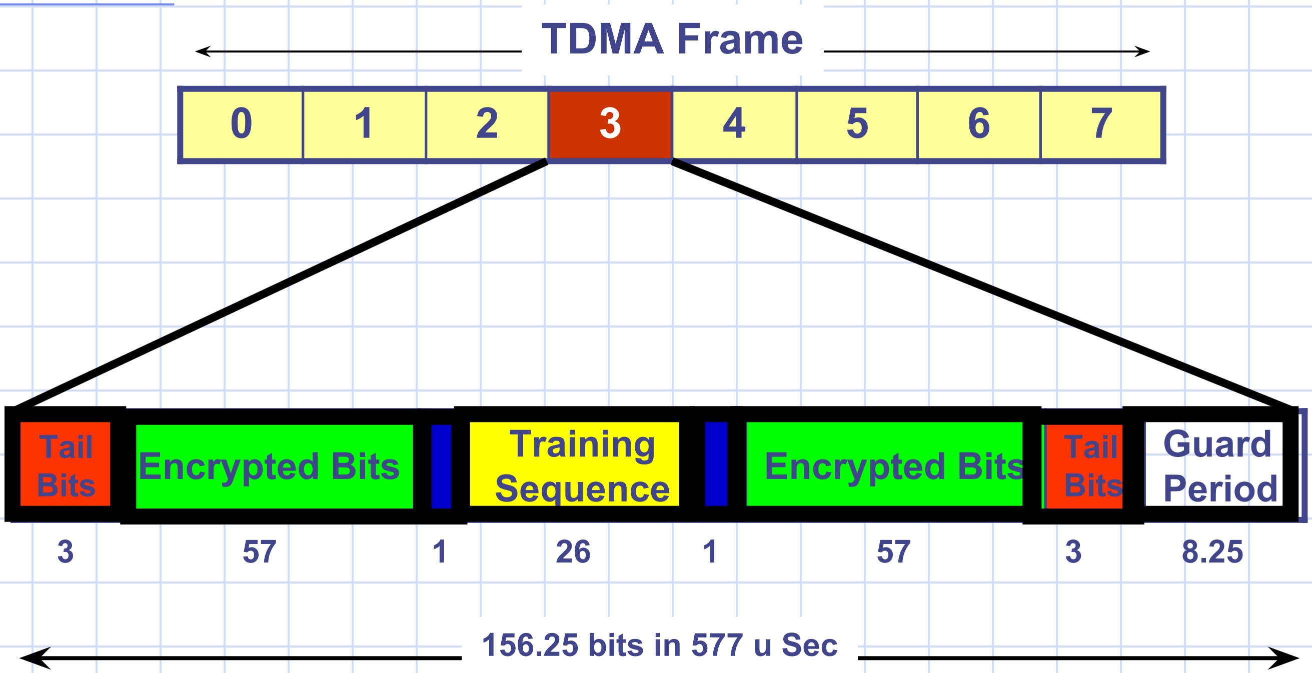
- A call setup before allocating a TCH
- Registration & Authentication.
- Transmission of SMS in idle mode.
- MS paging Response.

Cell Broadcast Control CHannel (CBCCH)

- It displays general information.
- It uses one of the SDCCH channels
- MS must be setup to receive this channel.

Dedicated Control Channels (DCCH)			
Logical Channel	*Direction*	*BTS*	*MS*
Stand alone Dedicated Control CHannel (SDCCH)	Uplink and downlink, point to point	The BTS switches to the assigned SDCCH, used for call set-up signaling. TCH is assigned on SDCCH. (SDCCH is also used for SMS messages to MS).	The MS switches to the assigned SDCCH. Call set-up is performed. The MS receives a TCH assignment information (carrier and time slot).
Cell Broadcast CHannel (CBCH)	DL,point to multi point, mapped on SDCCH	Uses this logical channel to transmit short message service cell broadcast.	MS receives cell broadcast messages.
Slow Associated Control CHannel (SACCH)	Uplink and downlink, point to point	Instructs the MS on the allowed transmitter power and parameters for time advance. SAACH is used for SMS during a call.	Sends averaged measurements on its own BTS (signal strength and quality) and neighboring BTS's (signal strength). The MS continues to use SACCH for this purpose during a call.
Fast Associated Control CHannel (FACCH)	Uplink and downlink, point to point	Transmits handover information.	Transmits necessary handover information in access burst

Normal Burst Structure



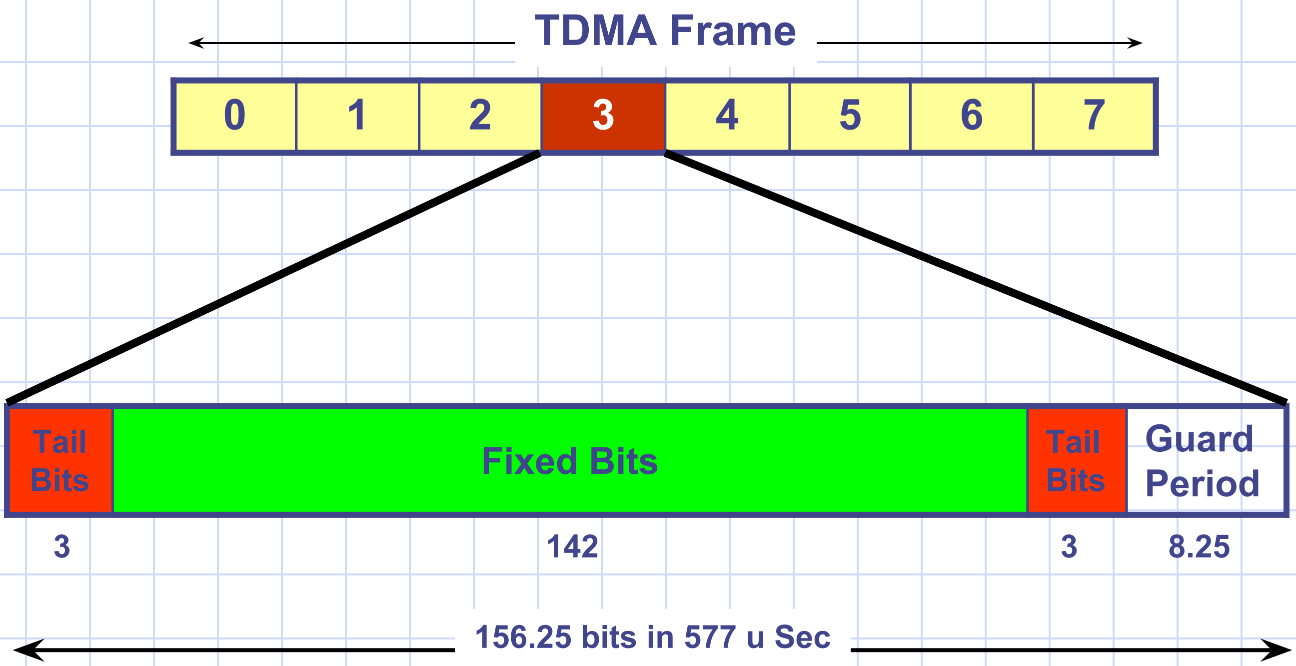
The training sequence is inserted in the middle of the burst to help the Viterbi equalizer to create the channel model.

The guard period of 8.25 bits length, which is equivalent to about 30 µs, is

It carries information of all logical channels except RACH, SCH and FCCH.

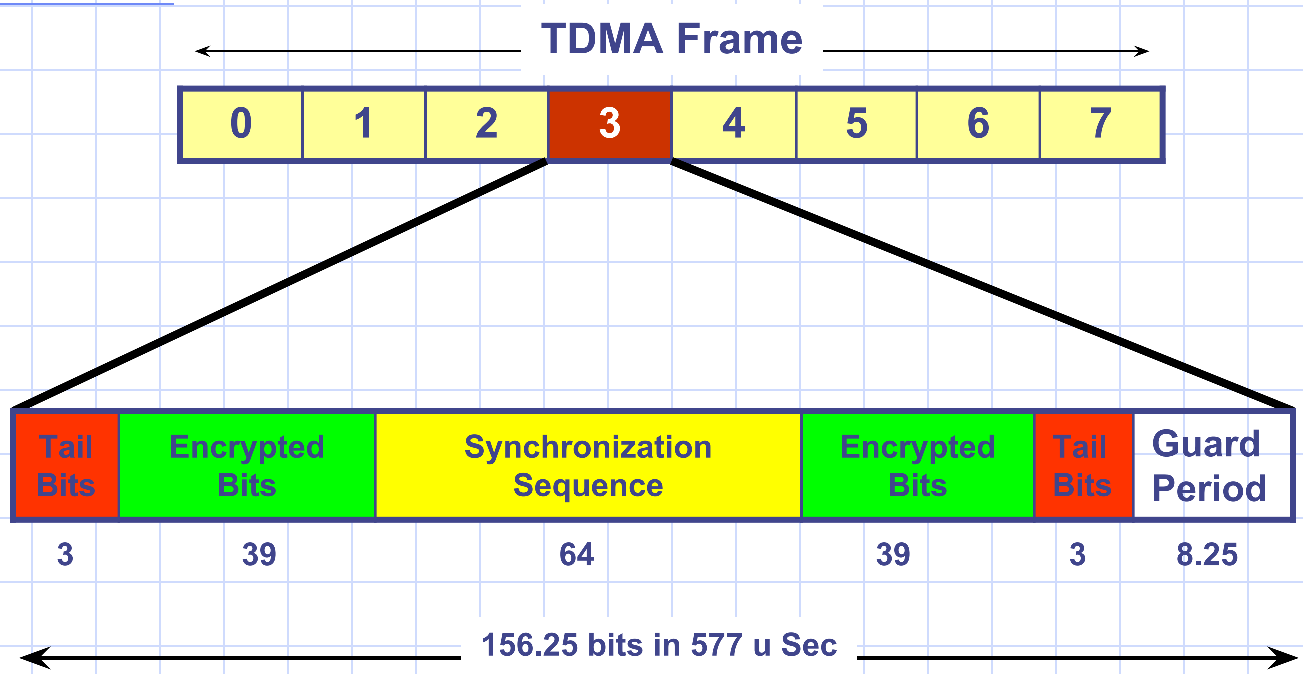
and they are always zeros

Frequency Correction Burst Structure



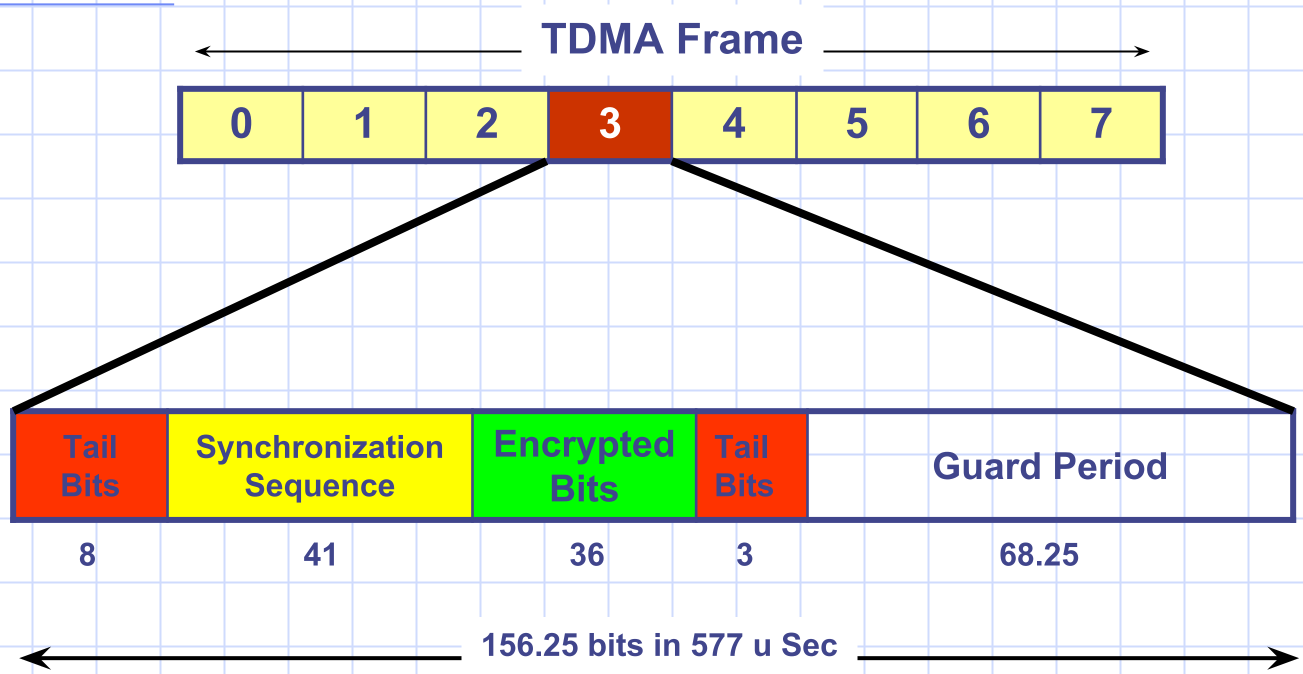
This is the one used by the channel (FCH) for frequency correction of the mobile. It consists of a long sequence of bits called the fixed bits which are all equal to zeros, leading to a constant frequency output from the GMSK modulator

Synchronization Burst Structure



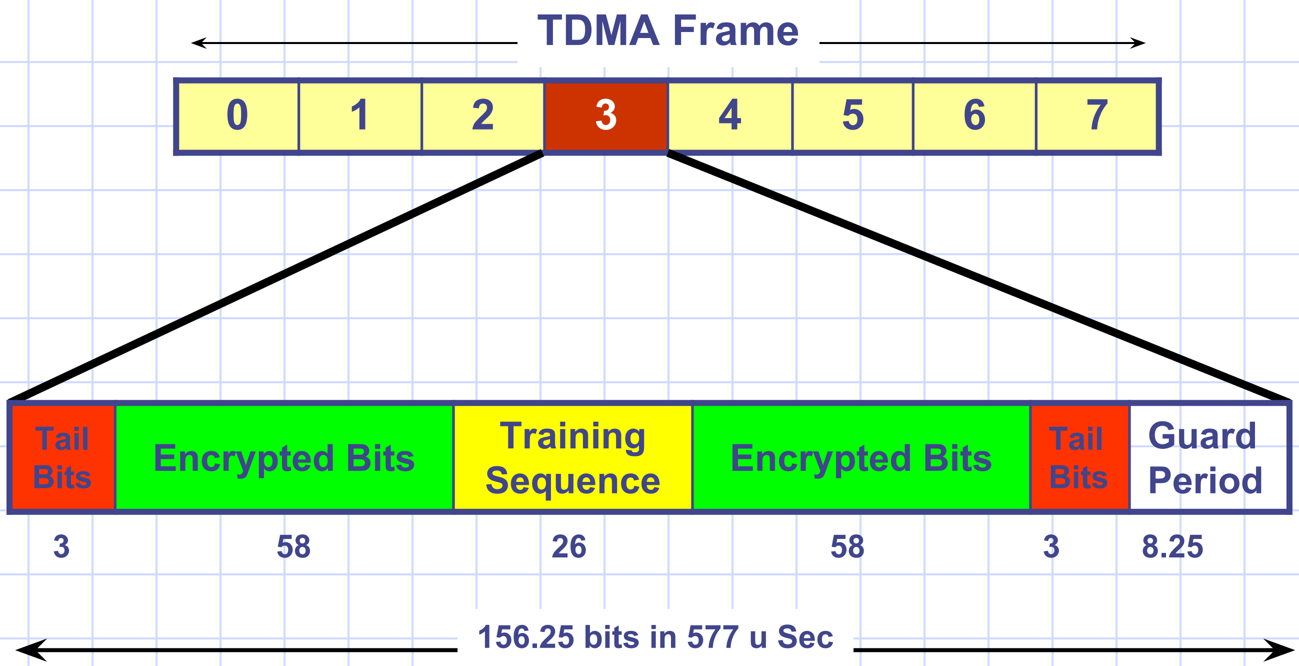
The SCH burst consists of a long synchronization sequence along with the important information being encrypted and divided into two blocks. The TDMA frame number is sent on the SCH channel, which carries also the Base station Identity code (BSIC). The TDMA frame number is used by the mobile to determine which control channels will be transmitted on that frame. It is used also as one of the input parameters to the algorithm that calculates the ciphering key Kc, which is in turn used for encryption of subscriber information transmitted on the air interface

Access Burst Structure



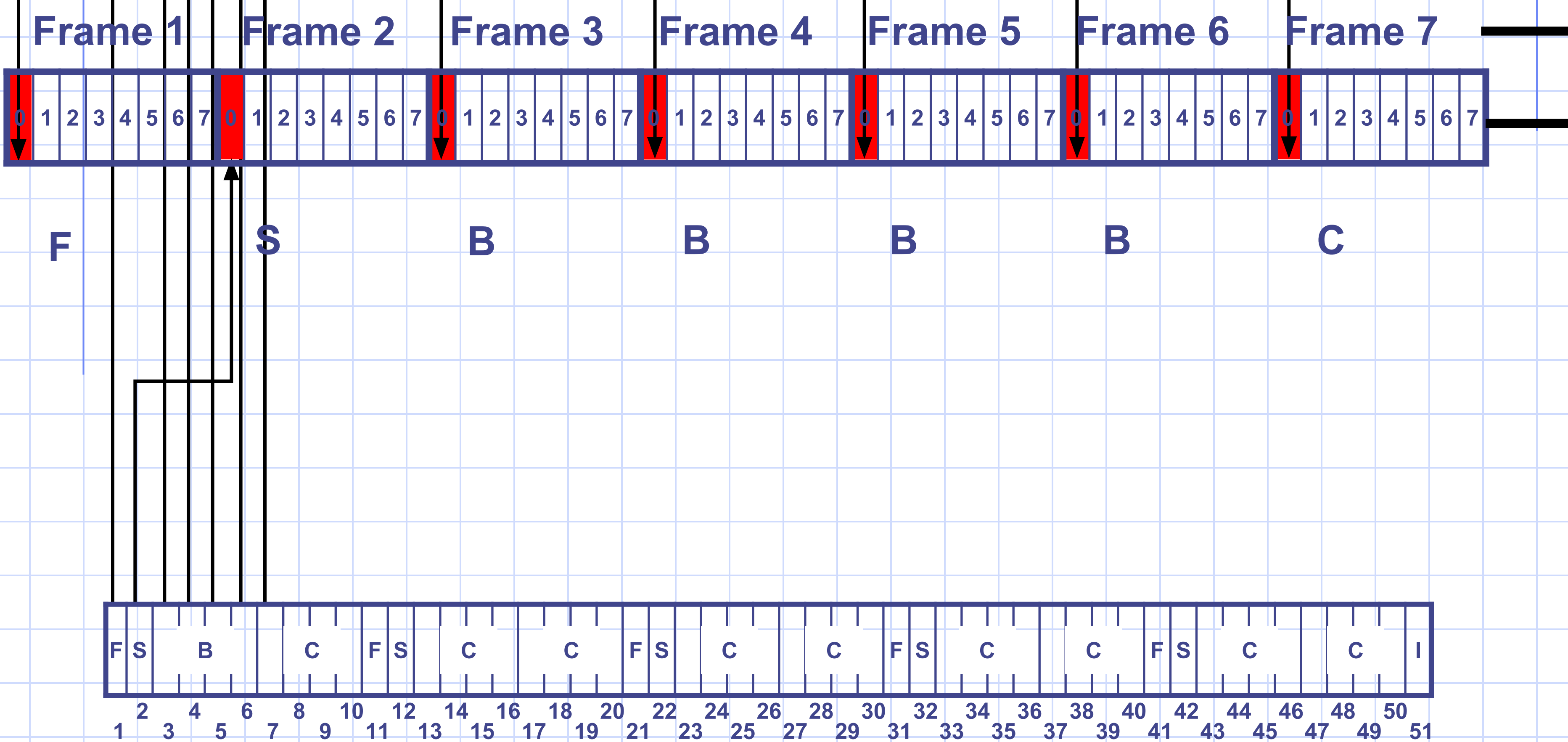
The Access Burst is used by the RACH channel. The mobile sends this burst when it does not know the distance to its serving BTS, which is the case when the mobile is switched on or after it makes a handover to a new cell. So this burst must be shorter in order to prevent it from overlapping with the burst on the next time slot

Dummy Burst Structure



The dummy burst is sent from the BTS when there is nothing else to be sent. It carries no information and it has the same structure of a normal burst with the encrypted bits replaced by a known bit pattern to the mobile

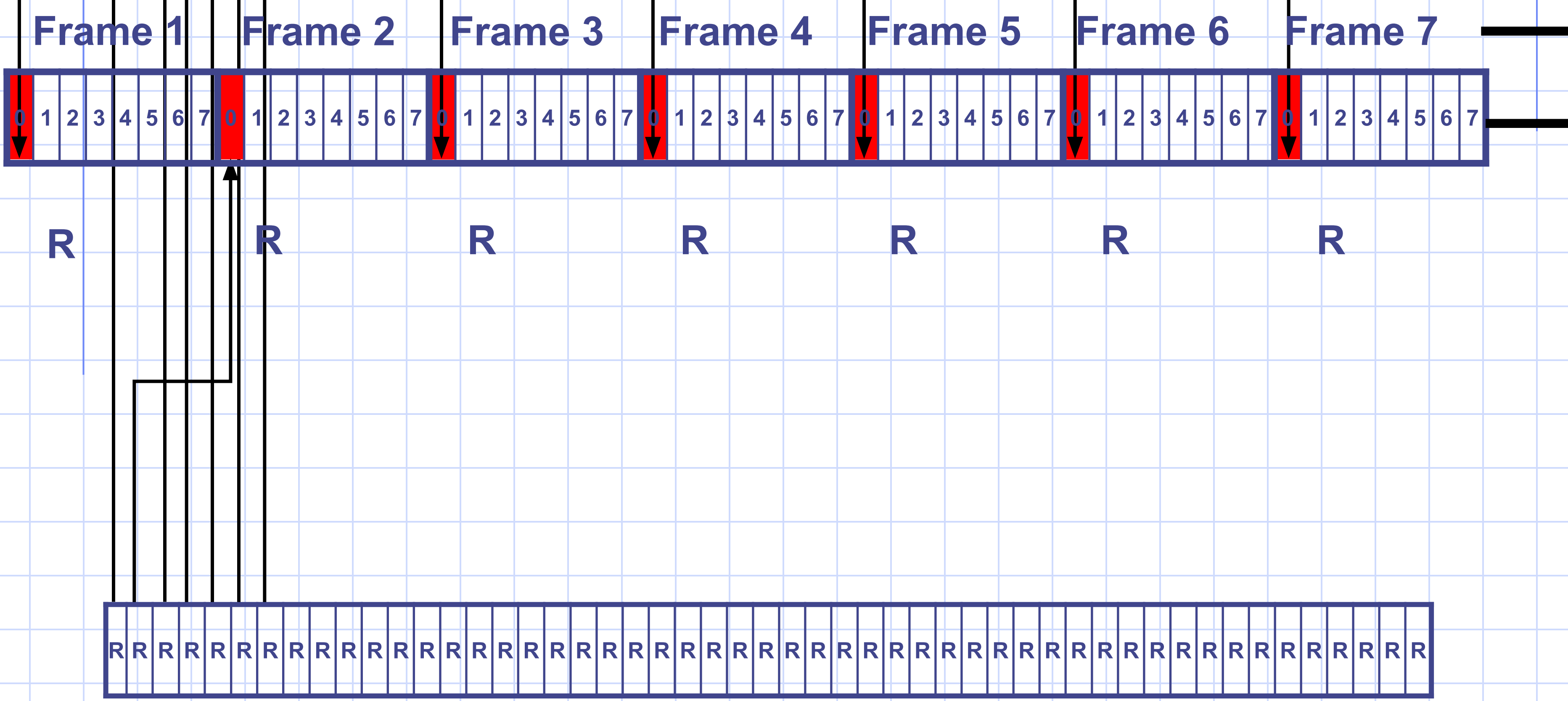
Mapping of Logical Channels onto timeslot 0 (Downlink)



Mapping of the 51 burst consumes 51 TDMA frame

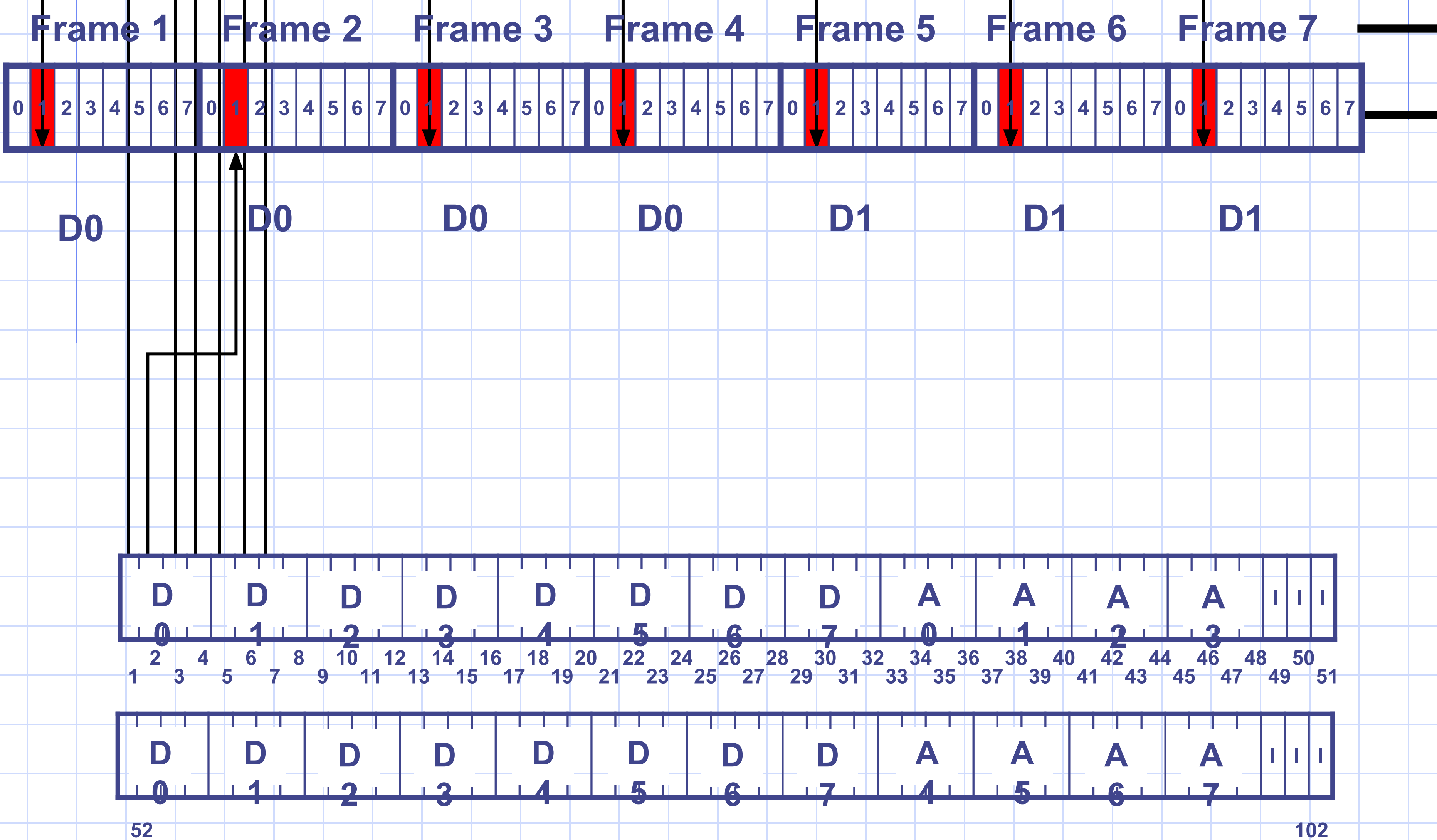
51 TDMA Frame = 1 Multi-frame

Mapping of Logical Channels onto timeslot 0 (Uplink)

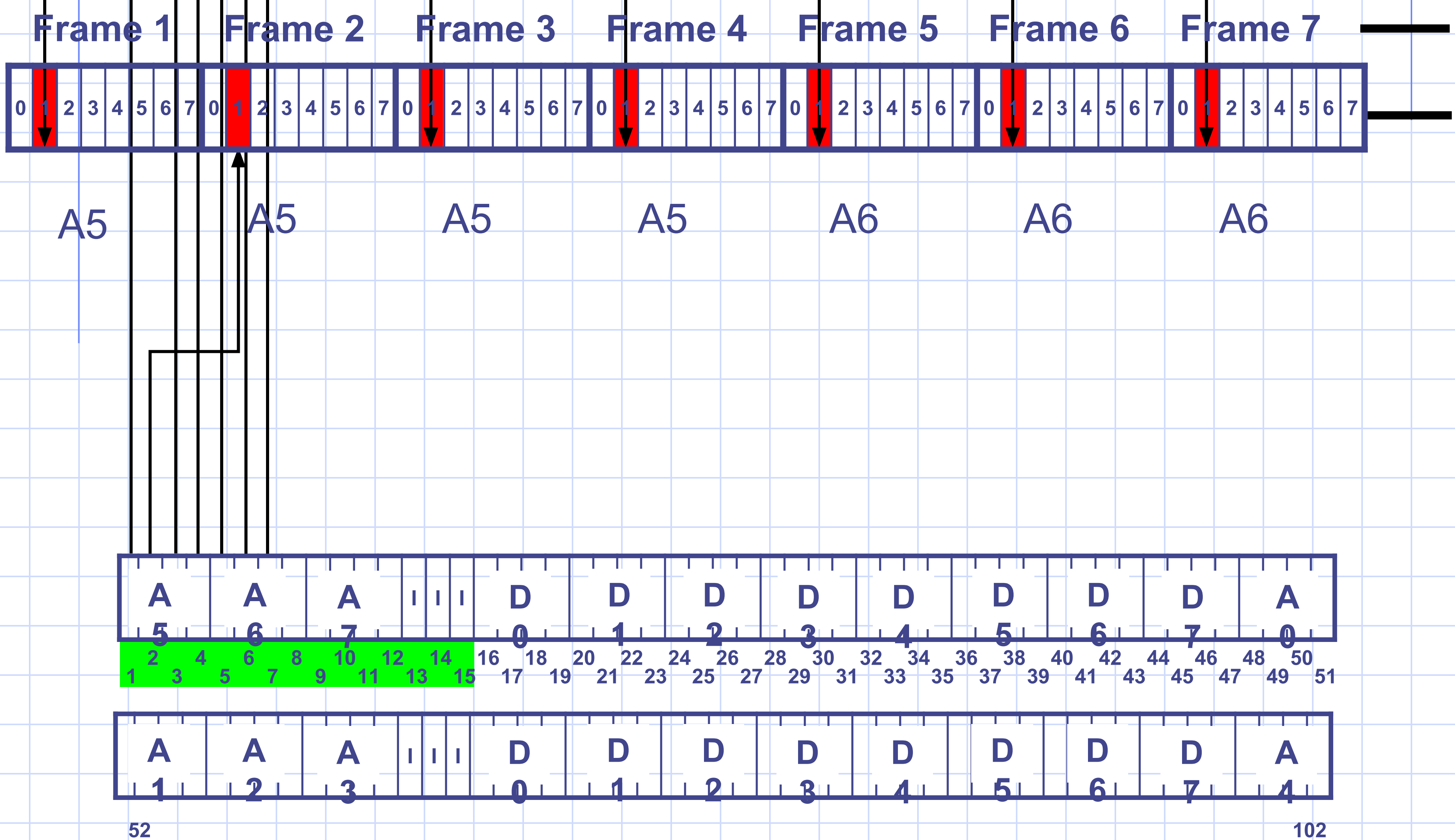


Time slot 0 in the uplink direction is reserved for the RACH channel which is used by the mobiles to make random access request to the system

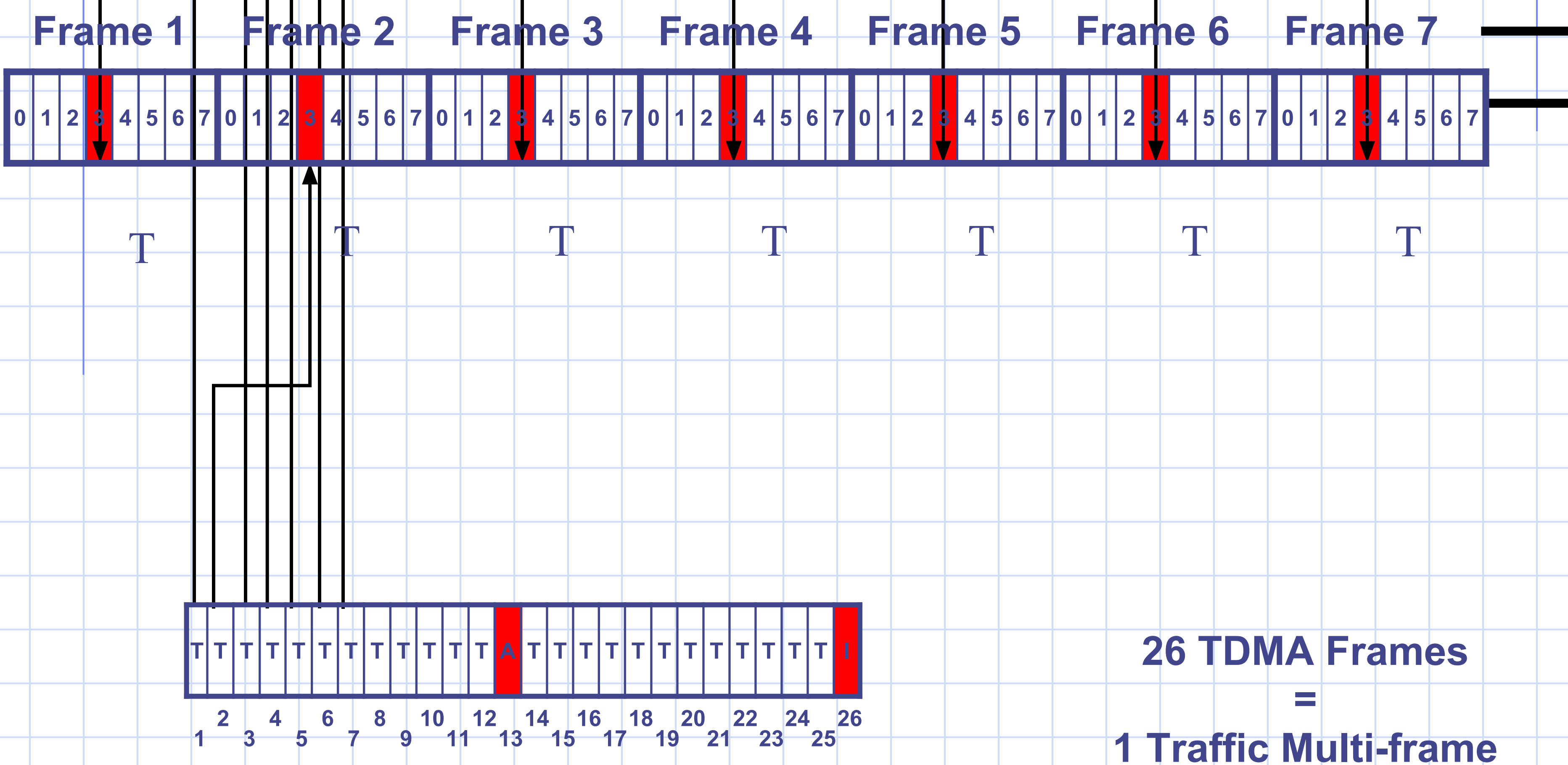
Mapping of Logical Channels onto timeslot 1 (Downlink)



Mapping of Logical Channels onto timeslot 1 (Uplink)



Mapping of Logical Channels onto timeslots 2 / 7



A (SACCH)

This contains the control signaling, an example of this is in order to change output power.

Air Interface

vodafone

GSM System Survey

51 TDMA Frames = 1 BCCH Multi-frame

	Downlink								Uplink							
	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
Frame □	F	D□	T						R	A□	T					
□	S	D□	T						R	A□	T					
□	B	D□	T						R	A□	T					
□	B	D□	T						R	A□	T					
□	B	D□	T						R	A□	T					
□	B	D□	T						R	A□	T					
□	C	D□	T						R	A□	T					
□	C	D□	T						R	A□	T					
□	C	D□	T						R	A□	T					
□□	C	D□	T						R	A□	T					
□□	F	D□	T						R	A□	T					
□□	S	D□	T						R	A□	T					
Frame □□	C	D□	A	I	A	I	A	I	R	I	A	I	A	I	A	I
□□	C	D□	T						R	I	T					
□□	C	D□	T						R	I	T					
□□	C	D□	T						R	D□	T					
□□	C	D□	T						R	D□	T					
□□	C	D□	T						R	D□	T					
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Frame □□	C	A□	A	I	A	I	A	I	R	D□	A	I	A	I	A	I
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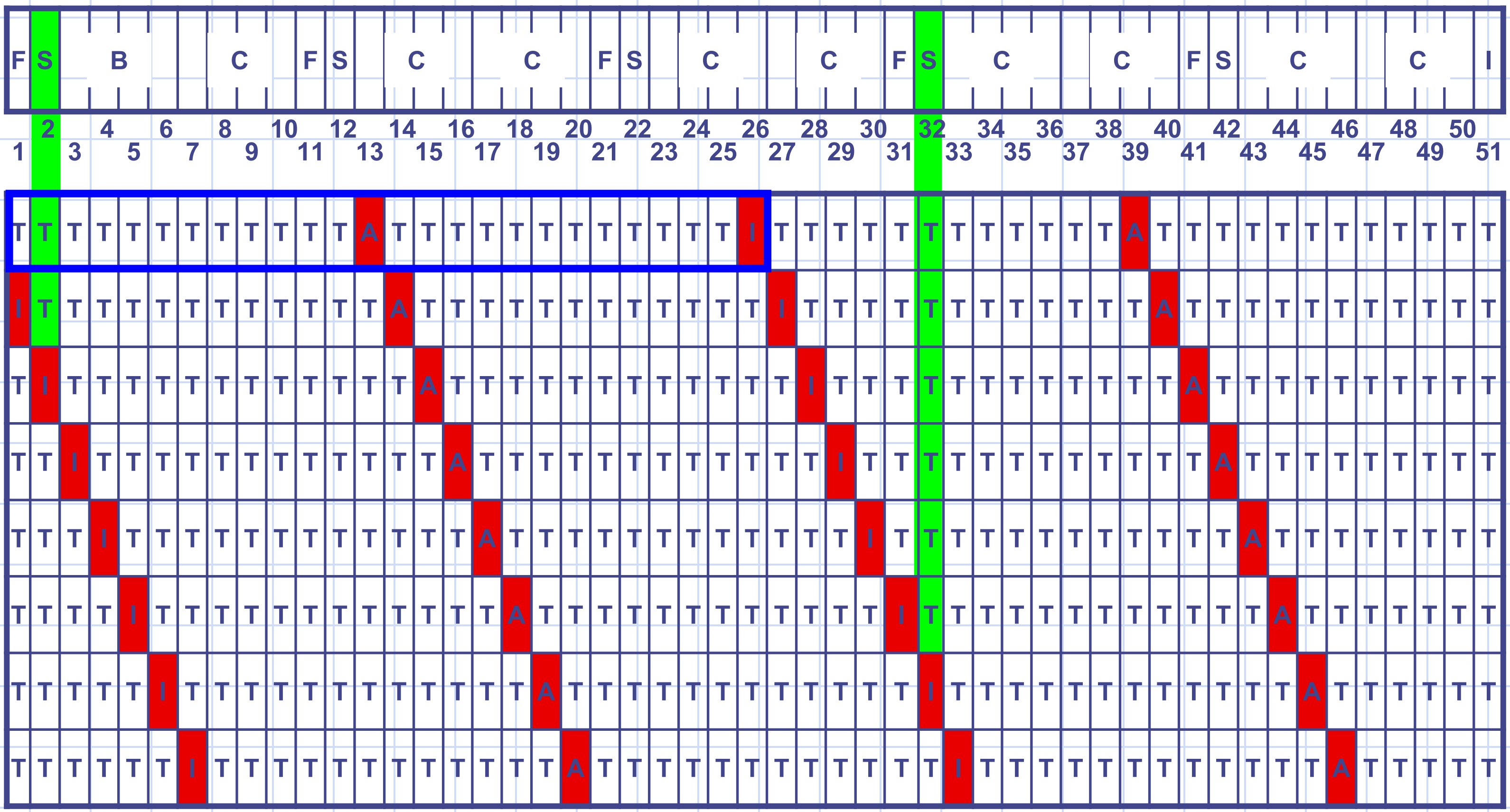
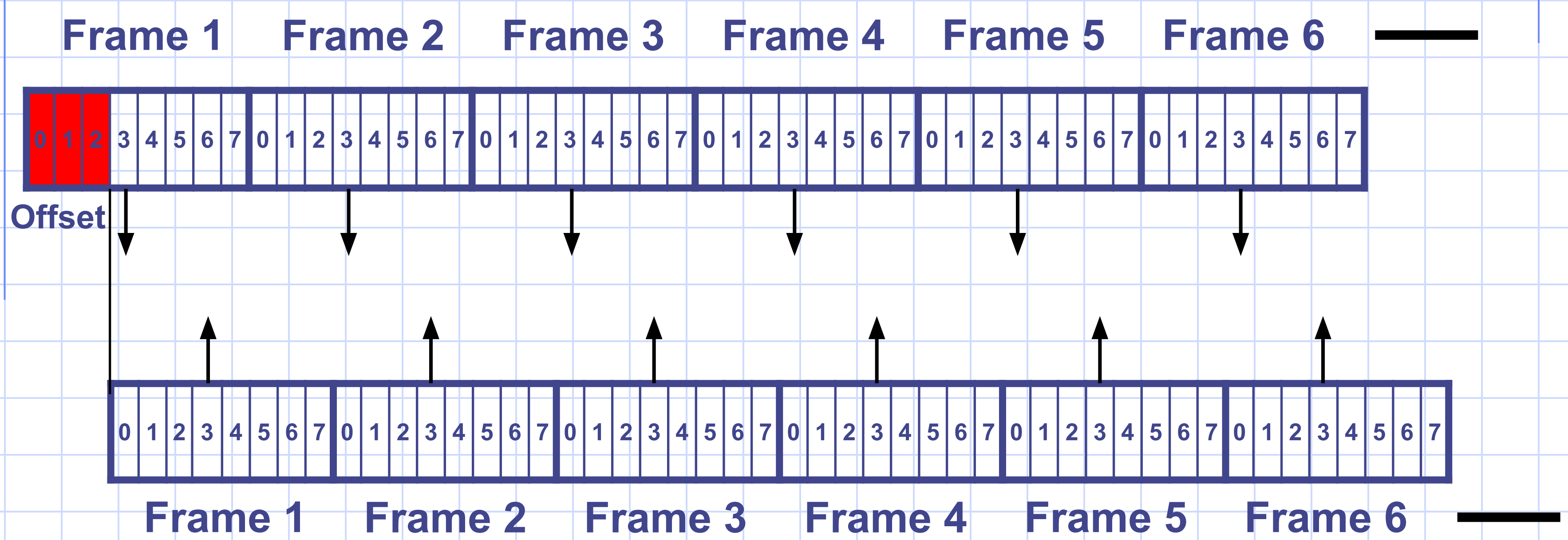
The Sliding Multi-frame

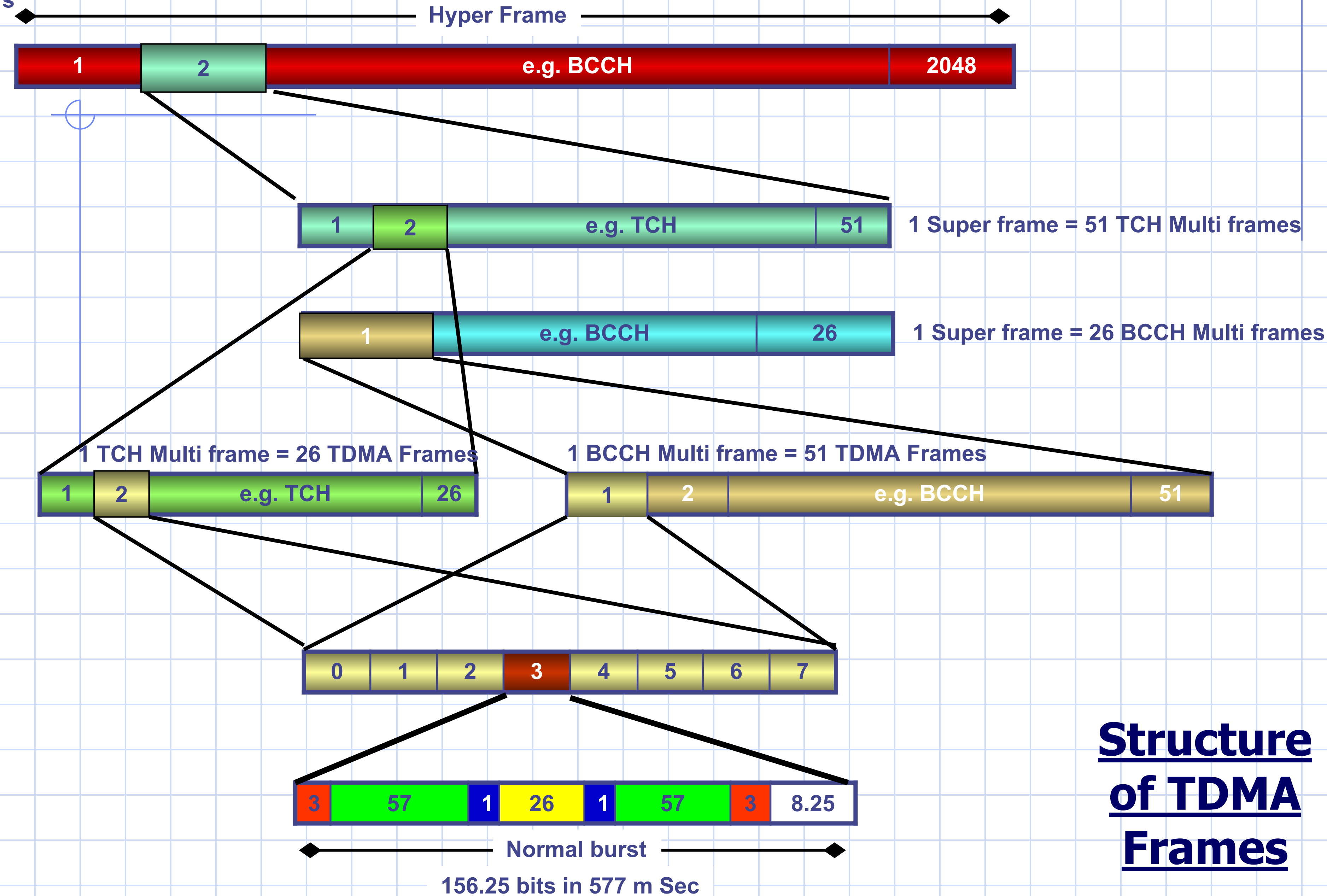


Traffic Channel Offset



Air Interface

1 Hyper frame = 2048 Super frames = 2,715,648 TDMA Frames = 3hrs 28 min and 53.76 s



vodafone

GSM System Survey

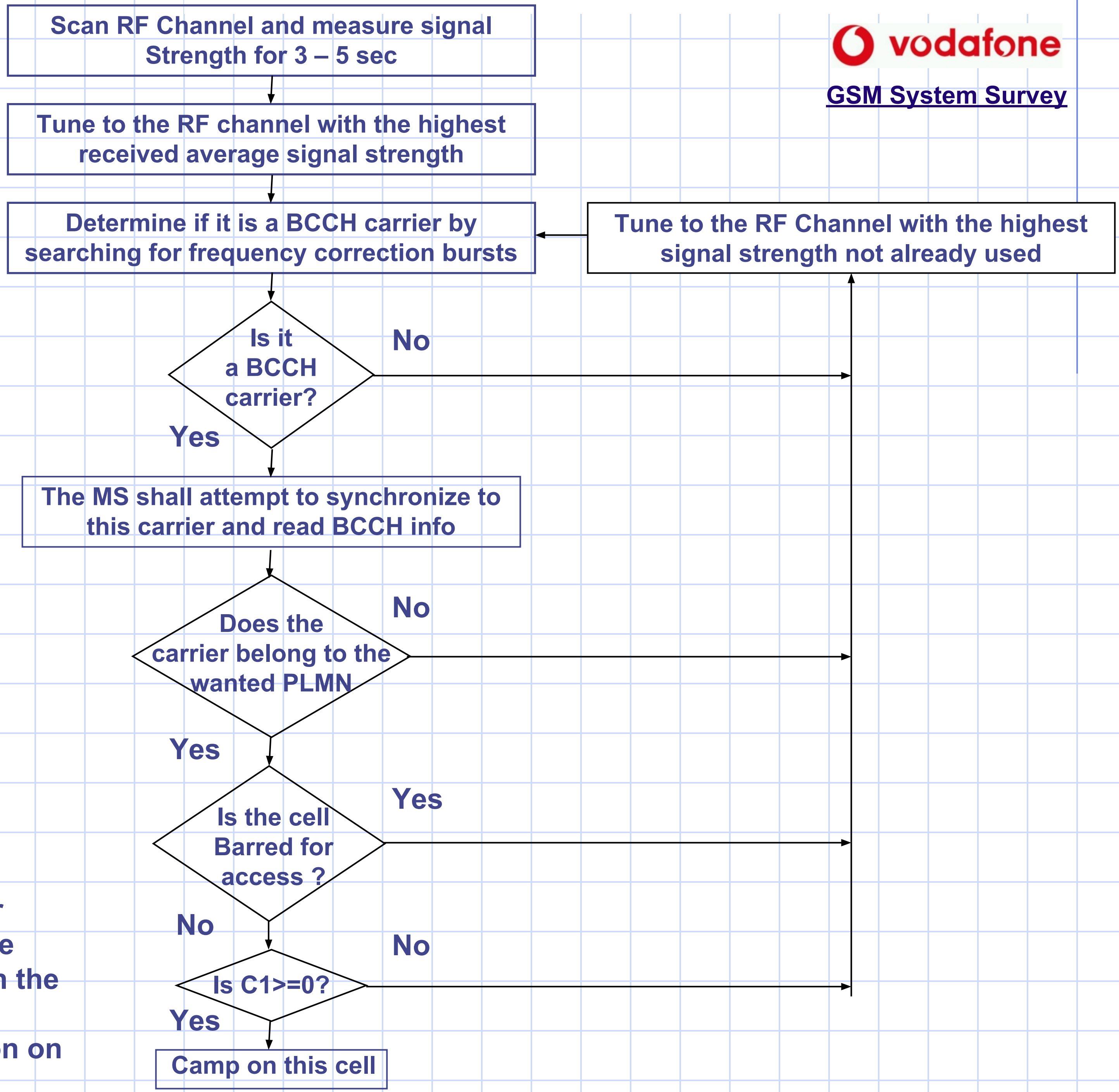
Structure of TDMA Frames

Air Interface

vodafone

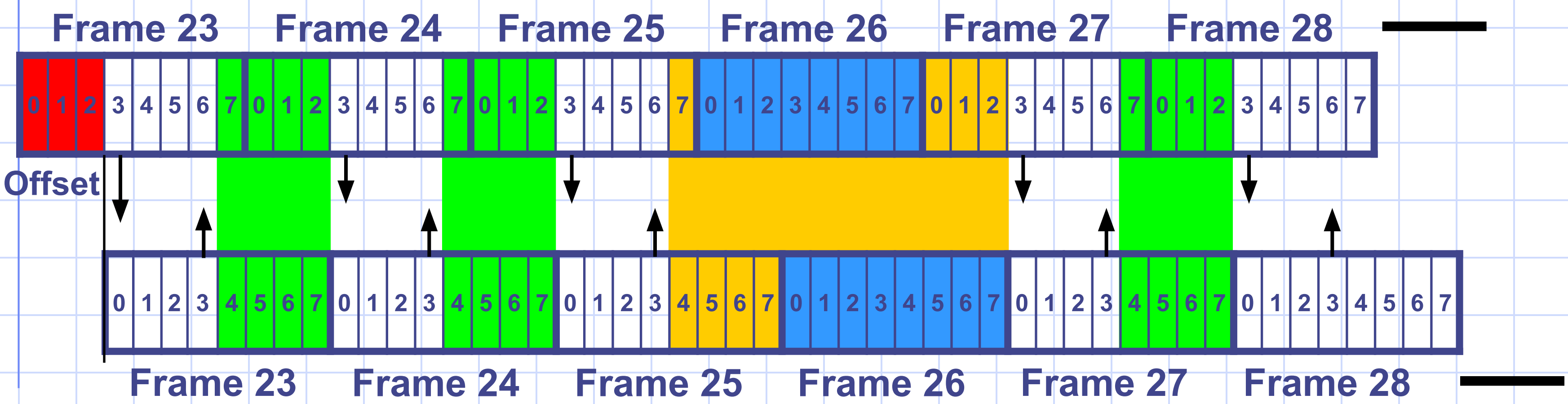
GSM System Survey

Measurement in idle mode



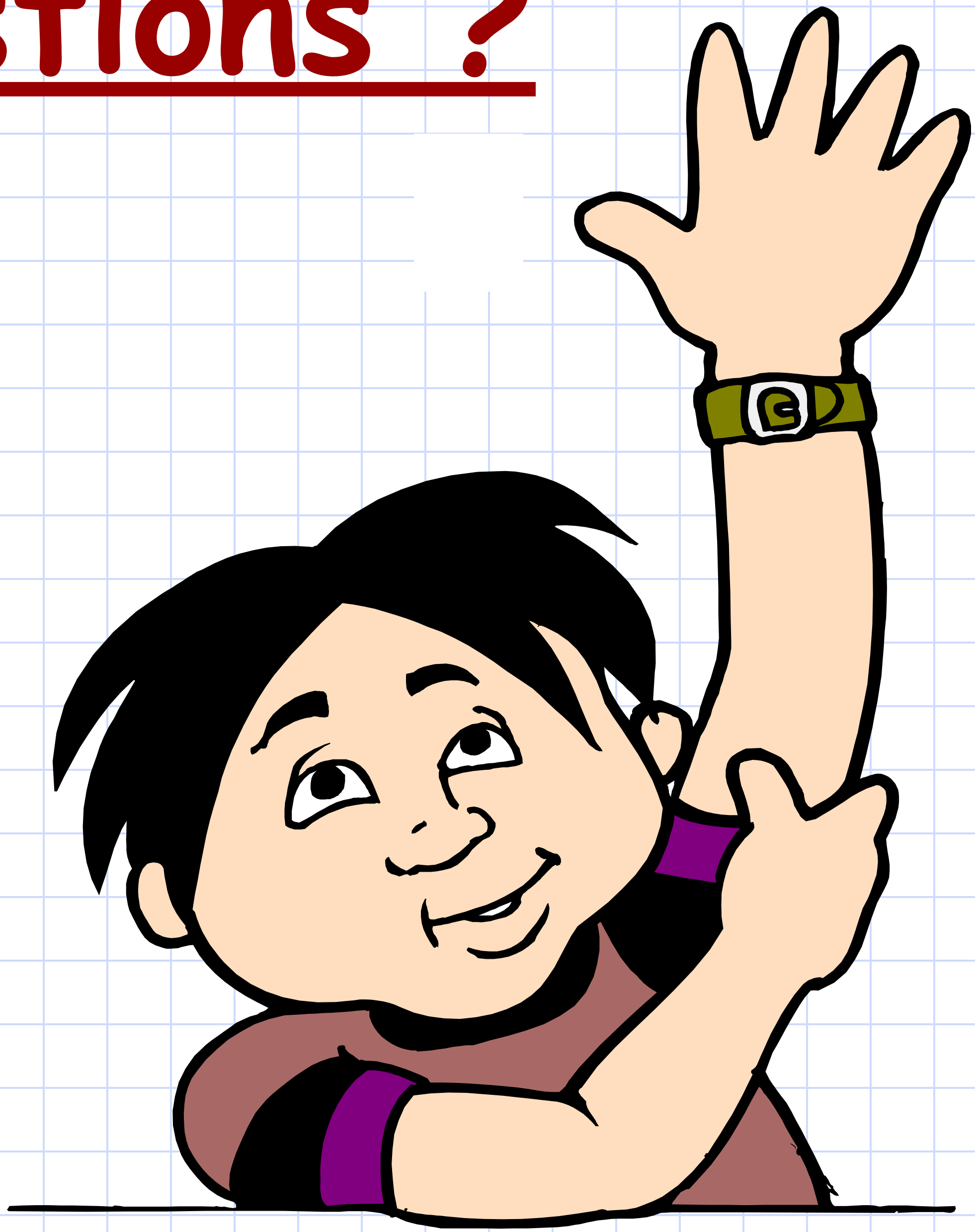
C1 (Path Loss Criterion Parameter) is a parameter used to make sure that the MS camps on the cell with the highest probability of successful communication on the uplink and downlink

Measurement in active mode



The mobile is informed on the SACCH channel which BCCH frequencies to be measured.
The mobile will try to measure the signal strength of these carriers one by one during the time between transmission and reception of the allocated traffic channel: the mobile transmits, measures and then receives, and so on.
To make sure that the measured carriers do not belong to co-channel cells, the mobile will have to check the identity of the adjacent cells by reading the BSIC value sent on the SCH of each cell. This will take place during the idle frame number 26.
The signal strength of the serving cell is measured during reception of the allocated traffic channel. Then the mobile will make a list of the strongest six carriers and their BSIC values along with the signal strength of its cell, and reports this list to the BSC via the uplink SACCH channel which is repeated once every 26 frames.

Questions ?



Chapter 7 : Traffic Cases

Communicate Anywhere

Chapter Objectives

By the End of this Chapter you will:

- Know How the Mobile Terminating call is done
- Know how the PLMN Coverage area is divided (MSC Coverage Areas & Location Areas)
- Know the different types of Location Updates
- Know the different types of Handover Procedures
- Know How the Mobile Originating call is done
- Know the different Traffic Cases a Roamer can have

Location Update

Why do we need to update our location data ?

Actually, the location update process is invited in aim to *exactly* identify your location within the network so that any incoming call goes directly to the called subscriber.

To fulfill this aim, one can say that we may update the system with the cell ID each time the subscriber changes his serving cell.

The MSC/VLR will now know the exact cell you are roaming in.
This will result in a huge amount of location update messages.

An extreme is never to make a location update and to be paged in all the network. This will cause huge amount of paging messages.

Do you have a compromising solution ?

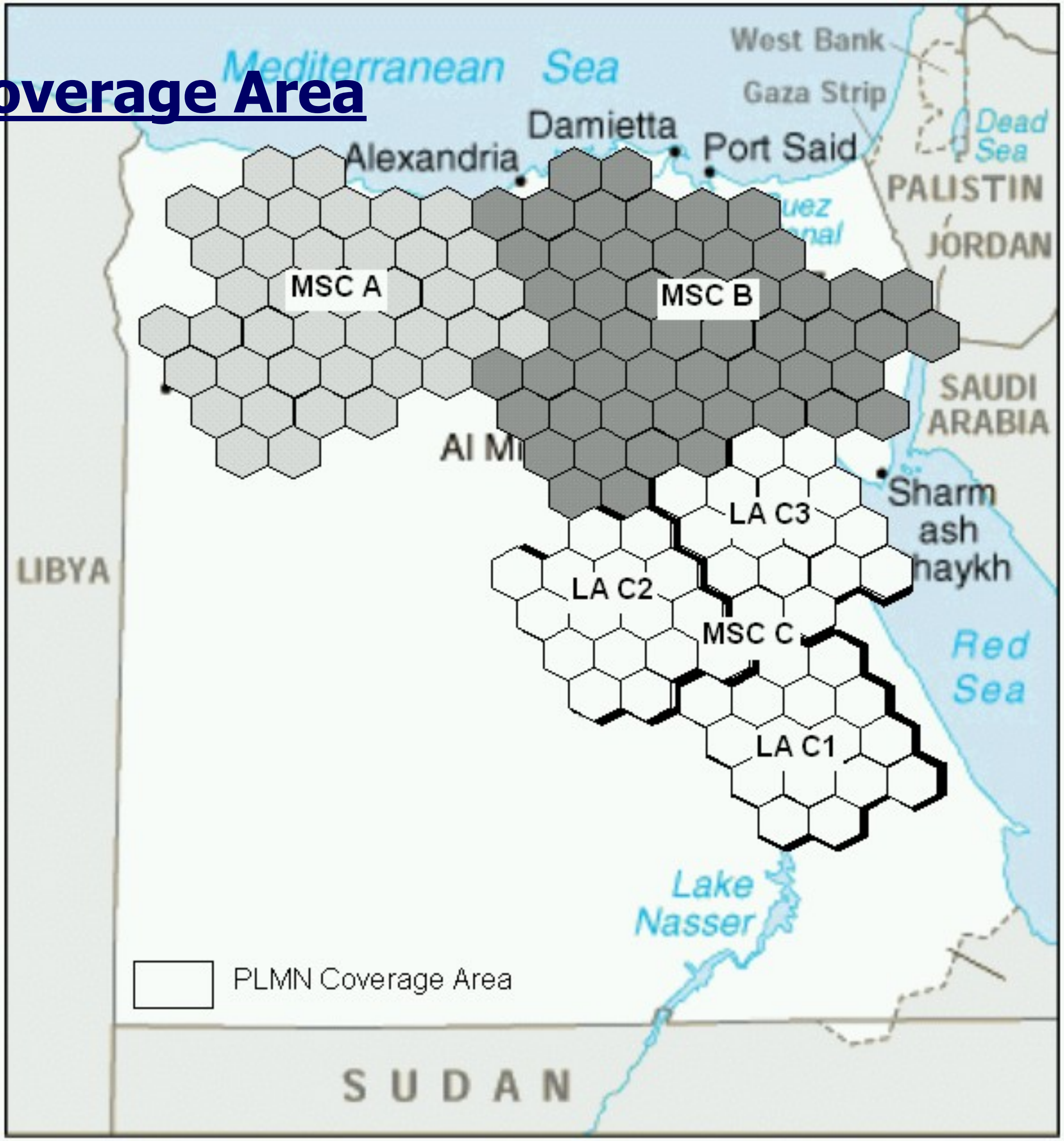
Location Area

Location area is a part of the MSC/VLR coverage area. Each group of adjacent cells is assigned a universal unique location area identity.

The mobile subscriber is only required to update the network with its new location every time it changes its Location Area.

Introducing the concept of Location area enables us to make an ***approximate*** estimation of your location.

MSC Coverage Area



Location Area Identity (LAI)

602 02 160
7

MCC MNC
LAC

Vodafone Egypt LAI

MCC : Mobile Country Code
MNC : Mobile Network Code
LAC : Location Area Code

Cell Global Identity (CGI)

602	02	1607	781
MCC	MNC	LAC	CID

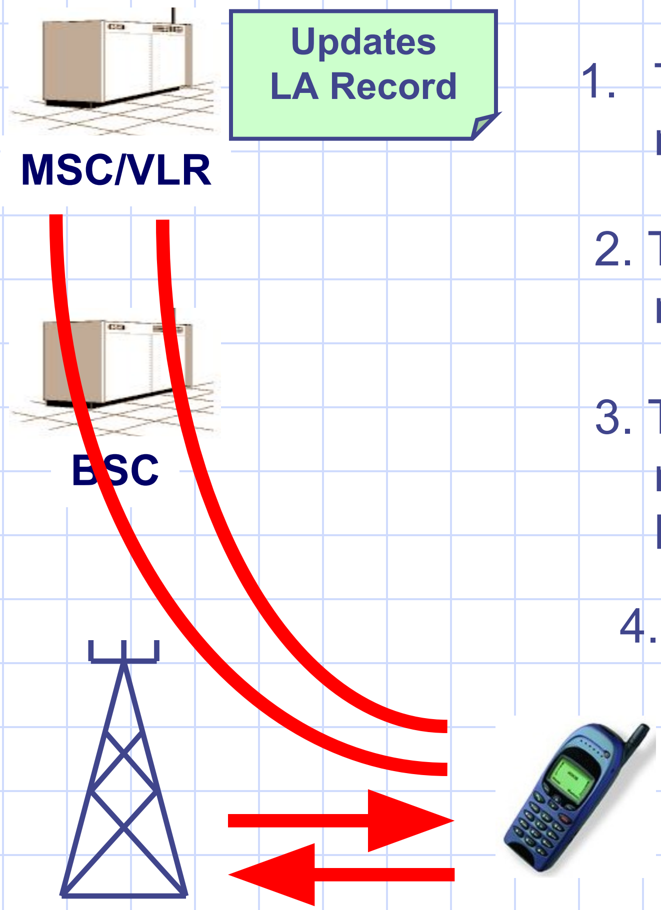
Vodafone Egypt CGI

MCC : Mobile Country Code
MNC : Mobile Network Code
LAC : Location Area Code
CID : Cell ID

Types of Location Update

1. Normal Location update within same MSC/VLR service area
2. Normal Location update between 2 different MSC/VLR service areas
3. IMSI attach/detach
4. Periodic Location Update

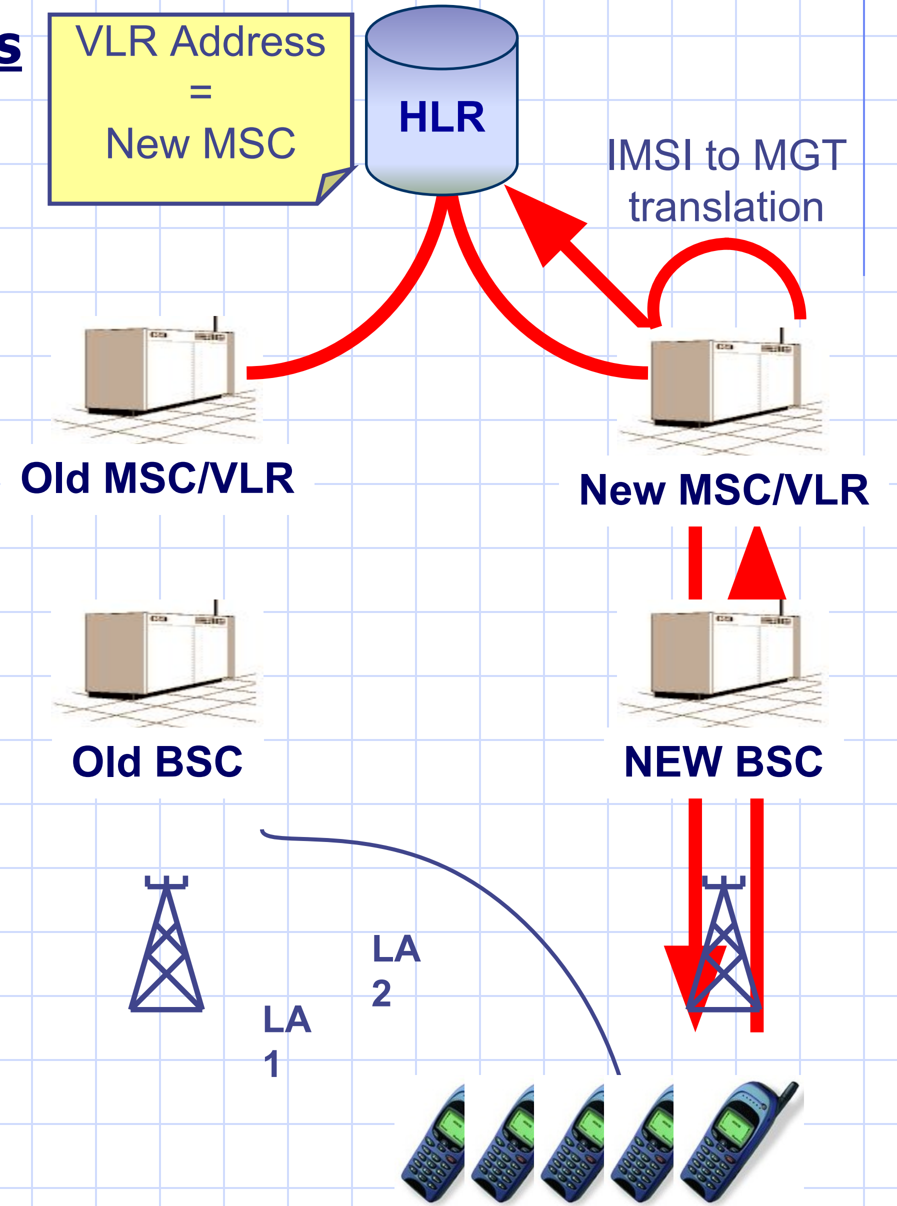
Normal Location within the same MSC/VLR Service area



1. The Mobile sends an allocation request message to the BTS
2. The BTS responds with the allocation message
3. The mobile sends a location update request message with its IMSI to the MSC/VLR
4. The MSC/VLR updates the location information and sends a Location Update confirmation message

Normal Location Update between 2 different MSC/VLR service areas

1. The mobile sends a location update request to the MSC.
2. The new MSC/VLR receives the IMSI and conclude the MGT.
3. The MSC/VLR sends a subscriber information request with the IMSI to the proper HLR
4. The HLR stores the address of the new MSC/VLR
5. The HLR sends the data to the new MSC/VLR and it is kept there
6. The HLR sends a location cancellation message to the old MSC/VLR to remove the data
7. The new MSC/VLR sends a location updating confirmation message to the mobile



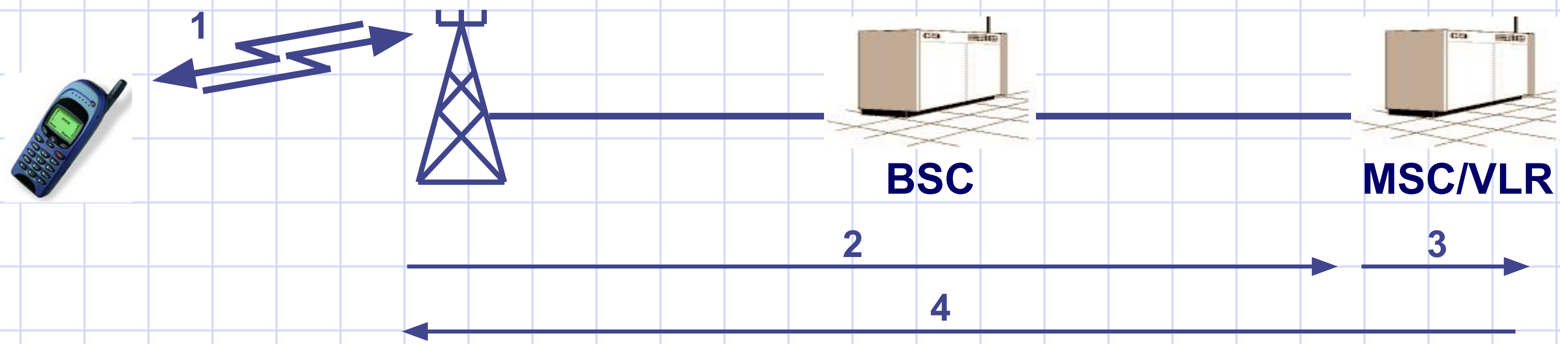
IMSI Detach

1. At power off, the MS asks for a signaling channel.
2. The MS uses this signaling channel to send the IMSI detach message to the MSC/VLR.
3. In the VLR, an IMSI detach flag is set for the subscriber. This is used to reject incoming calls to the MS.

IMSI Attach

IMSI attach is a complement to the IMSI detach procedure. It is used by the mobile subscriber to inform the network that it has re-entered an active state and is still in the same location area. If the MS changes location area while being switched off, a normal location update takes place.

1. The MS requests a signaling channel.
2. The MSC/VLR receives the IMSI attach message from the MS.
3. The MSC/VLR sets the IMSI attach in the VLR. The mobile is now ready for normal call handling.
4. The VLR returns an acknowledgment to the MS.



Periodic Location Update

Periodic location update is a routine task performed by the network if the MS doesn't make any location update (any of the previous 4 types) during a predefined period.

If the MS doesn't respond to this periodic location update, it will be marked as implicitly detached. (Temporarily out of service)

Handover

- Handover is to keep continuity of the call when the subscriber is roaming along the network moving from one cell to another and moving between different nodes in the network.
- During call, the MS is continuously measuring transmission quality of neighboring cells and reports this results to the BSC through the BTS.
- The BSC, being responsible on supervising the cells, is responsible of handover initiation.
- Good neighbor relations between cells is an important factor in keeping the network performance in the accepted level.

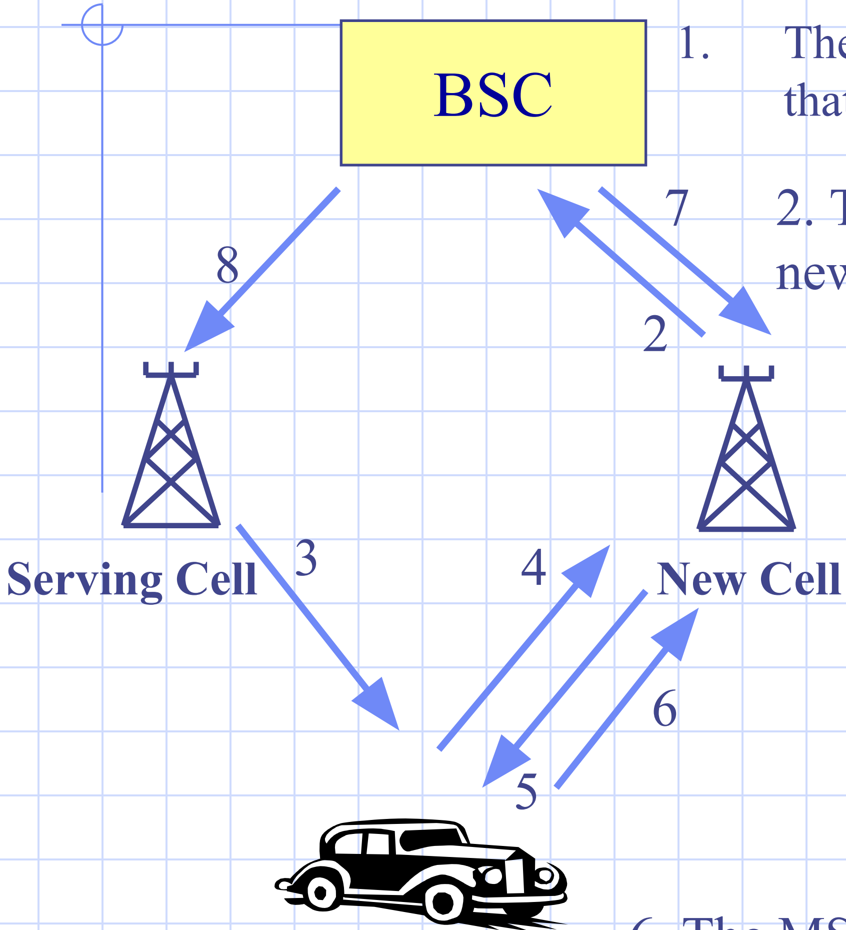
Types of Handover

1. Intra BSC Handover:
When the cell to which the call will be handed over belongs to the same BSC of the serving cell.

2. Inter BSC / Intra MSC Handover:
When the cell to which the call will be handed over belongs to the different BSCs but to the same serving MSC.

3. Inter MSC
When the cell to which the call will be handed over belongs to the different BSC and different MSC.

Intra BSC Handover



1. The BSC decides from the power measurement reports that the call must be handed over to another cell

2. The BSC checks for an vacant TCH in the new cell and orders this cell to activate the TCH

3. The BSC orders the serving cell to send a message to the MS telling the information of new TCH

4. The MS tunes to the new frequency and Sends handover access burst

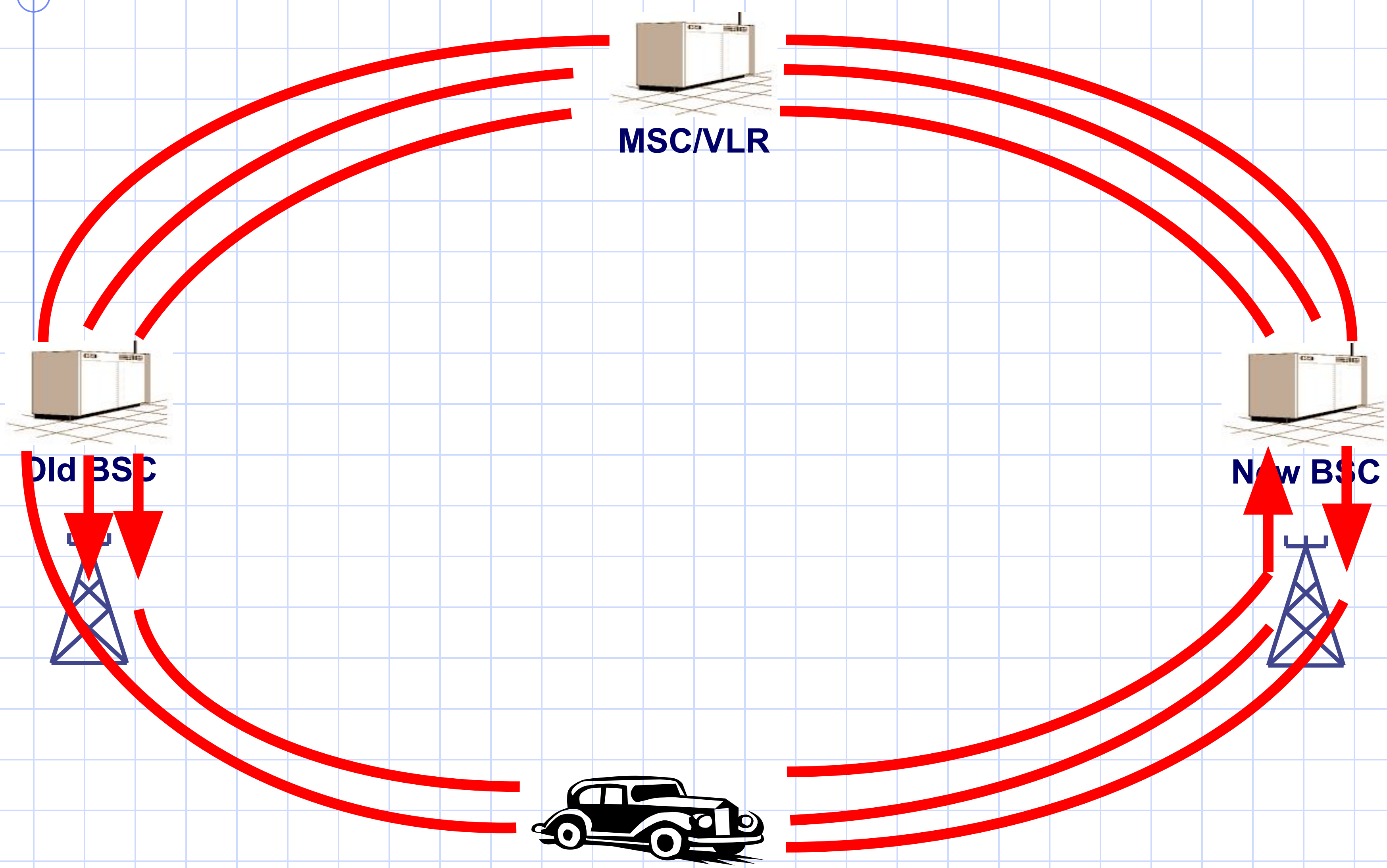
5. The new cell detects the handover burst and sends information about the suitable timing advance to the MS

6. The MS sends a HO complete message to the new cell

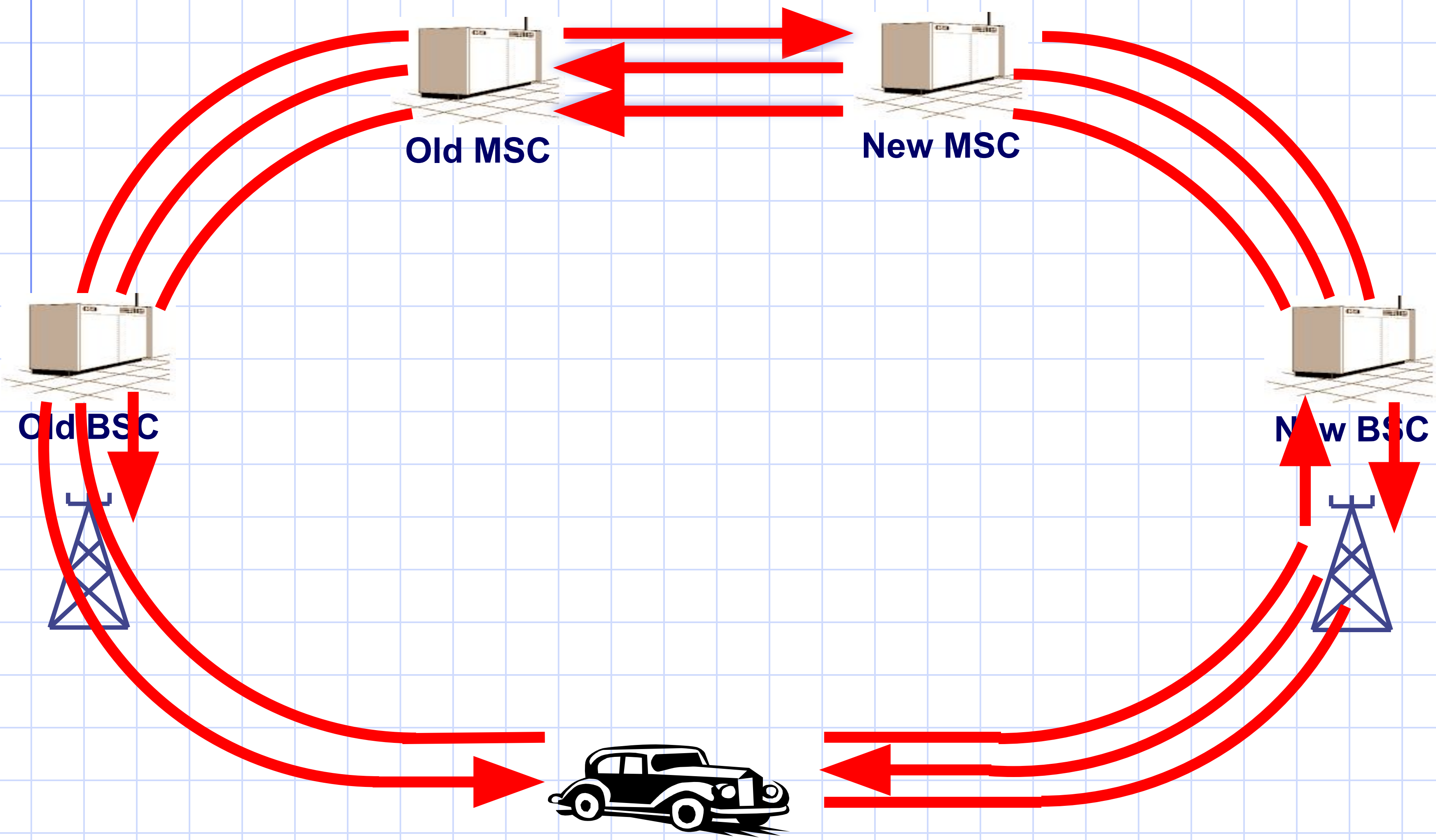
7. The new cell sends a message to the BSC that the handover is successful

8. The BSC orders the old Cell to release the TCH

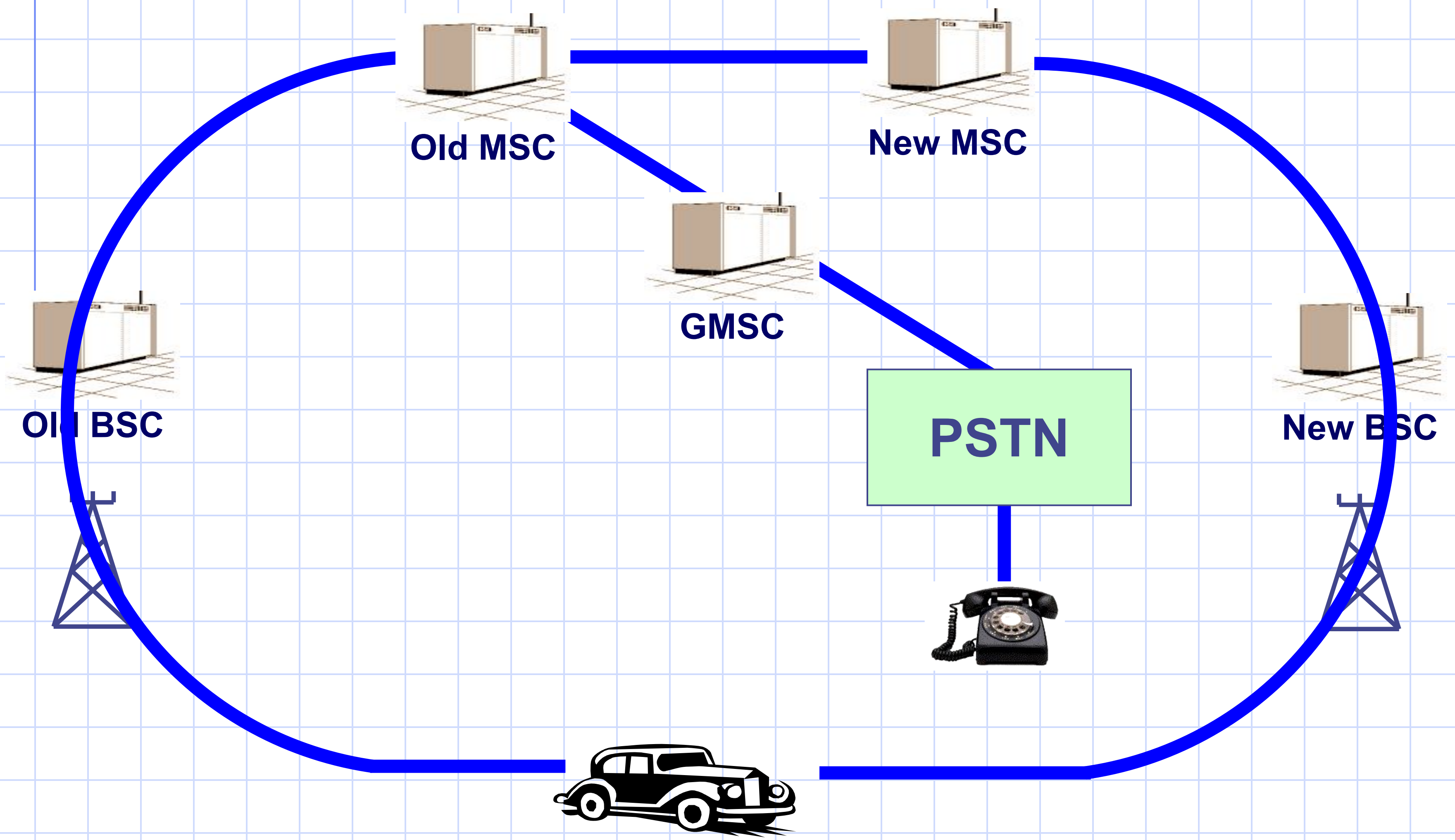
Inter BSC /Intra MSC Handover



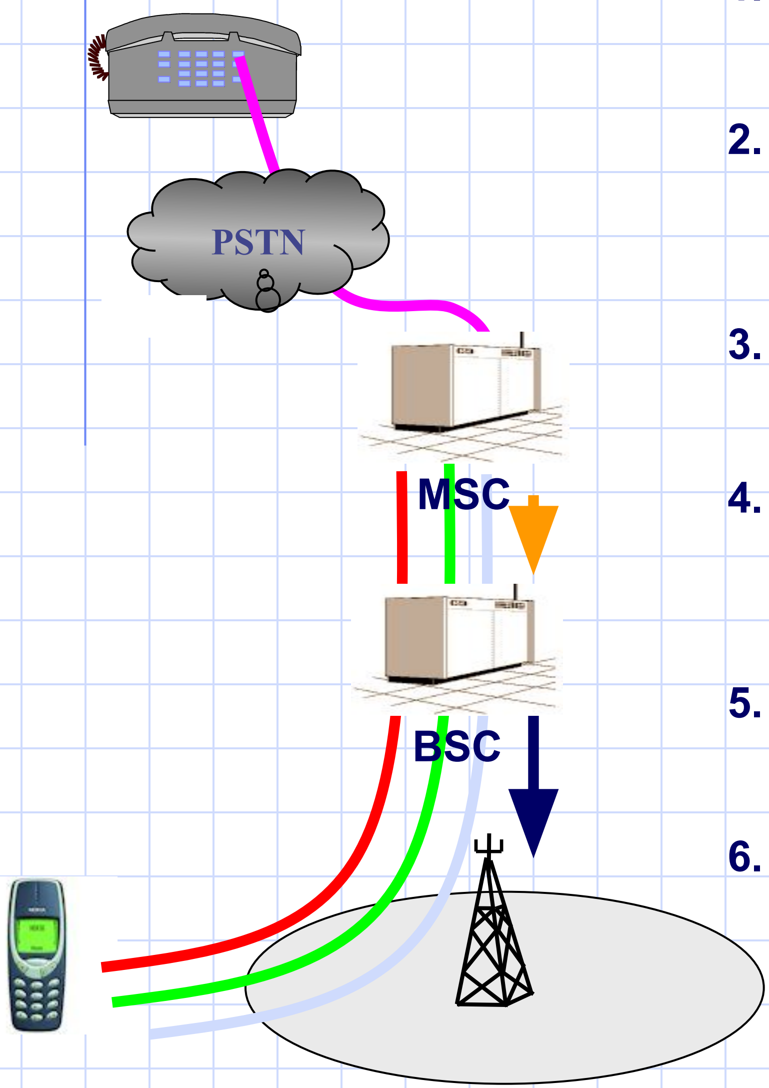
Inter MSC Handover



Inter MSC Handover

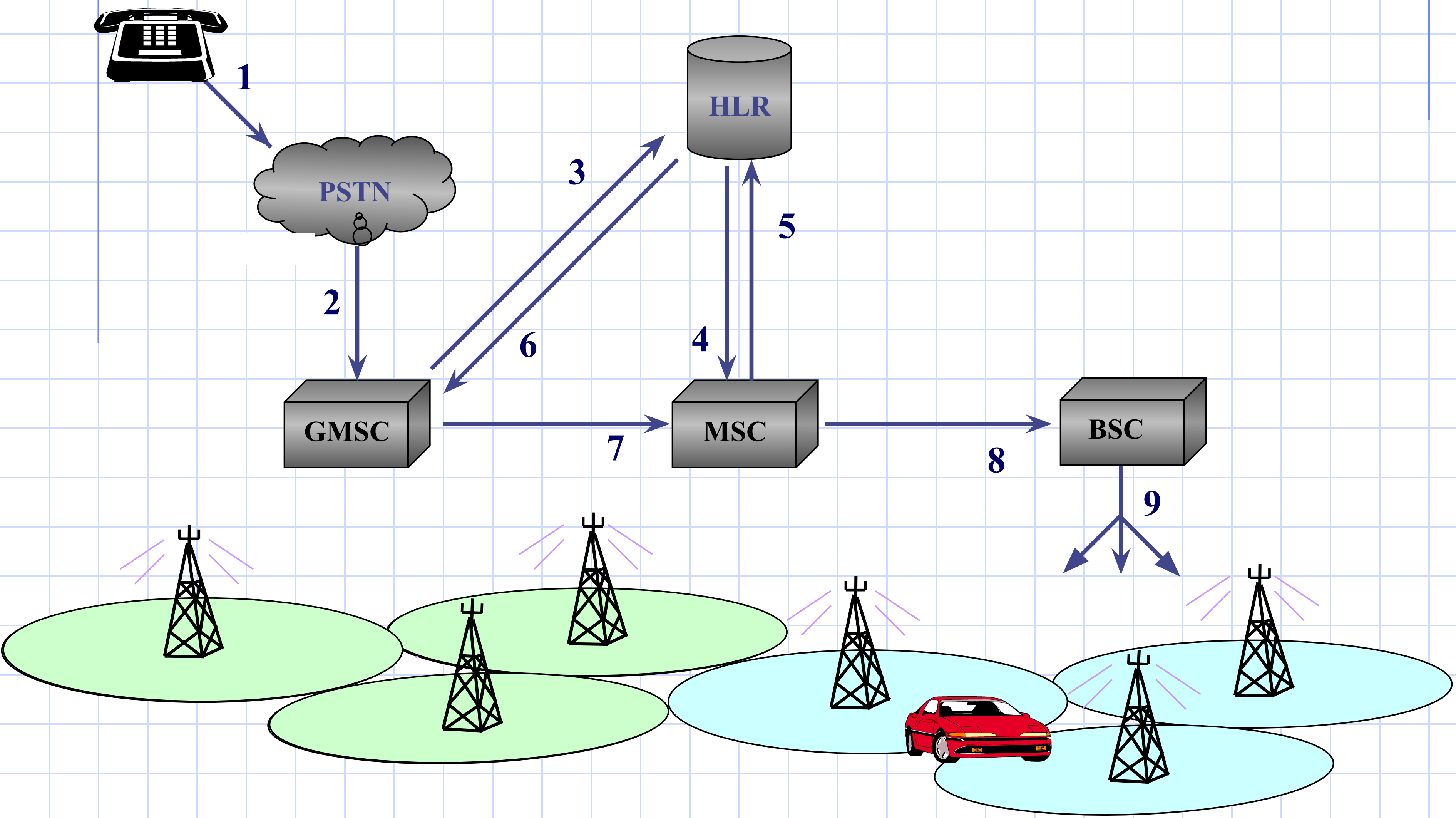


Mobile Originated Call

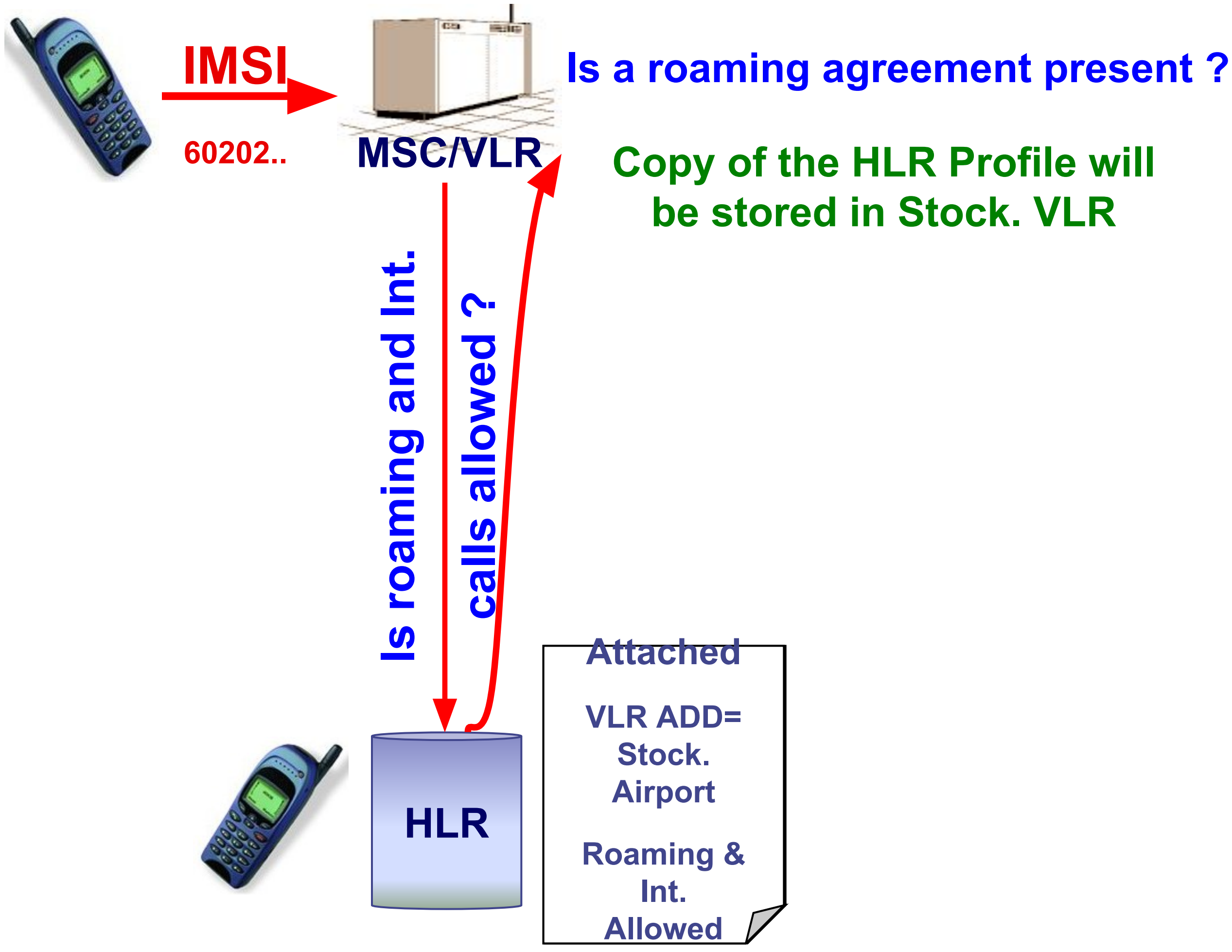


1. The mobile sends a call request along with its IMSI to its serving MSC/VLR which will mark the mobile as busy.
2. Authentication is performed by the MSC to verify the mobile access to the network, and then ciphering is initiated in order to protect the mobile call on the radio path.
3. The mobile sends a call setup message to the MSC with information about the call type, services required and the dialed number.
4. MSC checks the categories of the mobile subscriber to verify that he is authorized to use the required services, and then a link is established between the MSC and BSC.
5. BSC checks the mobile serving BTS for an idle traffic channel and then orders that BTS to seize this channel for a call.
6. The BSC informs the MSC when the traffic channel assignment is complete, and then the MSC/VLR starts to analyze the dialed number and sets up a connection to the called subscriber.

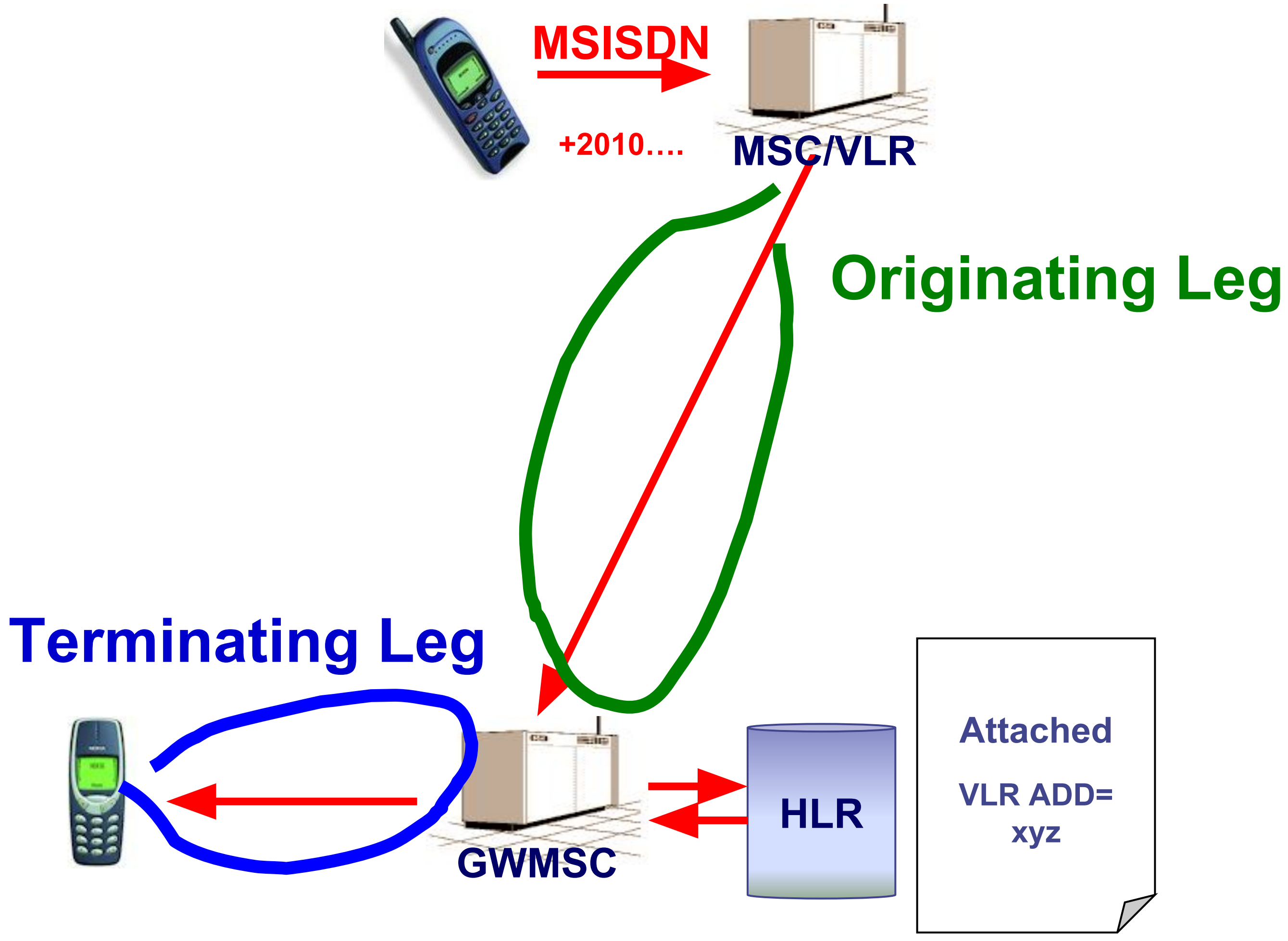
Mobile Terminating call



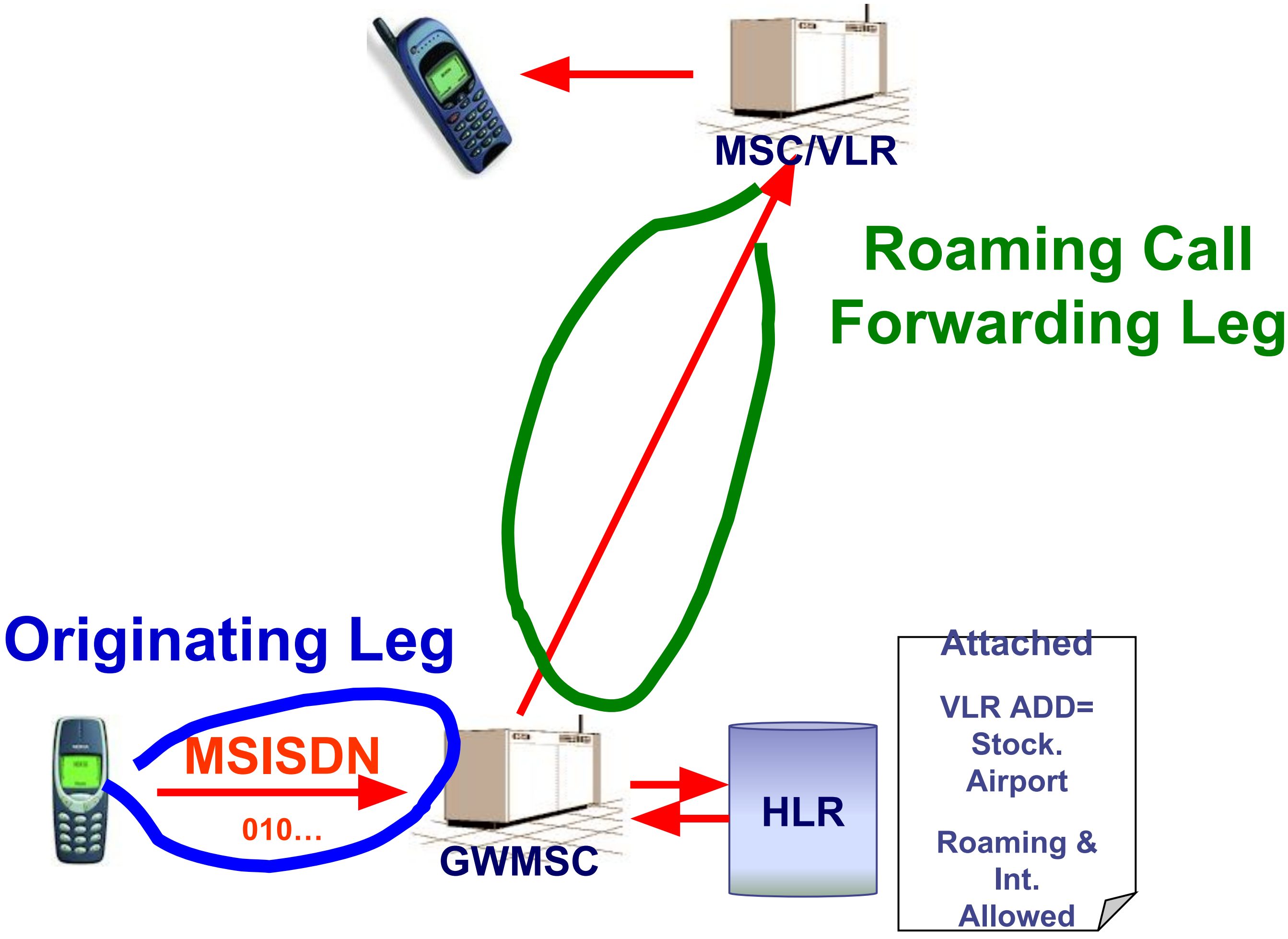
Roaming: Location Update



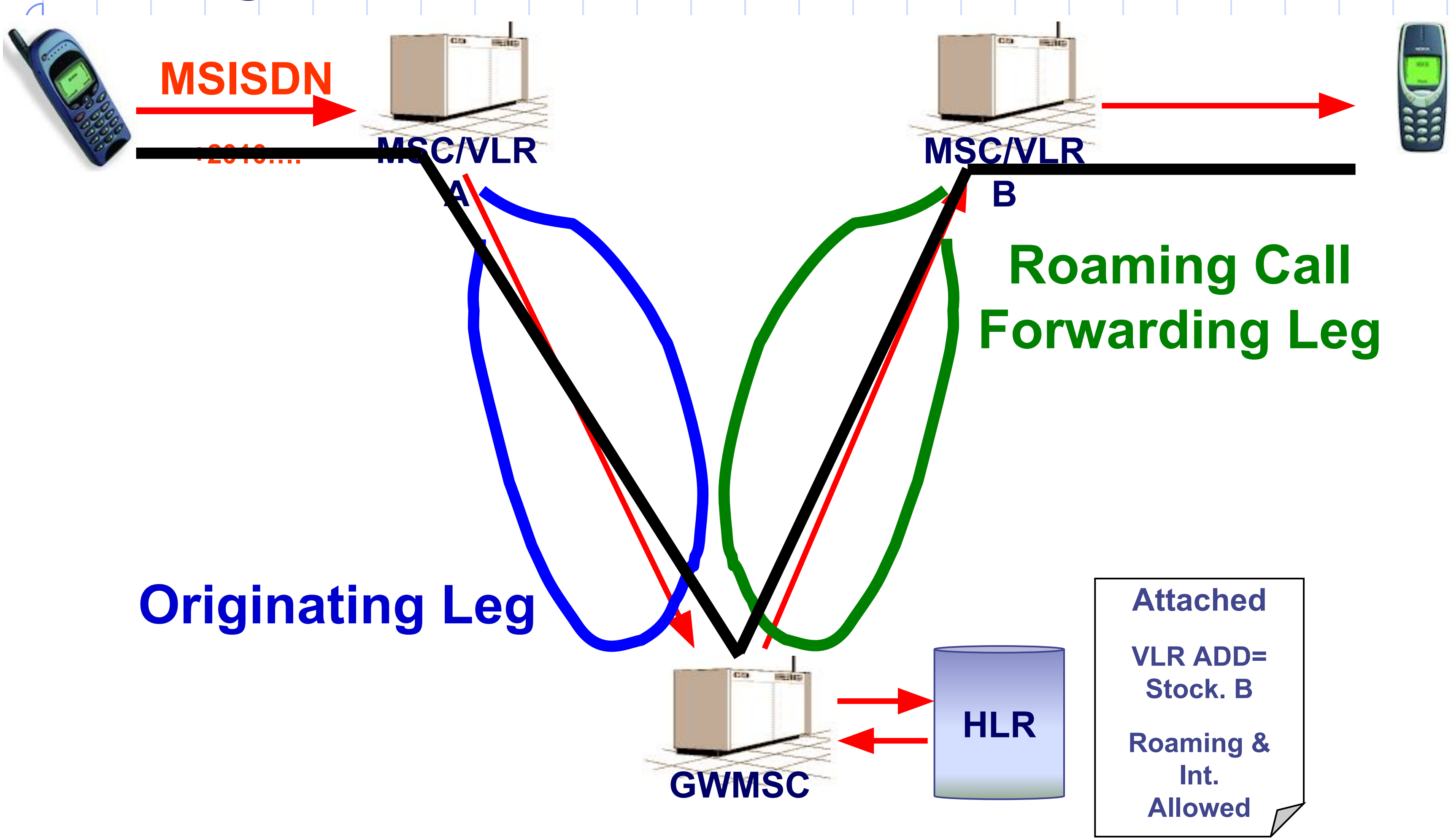
Roaming: Call to HPLMN



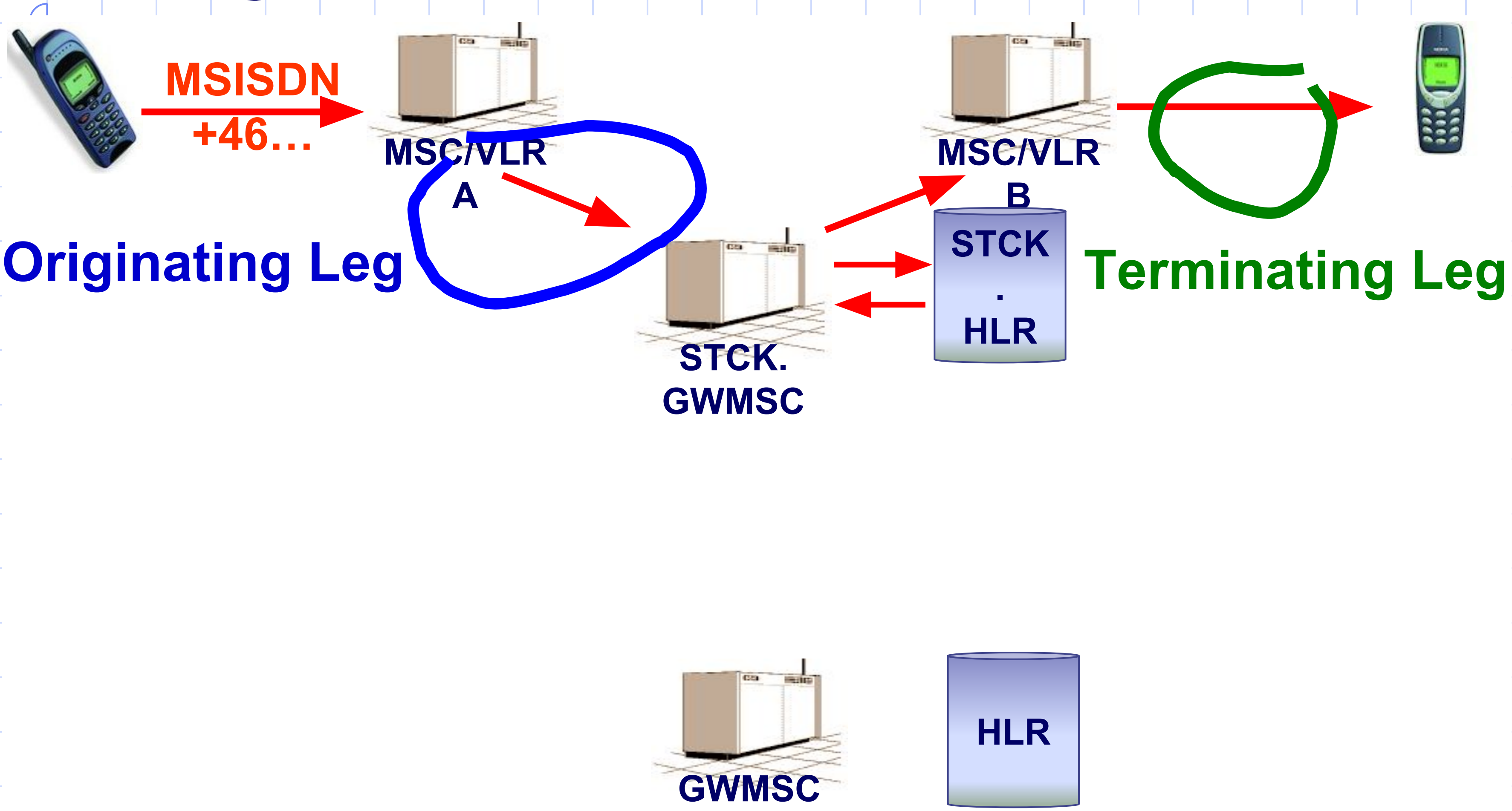
Roaming: Call from HPLMN



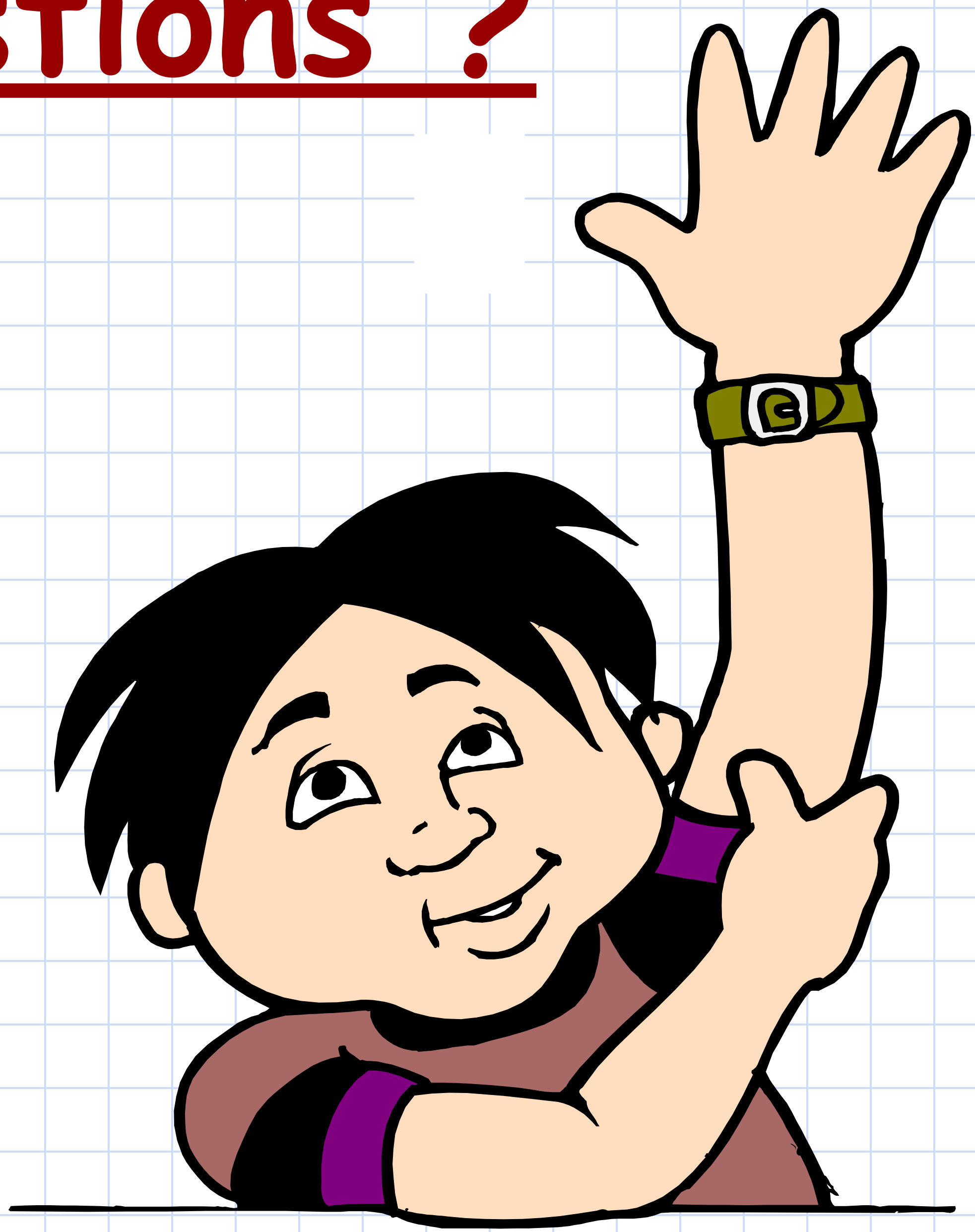
Roaming: Call from another Roamer



Roaming: Call to The visited PLMN



Questions ?



Chapter 8 : GSM Services

Chapter Objectives

By the End of this Chapter you will:

- Know how do Fax and data calls take place
- Know how does SMS is sent and received
- Know how the call of a Prepaid subscriber takes place

Introduction

Examples of GSM non-Speech services

- Fax calls
- Data calls
- Short messages service (SMS)

Conclusion GSM is a telecommunication network rather than a telephony network.

Fax and data Calls

Supported by :

PSTN **ISDN** **GSM**

The receiving node must know:

What is the service requested ? e.g.. (speech, fax or data)

How will the service be performed ? E.g.. (bit rate)

Fax and data Calls

Why do we have additional MSISDN for the same subscriber to be able to receive fax & data calls?

- Information sent at call setup from ISDN or GSM on "how" the service will be performed, is called Bearer Capabilities (BC).
- ISDN and GSM have different transmission requirements and different coding schemes, so they have different bearer capabilities referred to as ISDN-BC and GSM-BC.
- PSTN can not provide this type of information during call setup, so it can not distinguish between a telephony call and a fax or data call.
- An Additional MSISDN (AMSISDN) will be allocated to a mobile subscriber who has the service of receiving fax or data calls.

Fax and data Calls

For an MSC to be able to handle fax or data calls, it should be provided with a Data Transmission Interface **(DTI)** which is used for :

- ❖ rate adaptation.
- ❖ Protocol conversion.
- ❖ Providing modems.

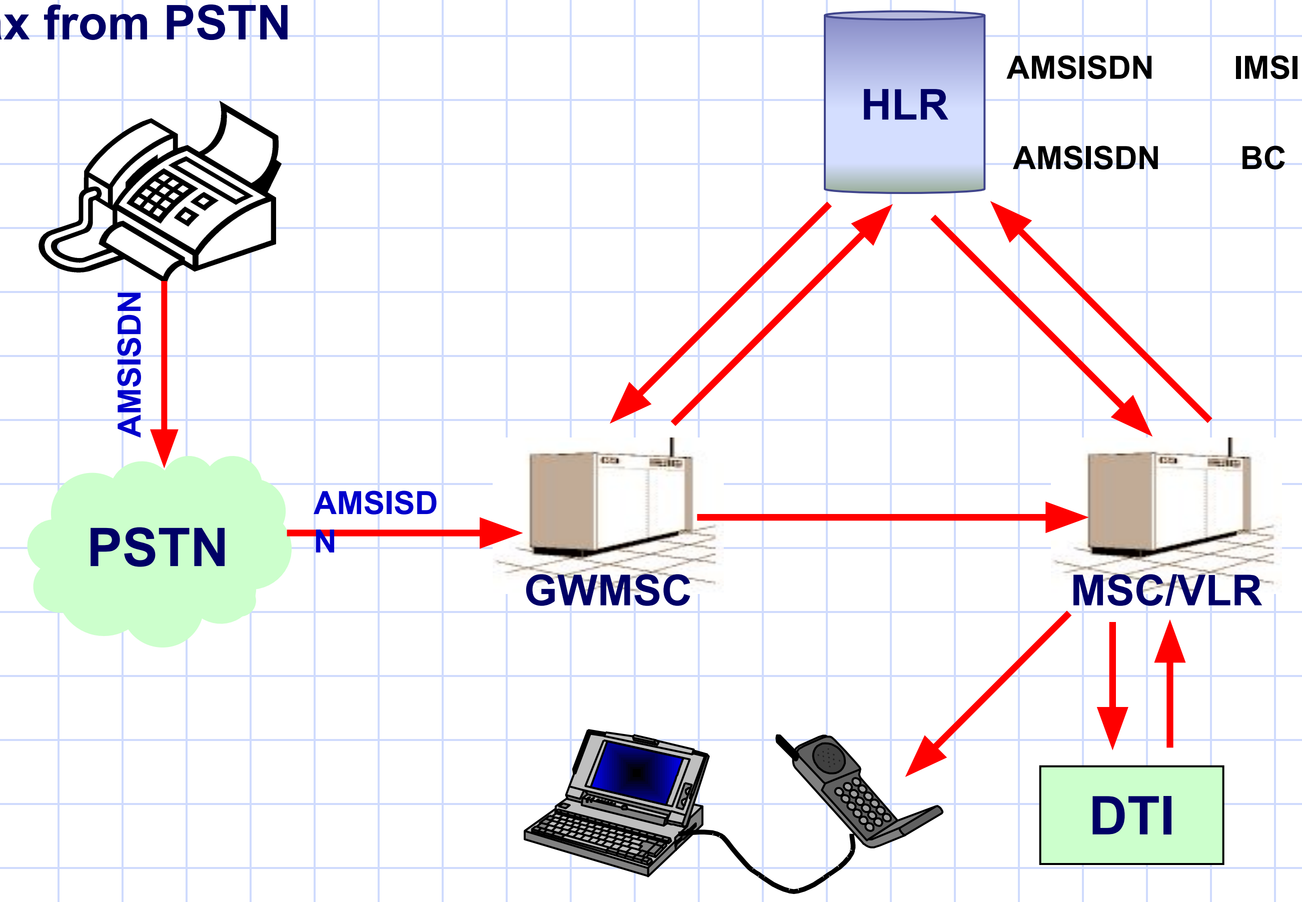
such that Fax , Data calls are to be established to/from mobile subscribers.

Standard Bit Rate in GSM = 9.6 KB/S
This rate can be increased into 14.4 KB/S

The High Speed Circuit Switched Data ***(HSCSD)*** uses 4 time slots to perform a data call thus increasing the rate into 57.6 KB/S

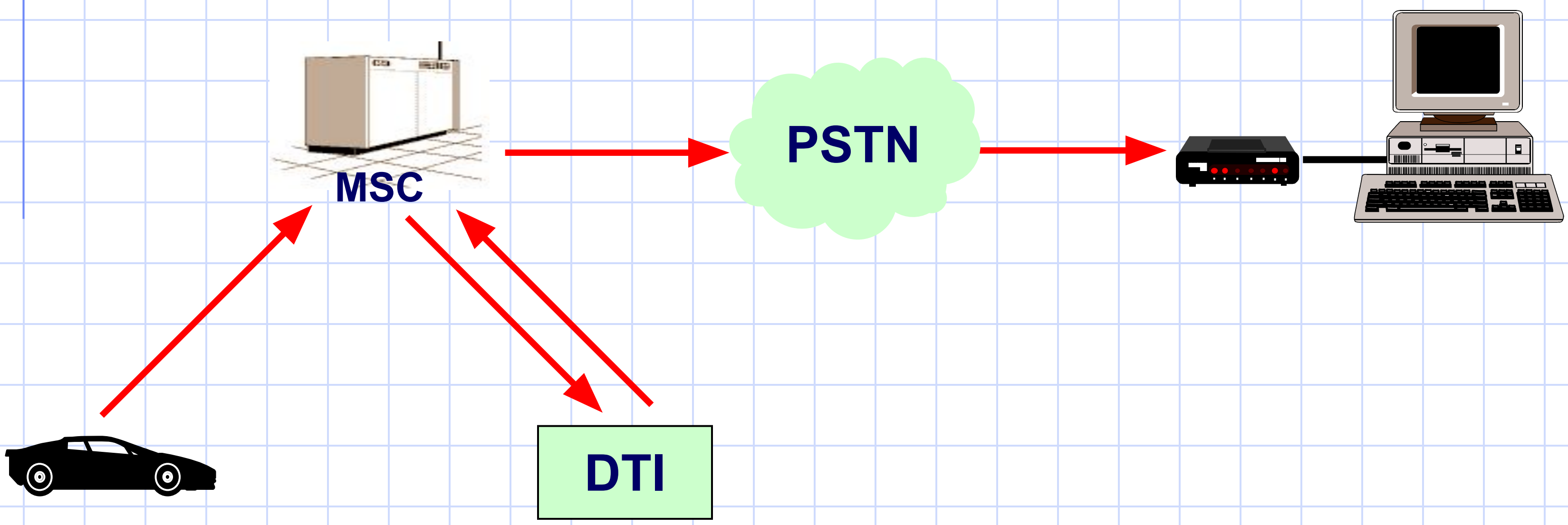
Fax and data Calls

Fax from PSTN



Fax and data Calls

GSM fax call



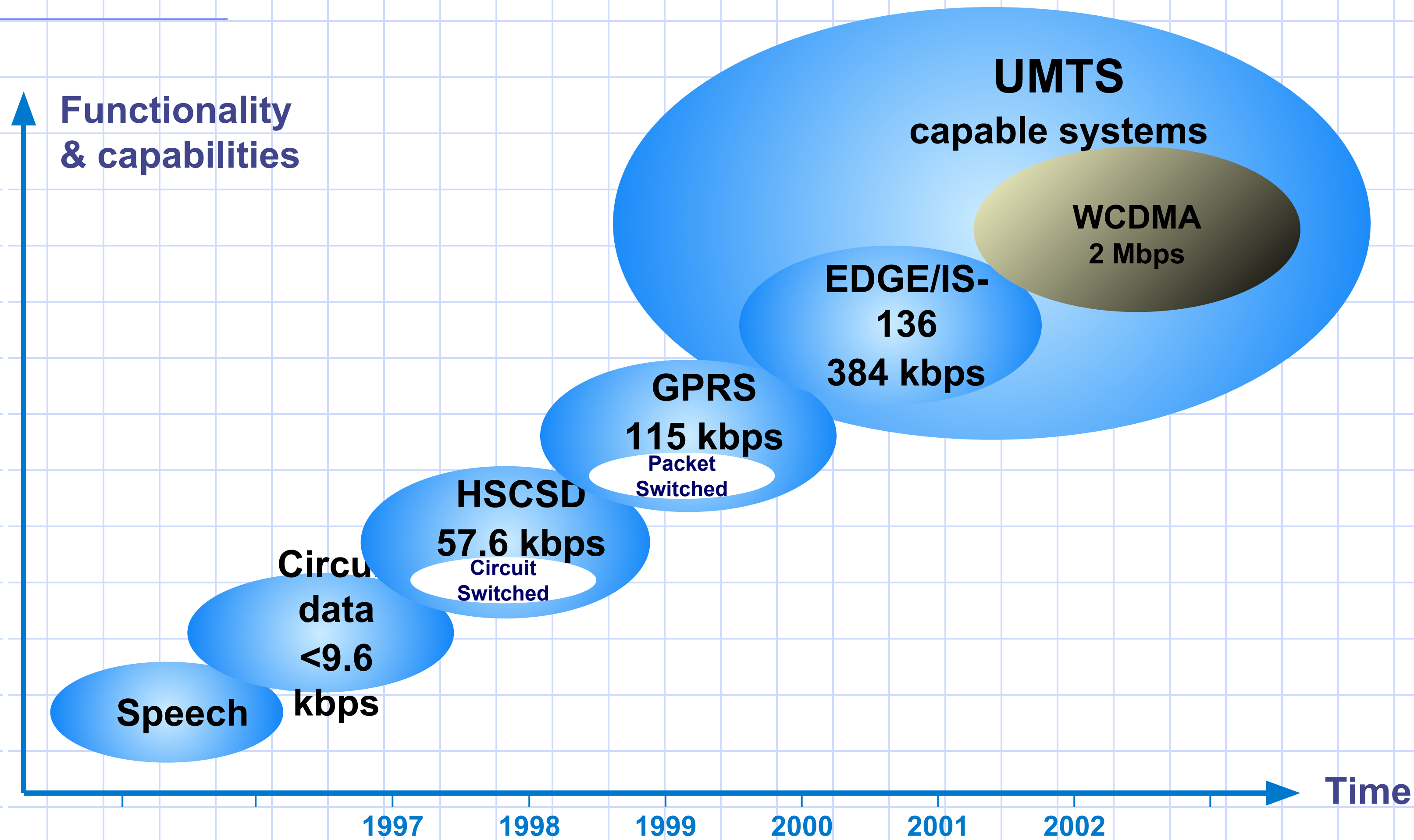
Future Enhancements

GPRS General Packet Radio Services
(Up to 171 Kbit/sec)

EDGE Enhanced Data Rates for GSM Evolution
(Up to 48 Kbit/sec per channel)

UMTS Universal Mobile telecommunication System
(Up to 2 MB)

GSM Evolution



Short Message Service

The Short Message Service (SMS) allows a mobile subscriber to send and receive text messages composed of 160 characters at most.

The short messages sent or received are handled by the Short Message Service Center (SMSC), which consists of three parts :

- Service Center (SC)
- SMS GMSC (SMS-GMSC)
- SMS inter-working MSC (SMS-IWMSC)

Short Message Service

Service Center

- Handles the delivery of short messages to/from Short Message Entities (SME), which can be any originator or receiver of short messages such as mobile, fax, … etc.
- Stores the short messages.
- Create billing files.
- Monitors system events and alarms.

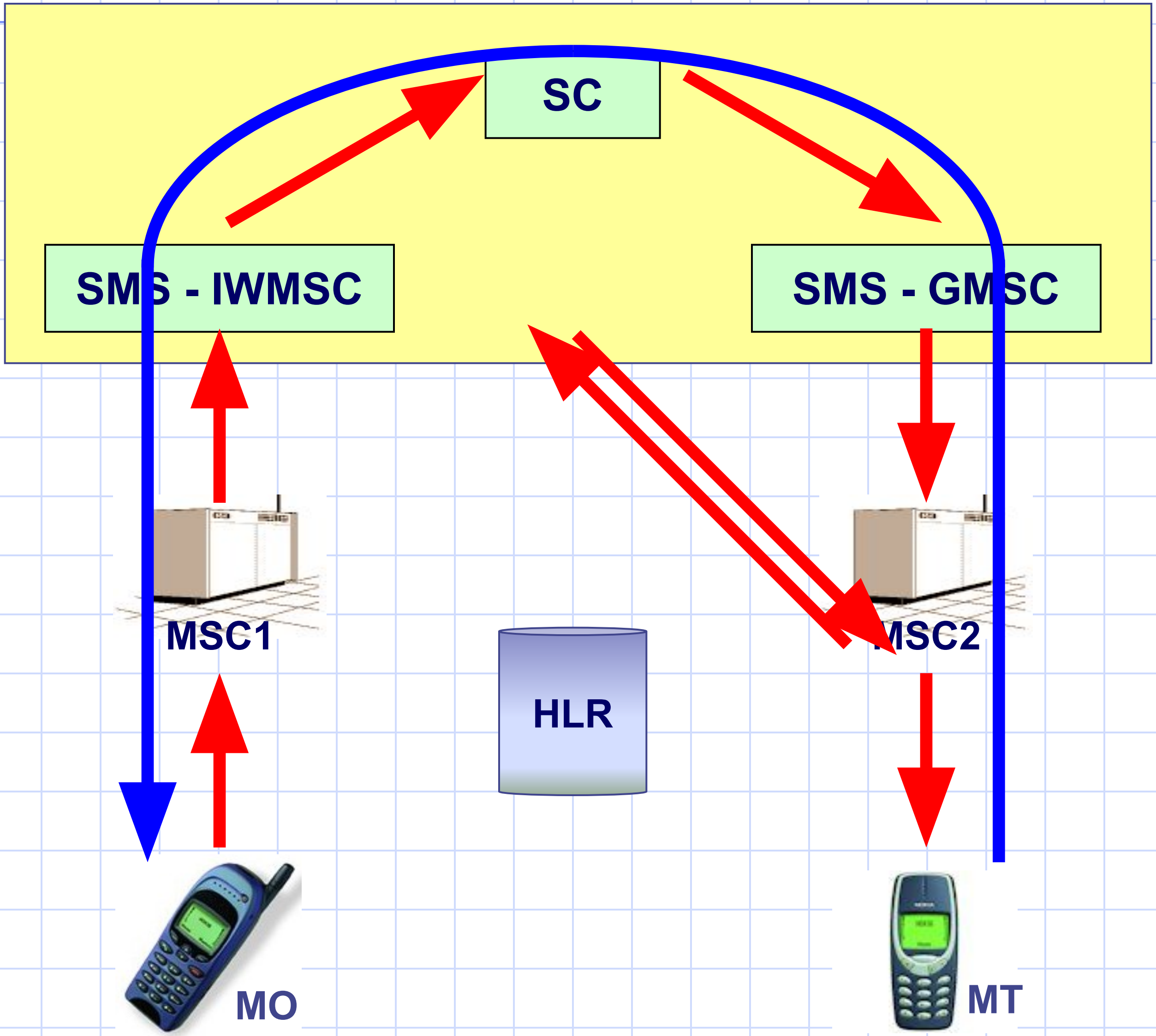
SMS-GMSC

- Interrogates the HLR to determine the location of a mobile subscriber.
- Forwards the short message to a mobile subscriber via its serving MSC.

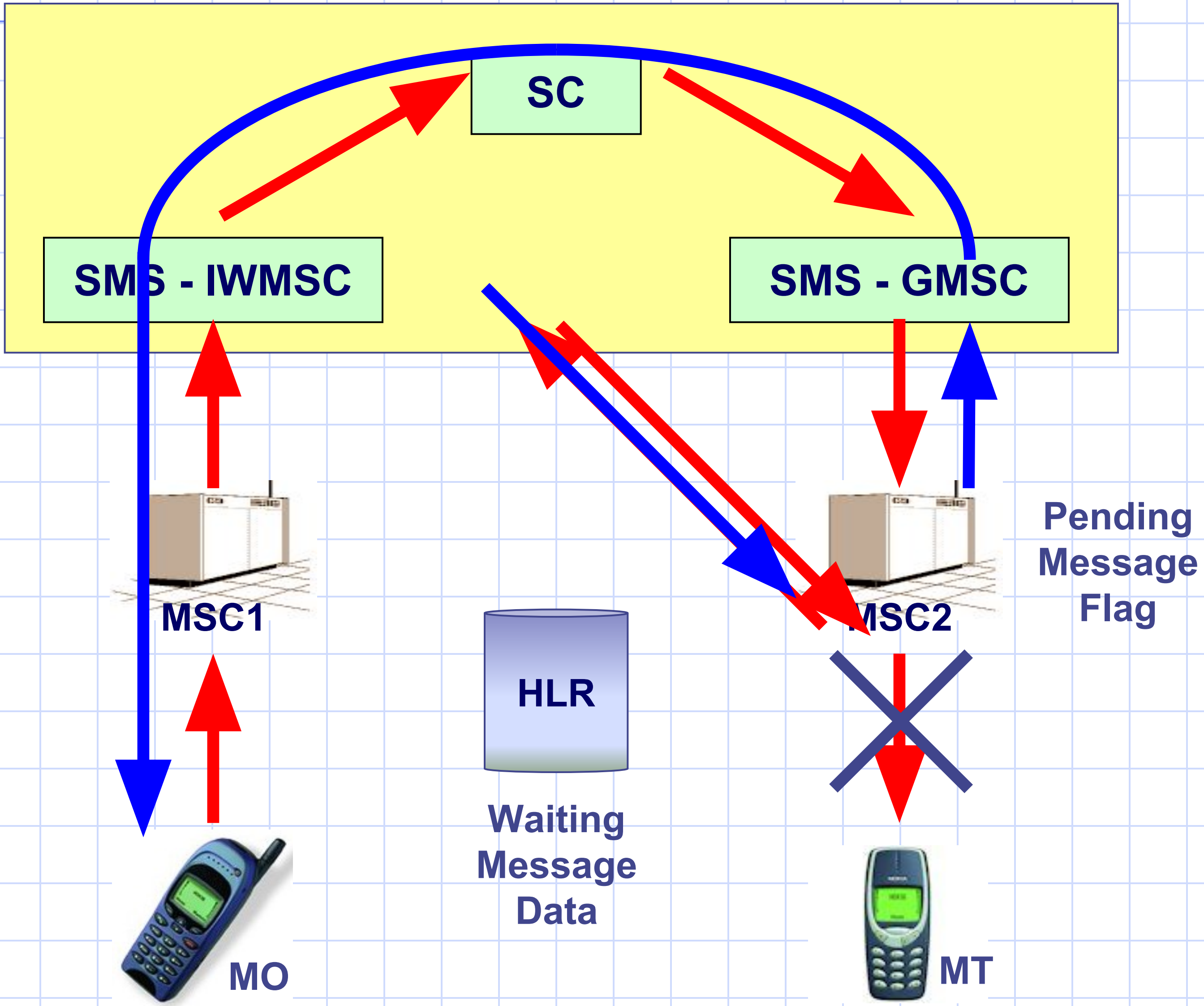
SMS-IWMSC

- Receives the mobile originated short message from any MSC in the network.
- Receives an alert message from the HLR to inform the SC that a mobile subscriber who was absent during a previous short message delivery attempt is reachable again.

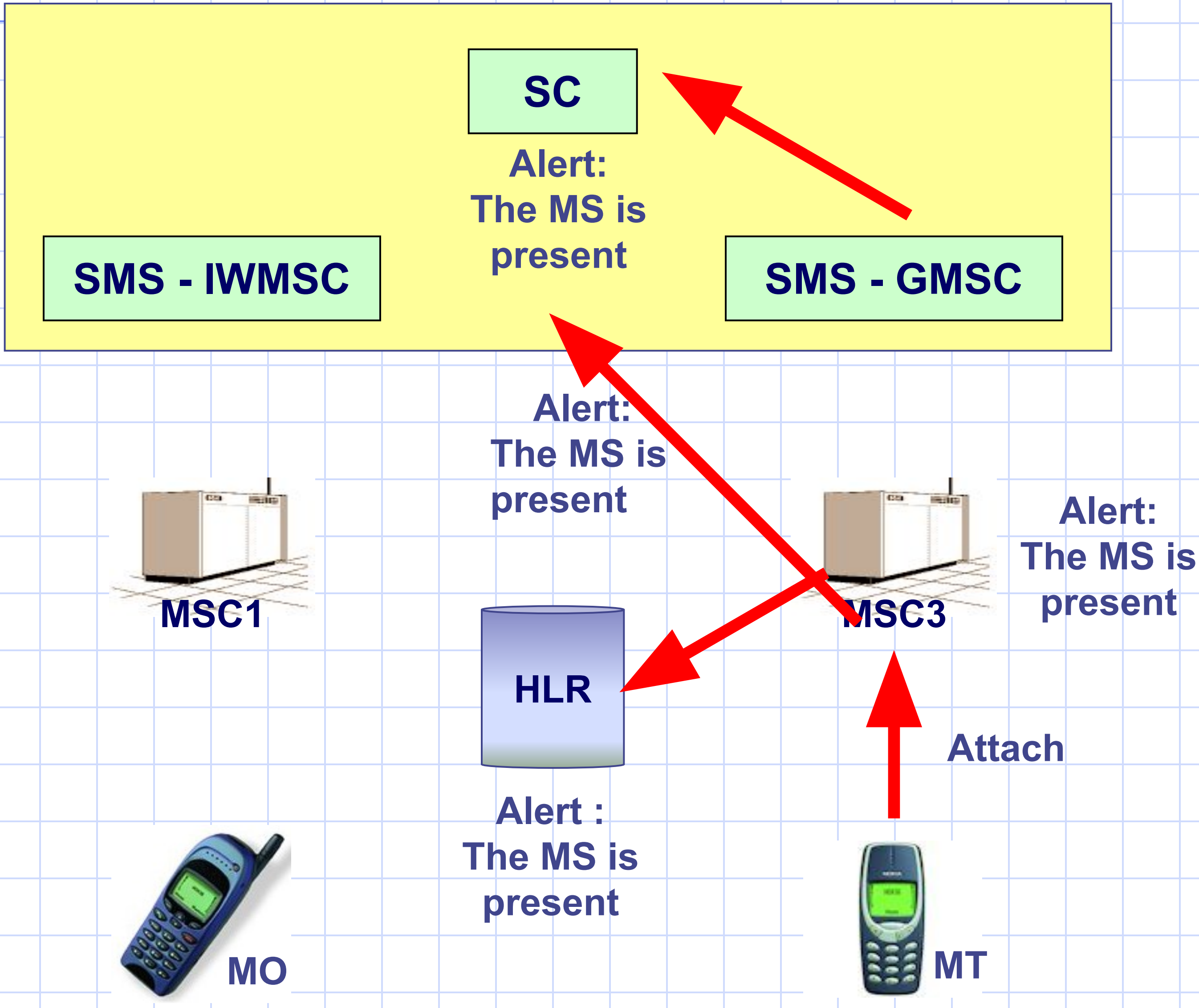
Mobile Originated SMS



Unsuccessful Message Transfer



The Mobile is present once more



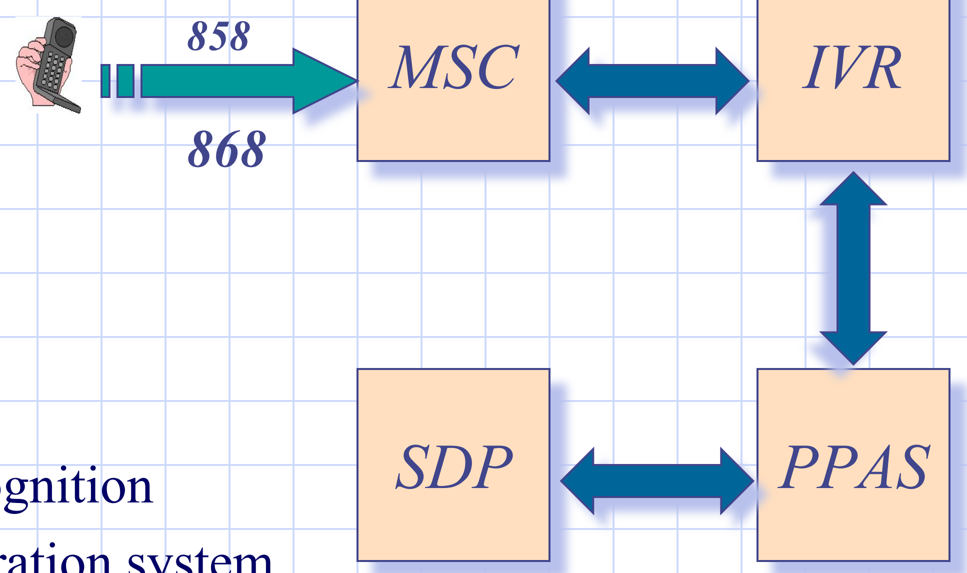
Advanced GSM Services

WAP Wireless Applications Protocol

CAMEL Customized Application of Mobile Enhanced Logic

The Pre-payment system

A pre-paid subscriber

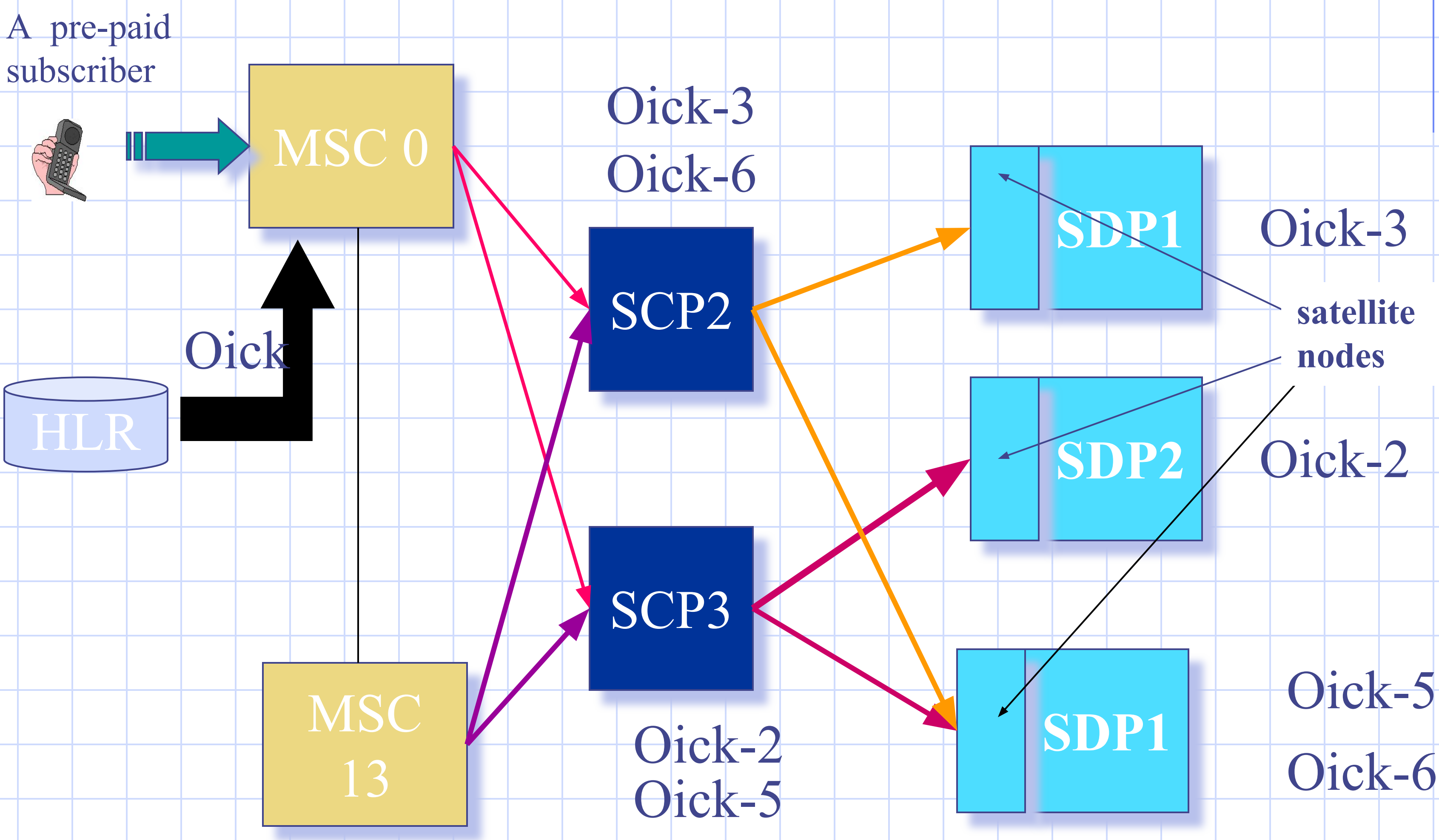


Charging 858

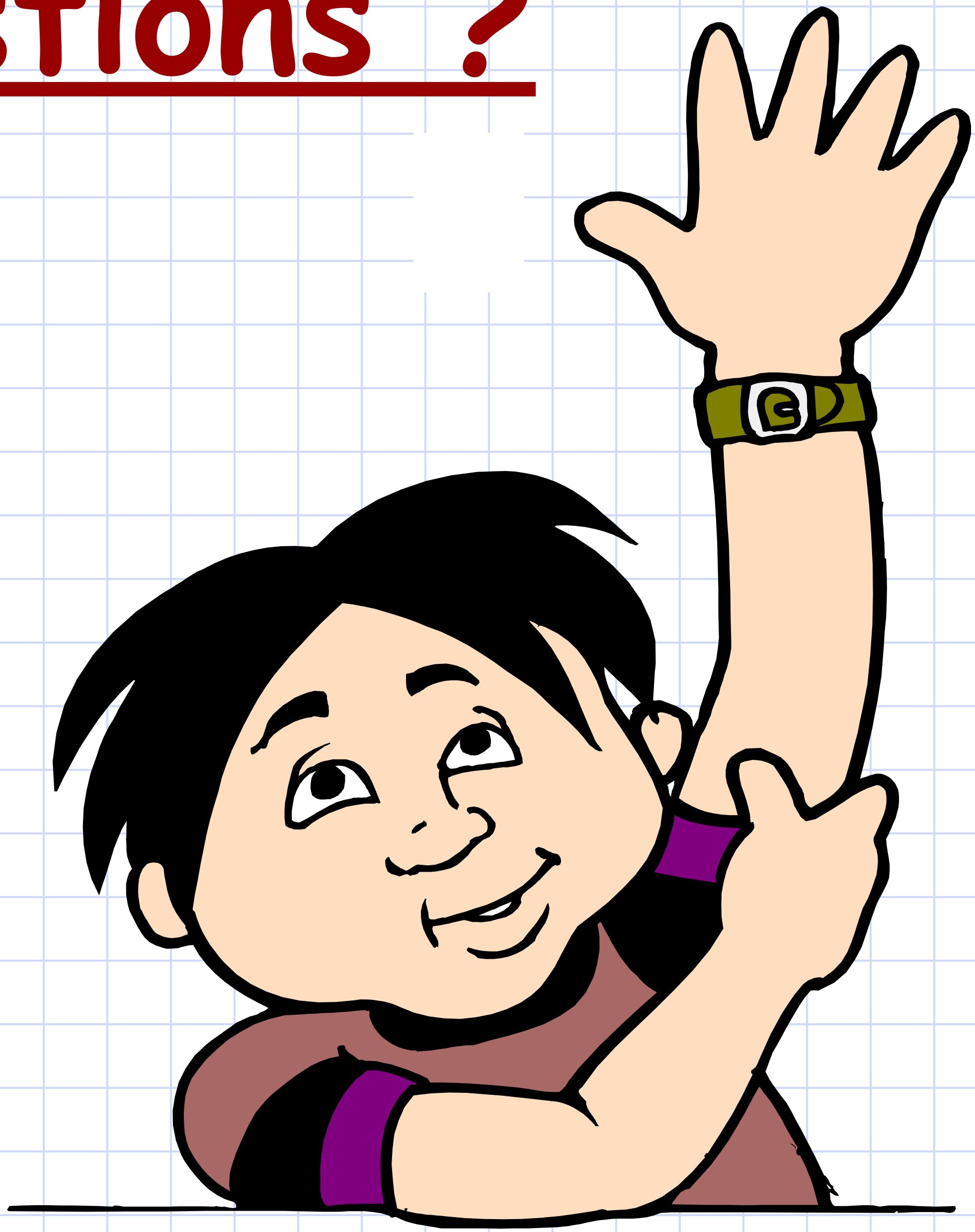
Balance inquiry 868

IVR :Interactive Voice Recognition
PPAS:Prepayment Administration system.
SDP :Service Data Point.

Pre-payment Originating call



Questions ?



Thank You For your time